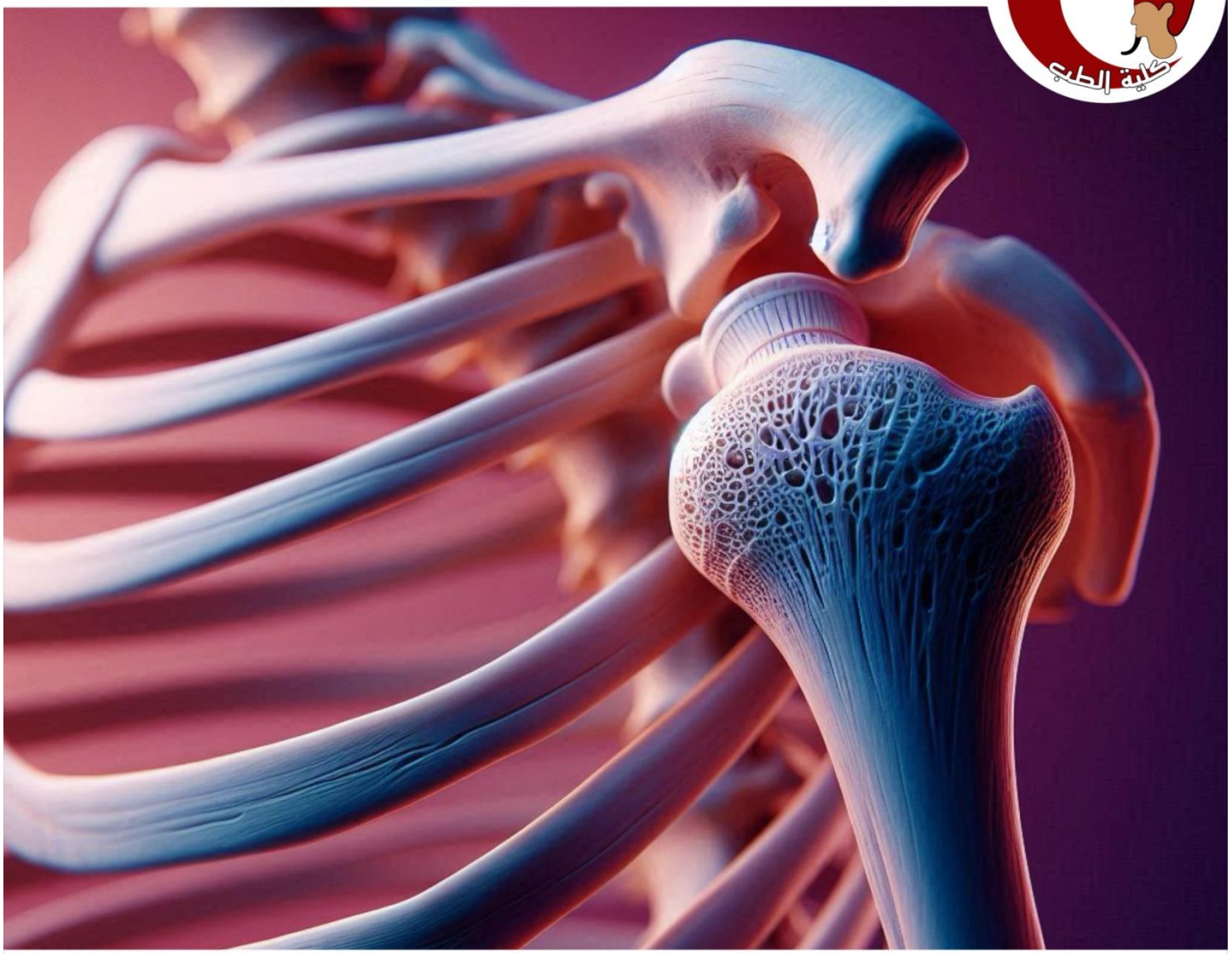
Sohag university
Faculty of medicine

General Anatomy & embryology

Anatomy department



GAD-Block

2024/2025

Chapter	Title	Page
1	**General Anatomy**	2
	Skin	7
	Fascia	10
	Skeletal System	12
	Joints	19
	Muscular System	24
2	**Systemic Anatomy**	28
	Cardiovascular System	29
	Respiratory System	33
	Digestive System	37
	Nervous System	47
	Urinary System	50
	Genital System	52
3	**General Embryology**	56
	1st week of development	74
	2nd week of development	82
	3rd week of development	90
	4th week of development	100
4	**Twin Development**	145

Chapter 1

General Anatomy

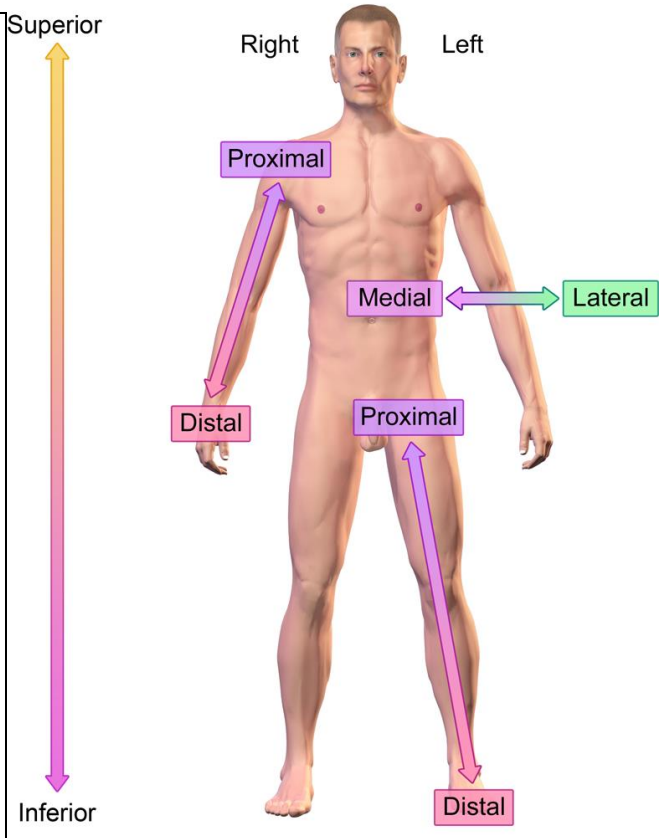
General Anatomy

Human anatomy :

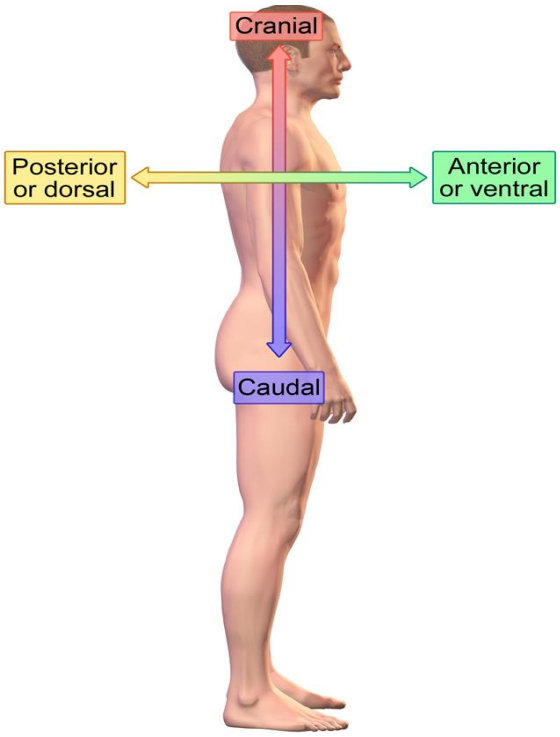
is the study of the structure of the body, from the macroscopic picture of view

A -Anatomical Position

- The body is standing,
- Head, eyes, and toes directed forward.
- Upper limbs by the sides with the palms facing to the front.
- Lower limbs close together with the feet parallel and the toes directed to the front.



Anterior view



Lateral view

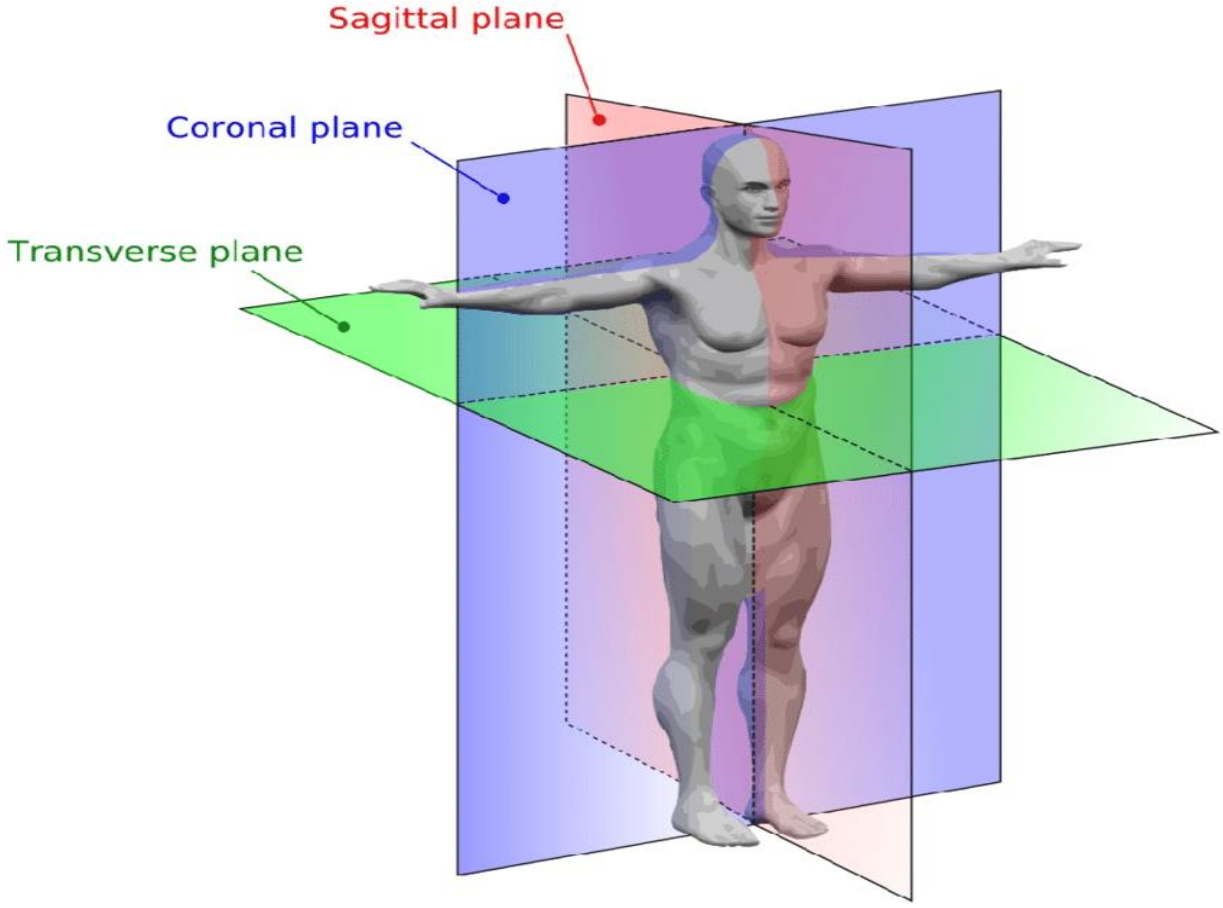
B. Anatomical terms

- ✓ **Anterior** ►Towards the front.
- ✓ **Posterior (Dorsal)** ►Towards the back
- ✓ **Superior (cranial)** ►Towards the head.
- ✓ **Inferior (Caudal)** ►Towards the feet.
- ✓ **Proximal (nearest)** ►closer to the root. The elbow is proximal to the wrist on the upper limb.
- ✓ **Distal** (Remote) ► away from the root. The elbow is distal to the shoulder joint on the upper limb.
- ✓ **Median** ►A point or area on the midsagittal plane (median plane).
- ✓ **Medial** ►A point or area that is closer to the midsagittal plane than another.
- ✓ **Lateral** ►A point or area that is more distant from the mid-sagittal or median plane. For example the outer side of the elbow is lateral when compared with, the inner side.

✓ **Superficial**: near the surface

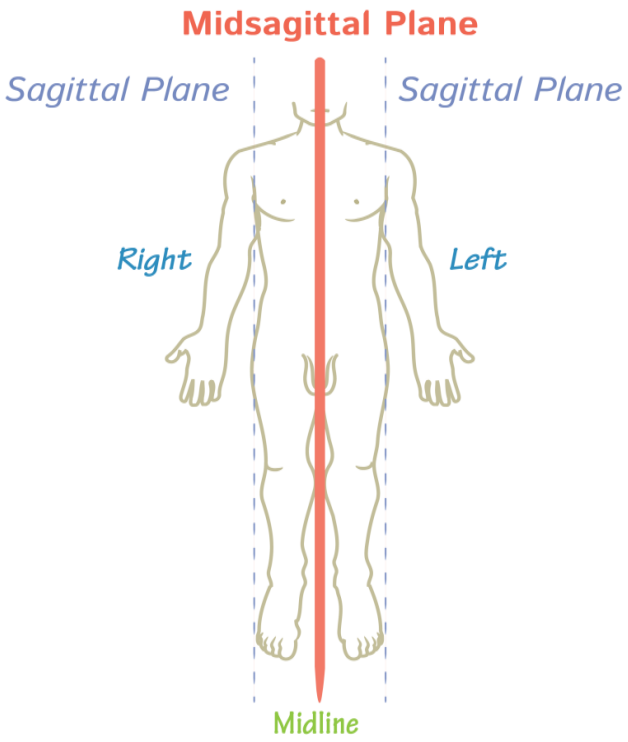
✓ **Deep**: away from the surface

C - Planes of the Body



- **Frontal or Coronal Plane**: - A plane that divides the body into an anterior and a posterior portion.
- **Transverse Plane**: - A plane that divides the body or limbs into upper and lower parts.

- **Median (midsagittal) plane** is the vertical plane passing longitudinally through the center of the body, dividing it into right and left equal halves.



- **Sagittal planes** are vertical planes passing through the body parallel to the median plane.

How to Study Anatomy???

Regional anatomy

is based on the organization of the body into parts: the head, neck, trunk (subdivided into thorax, abdomen, pelvis/perineum, back) and paired upper and lower limbs.

Systemic anatomy:

based on the organization of the body into systems (**A system is a collection of elements or components that are organized for a common purpose**).



The integumentary system -The skeletal system -The muscular system-The articular system -The nervous system-The cardiovascular system-The lymphatic system-The digestive system-The respiratory system -The urinary system and The reproductive system.

Clinical (applied) anatomy:

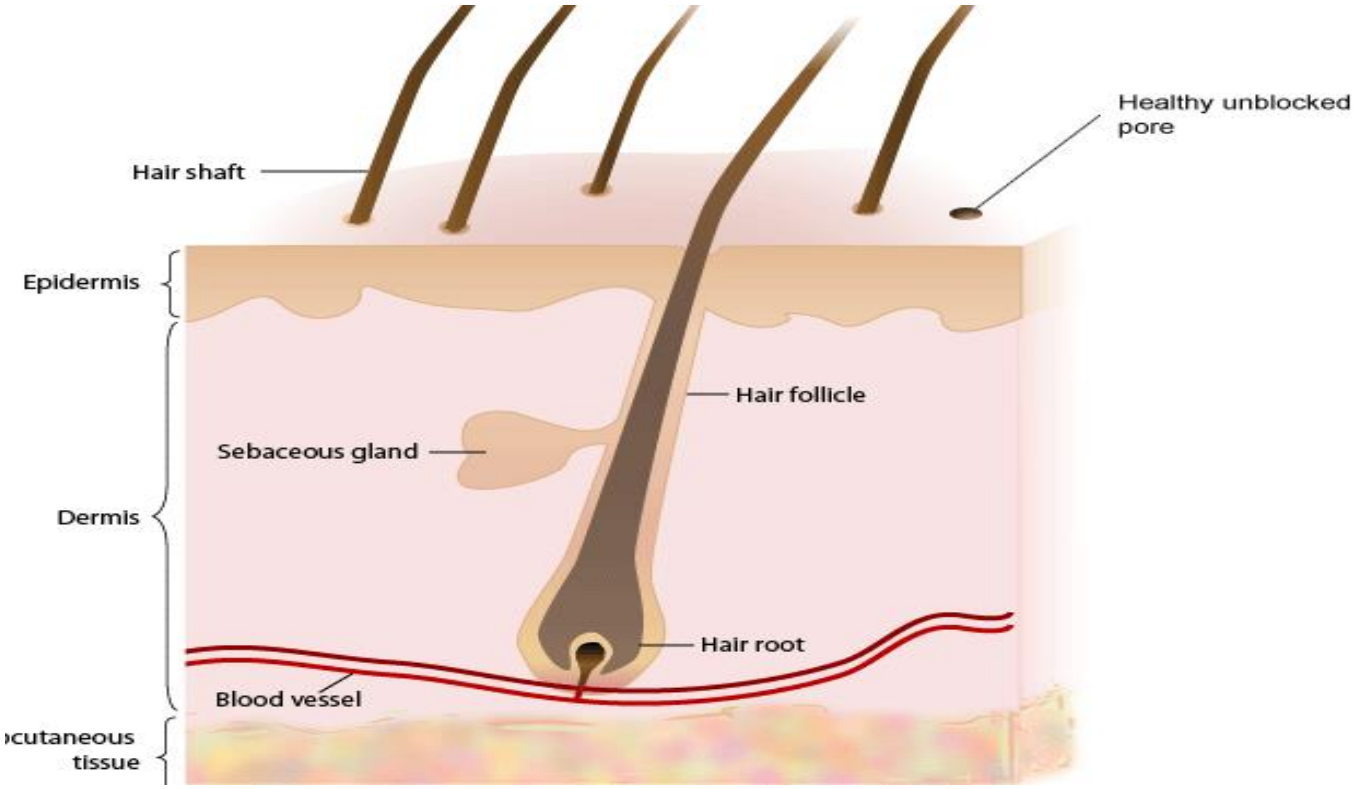
aspects of the structure and function of the body important in the practice of medicine.

Skin

Skin is the outside covering of body tissue, accounts for 12% of body weight.

The skin consists of a **superficial** cellular layer, the **epidermis**, which creates a tough protective outer surface

, and a basal **(deep)** layer, the **dermis**.



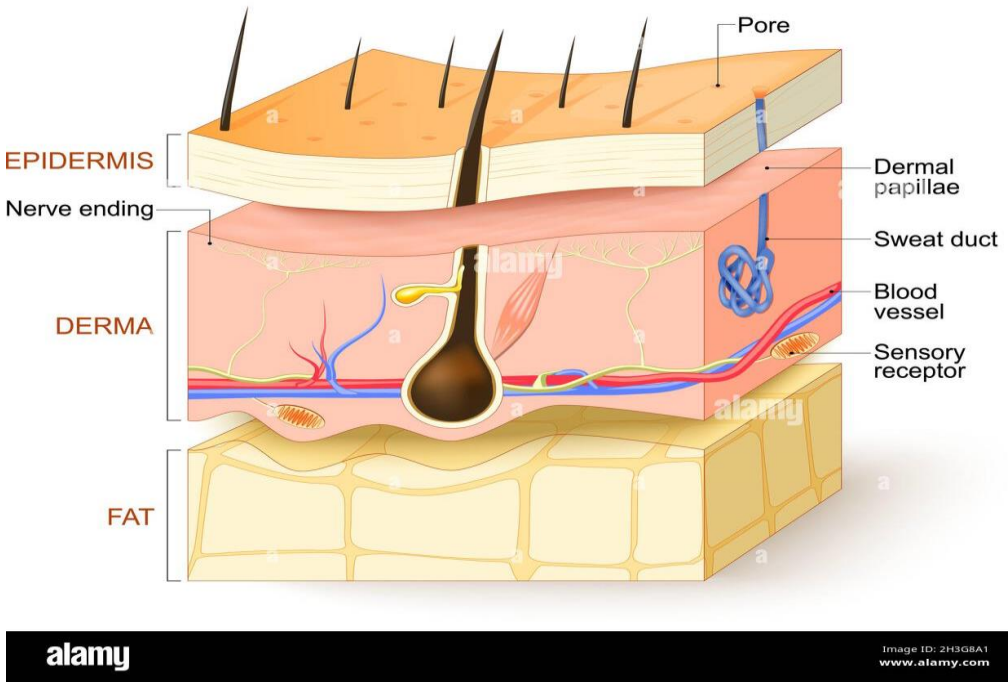
The epidermis

- A keratinized epithelium composed of keratin (a fibrous protein)
- Continuously rubbed away with replacement of new cells from the basal layer.
- Avascular (no blood vessels or lymphatics) and is nourished by the vessels in the underlying dermis.
- Contains nerve endings that are sensitive to touch, irritation (pain), and temperature.

The Dermis

- Formed by a dense layer of collagen and elastic fibers.
- The deep layer of the dermis contains hair follicles, with their associated smooth muscle and sebaceous glands.
- In addition, sweat glands, lymphatic vessels and blood vessels are present also in the dermis

Skin layers



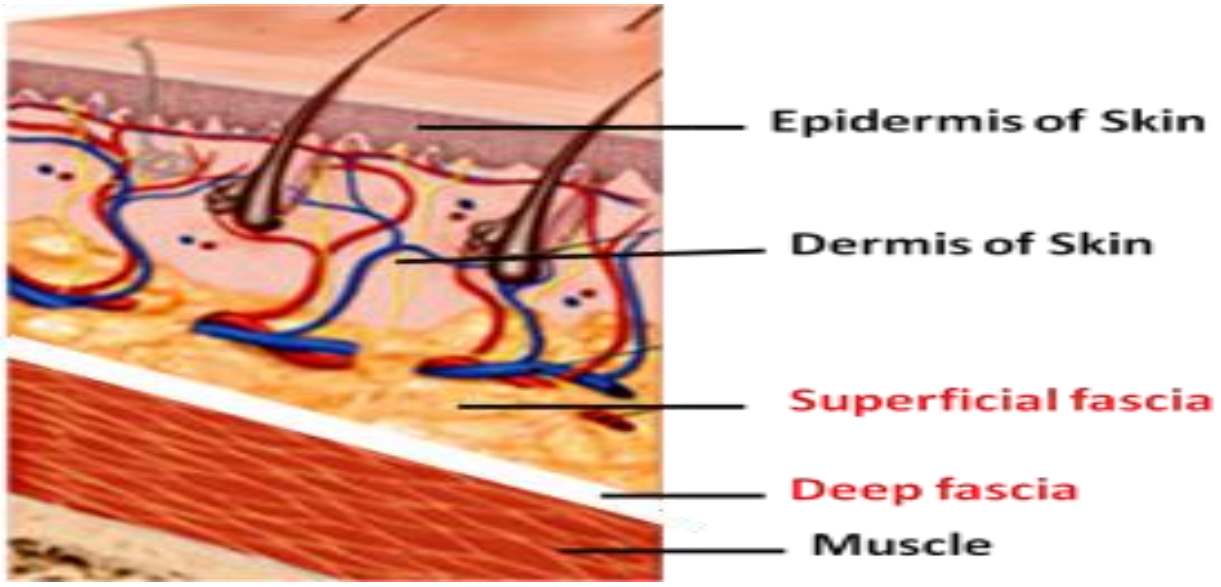
Skin functions

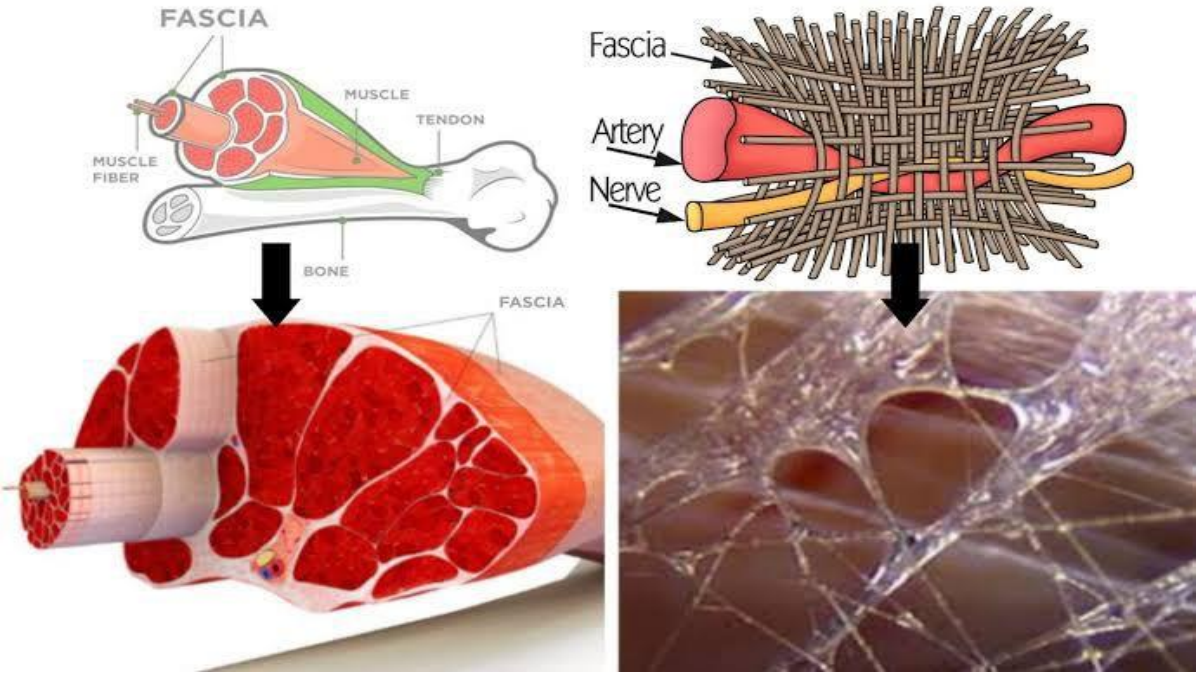
- **Protection** for the body from environmental effects.
- **Sensation**: contains a variety of nerve endings that jump to heat and cold, touch, pressure, vibration, and tissue injury;
- **Heat regulation** through sweat glands, blood vessels, and fat.
- **Storage** : acts as a storage center for lipids and water
- **Absorption**: Oxygen, nitrogen and carbon dioxide can diffuse into the epidermis in small amounts,

Fascia

A fascia is a connective tissue that surrounds muscles, blood vessels, and nerves, binding those structures together.

It consists of **a superficial fascia** and **a deep fascia**





	Superficial fascia. (subcutaneous layer)	Deep fascia
Composition	- Immediately deep to the skin - composed of areolar connective tissue and adipose tissue	-Composed of dense irregular connective tissue that lines the body wall and limbs - holds muscles together, separating them into functional groups
Function	-Stores water and fat. - Reduces the rate of heat loss - provides mechanical protection against traumatic blows. - provides a framework for nerves and blood vessels to enter and to exit muscles	- allows free movement of muscles - supports nerves, blood vessels and lymphatic vessels - fills spaces between the muscles

NOTE

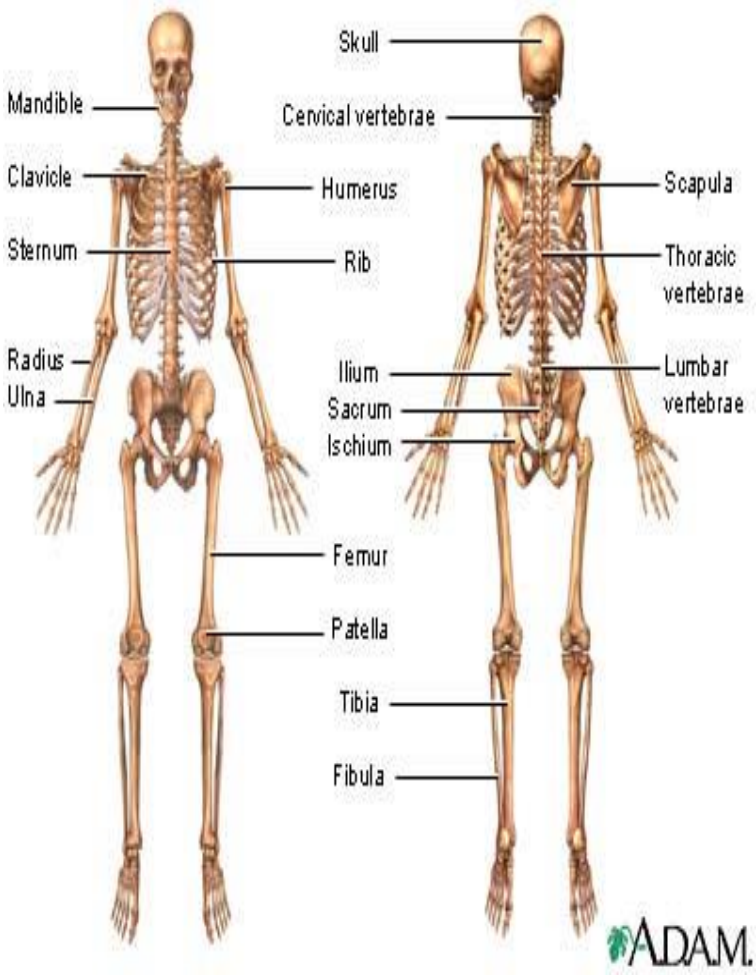
Superficial fascia is present, but does not contain fat, in the eyelids, ear, scrotum, penis and clitoris. It is very dense in the scalp, in the palms of the,hands and soles of the feet .

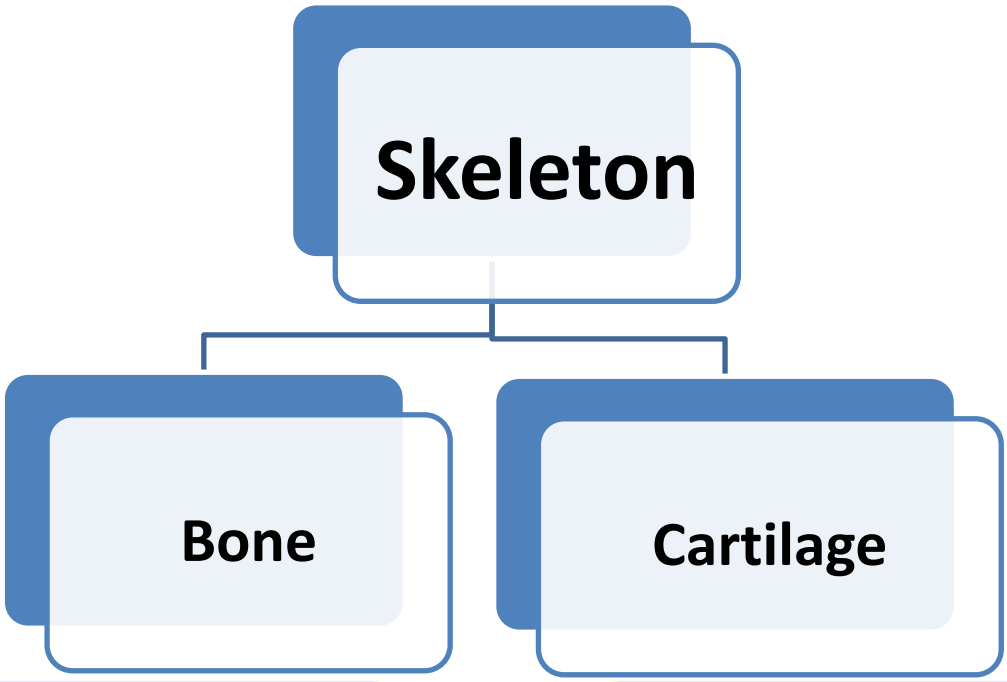
The Skeletal System

The skeleton of the body is composed of **bones and cartilages** and has two main parts

The axial skeleton consists of the bones of the head (cranium), neck (cervical vertebrae), and trunk (ribs, sternum, vertebrae, and sacrum).

The appendicular skeleton consists of the bones of the limbs, including those forming the pectoral (shoulder) and pelvic girdles





Defenition :

a living tissue, form of connective tissue that makes up most of the skeleton

functions :

- Protection for vital structures.
- Support for the body.
- The mechanical basis for movement.
- Storage for salts (e.g., calcium).
- A continuous supply of new blood cells (produced by the bone marrow within

Defenition :

is a semi rigid connective tissue that forms parts of the skeleton where more flexibility is necessary (e.g., the costal cartilages that attach the ribs to the sternum).

functions :

- fibrous connective tissue that surrounds cartilage element is perichondrium.

-The periosteum and perichondrium help in nourishing the tissue

Types of Bones in the Human Body

There are 6 types of bones in the human body.

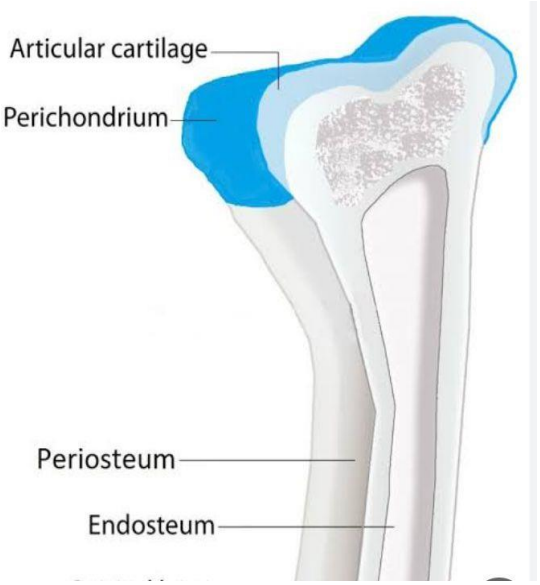
- ☒ Long Bones (bones of limbs)The Long Bones are those that are longer than they are wide, The long bones in the upper limb include (clavicle – humerus-radius and the ulna) in the lower limbs include (femur-tibia and fibula) in addition to the metacarpals and metatarsals of the hands and feet and the phalanges of the fingers and toes.

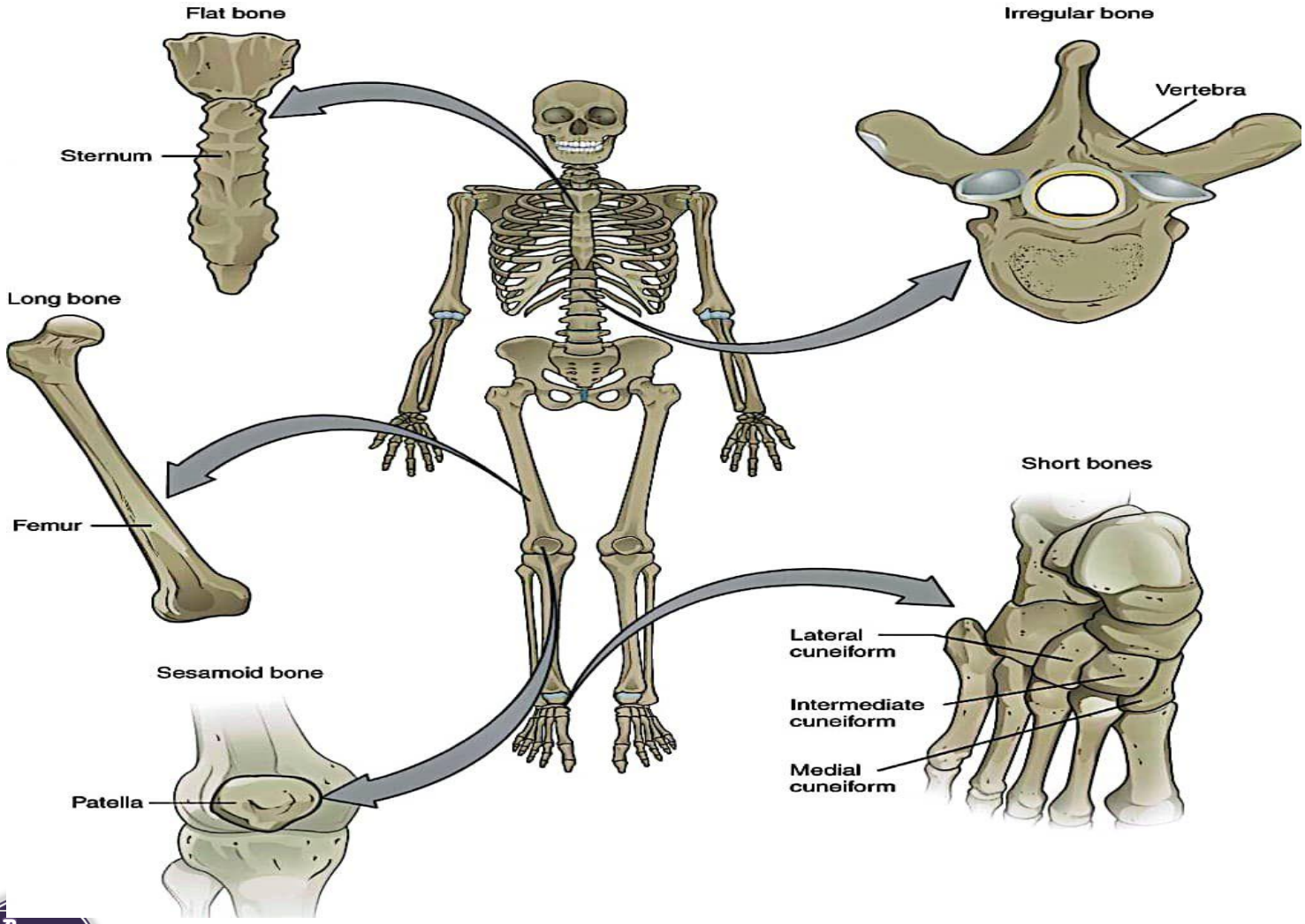
- ☒ **Short Bones**: Bones that is as wide as they are long. Examples of these bones include the tarsals in the foot and the carpals in the hand.
- ☒ **Flat Bones:** The classic example of a flat bone is the Scapula (shoulder blade). The Sternum (breast bone), and the skull.
- ☒ **Irregular Bones:** These are bones in the body which do not fall into any other category, due to their non-uniform shape. Good examples of these are the Vertebrae and Sacrum.
- ☒ **Sesamoid Bones**: usually short or irregular bones, imbedded in a tendon (for protection). The most obvious example of this is the Patella (knee cap).

Pneumatic bones: are bones which contain air filled cavities as bones enclosing Paranasal sinuses

Note

The periosteum is richly supplied with sensory nerves that carry pain fibers so it is sensitive to pain while bone itself is not sensitive to pain .





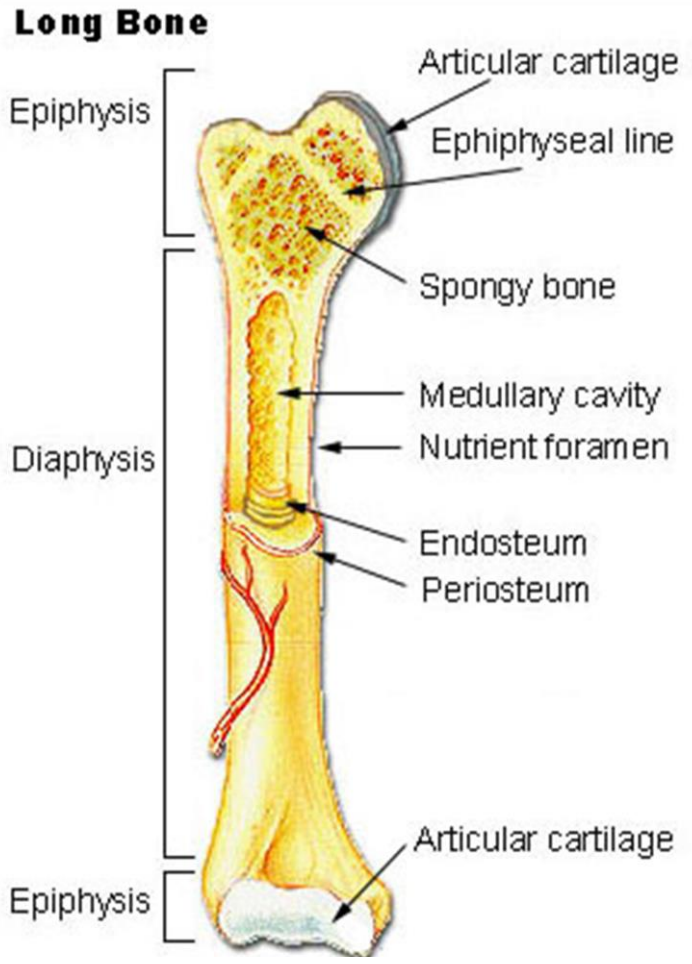
Bone Composition

1- **Compact bone**: dense, hard bone and forms the protective external surface of all bones.

2-**Spongy bone** is inside the compact bone and is full of tiny holes

- ✓ The outer part of a long bone is made of compact bone.
- ✓ Inside this is the spongy (cancellous) bone which contains bone marrow.
- ✓ The interior part of a long bone consists of the medullary cavity.
- ✓ The ends of epiphyses are covered with a hyaline cartilage ("articular cartilage").
- ✓ Nutrient arteries pass through the shaft of a long bone via nutrient foramina
- ✓ Nerves accompany the blood vessels supplying bones.

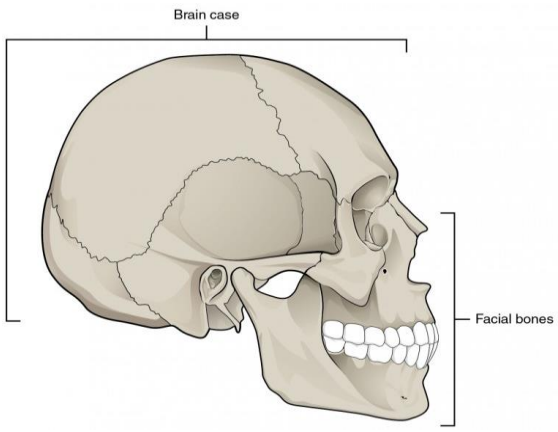
A long bone consists of a long shaft (diaphysis) with two bulky ends or extremities (epiphyses).



Tha Axial skeleton

1-The skull

consists of 22 individual bones, 21 of which are immobile and united into a single unit. The 22nd bone is the mandible (lower jaw)



The skull consists of the rounded brain case (cranial –cerebral bones) that houses the brain and the facial bones

2-The vertebral column:

It protects the spinal cord in its spinal canal.

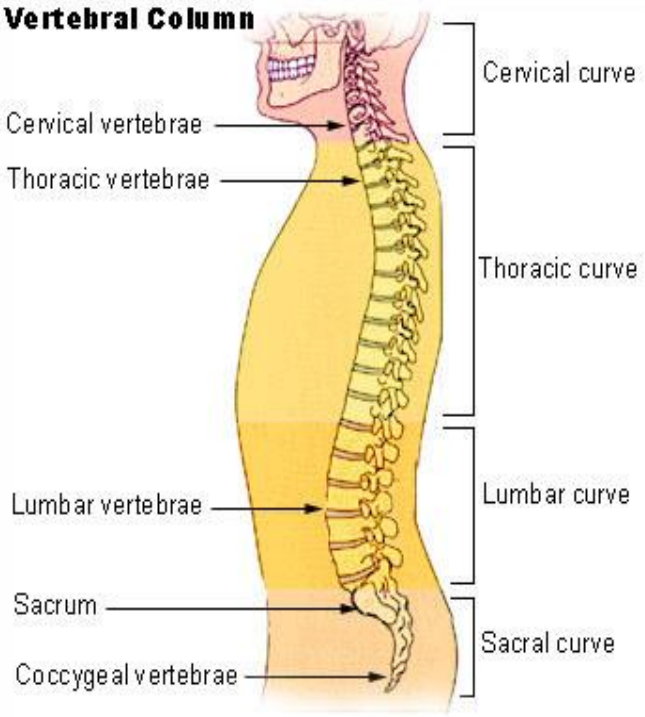
The vertebral column consists of 33 vertebrae:

Cervical: 7 vertebrae (C1–C7)

Thoracic: 12 vertebrae (T1–T12)

Lumbar : 5 vertebrae (L1–L5)

Sacrum: 5 sacral (fused) vertebrae



(S1–S5)

Coccyx: 4 (3–5) (fused) coccygeal vertebrae (Tailbone)

3-**The thoracic cage**,

also known as the rib cage, encloses the thorax.

It is formed by the 12 thoracic vertebrae, 12 pairs of ribs and associated costal cartilages and the sternum.

All ribs are attached in the back to the thoracic vertebrae.

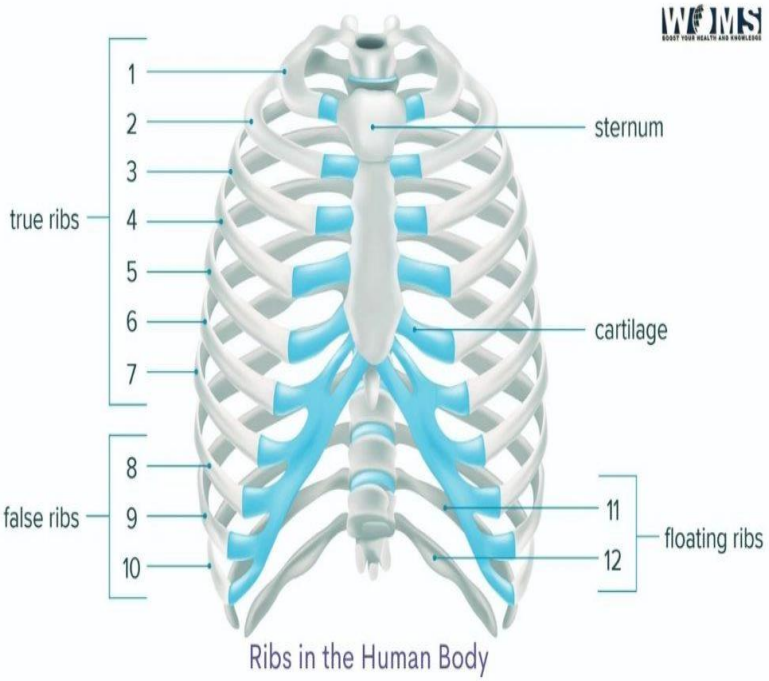
upper seven (true ribs) are attached to the sternum by means of costal cartilage

The 8-10 (false ribs) are joining with the costal cartilages of the ribs above.

The 11th and 12th ribs are known as floating ribs as they do Thenot have any anterior connection to the sternum.

The spaces between the ribs are known as intercostal spaces.

Costal cartilage is only found at the anterior ends of the ribs



Ribs in the Human Body

1 Joints

1) DEFINITION

A joint is the point where two or more bones meet.

2) 3 Types Of the joint

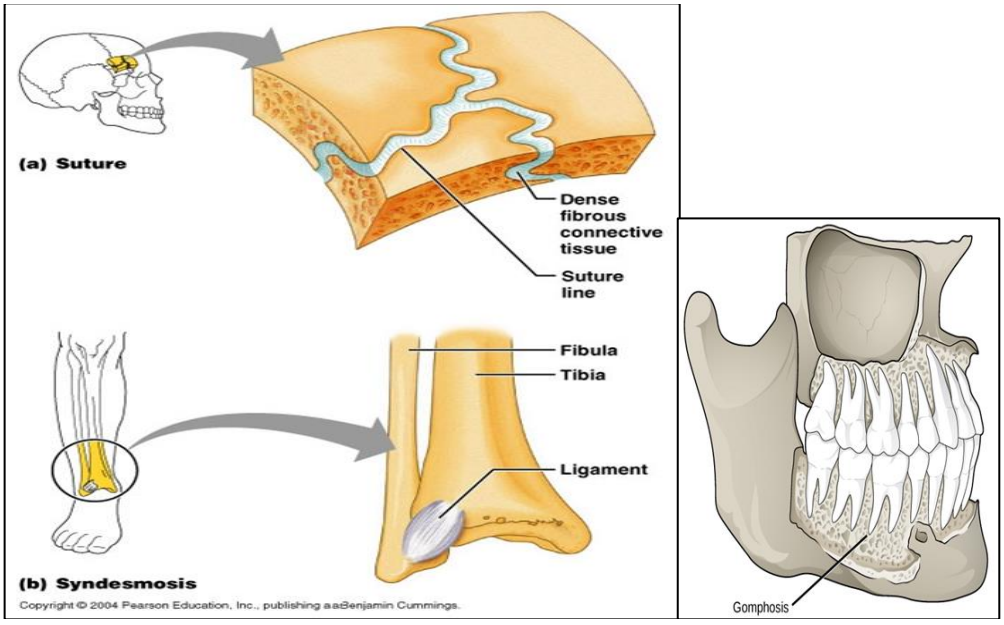
are classified according to the manner or type of material by which the articulating bones are united.

Fibrous joint	Cartilaginous joint	Synovial joint

✓ **Fibrous joints (immoveable)**

The articulating bones of fibrous joints are united by fibrous tissue.

Examples are joints between skull bones (sutures a) and where the teeth are held to their bony sockets **(gomphosis**) and at both the radioulnar and tibiofibular joints (**syndesmosis B**)

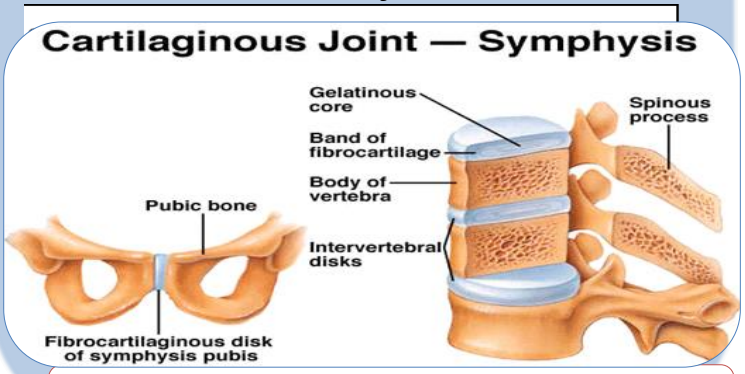


✓ **Cartilagenous joints (partially moveable**

The articulating structures of cartilaginous joints are united by hyaline cartilage or fibrocartilage.

Primarycartilaginous joints(synchondrosis) are united by hyaline (transparent) cartilage. These joints permit growth of the length of the bone and allow slight bending during early life, until the epiphysial plate converts to bone and the epiphyses fuse with the diaphysis, examples are growing bones.

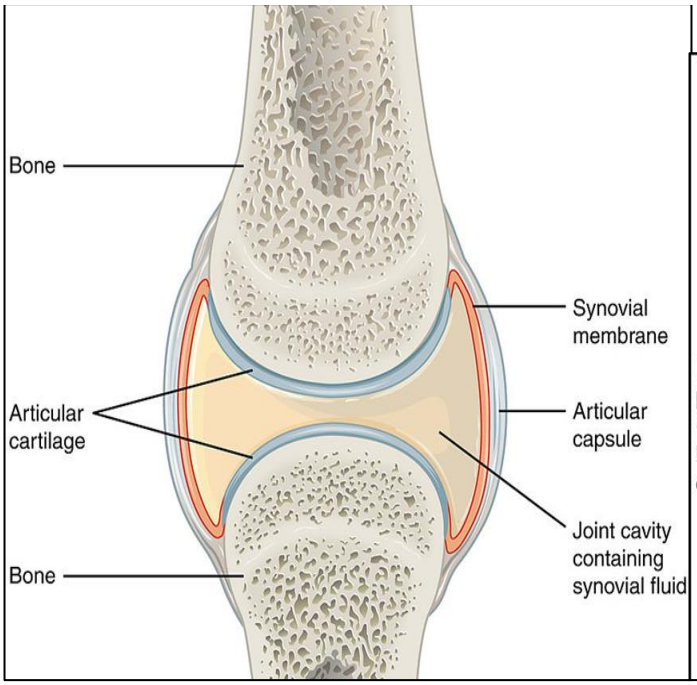
Secondary cartilaginous joints (symphyses) are strong, slightly mobile joints, united by fibrocartilage (mixture of white fibrous tissue and cartilaginous tissue), examples are intervertebral discs and midline joints

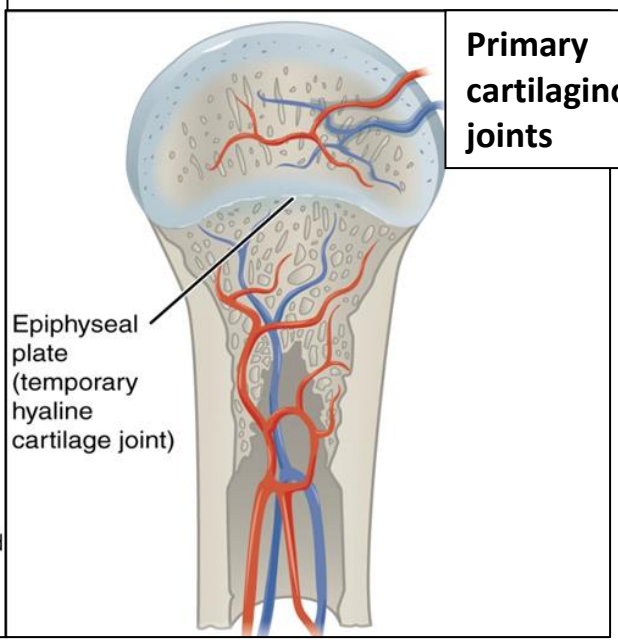


Secondary cartilaginous joint

- **3.Synovial Joints=diarthrosis(freely moveable)**

- The most common type of joints within the human body.
- **Synovial joints contain the following structures:**
  - **Articular cartilage**: the bones ends are covered by a layer of hyaline cartilage
  - **Synovial cavity** a potential space that contains a small amount of synovial fluid that serves the function of nourishing the articular cartilage and lubricating the joint surfaces
  - **Joint capsule**: the fibrous capsule that surround the joint and continuous with the periosteum of articulating bones
  - **A synovial membrane** which secretes synovial fluid, it covers the inner layer of the capsule and articular bones except these covered with articular cartilage.

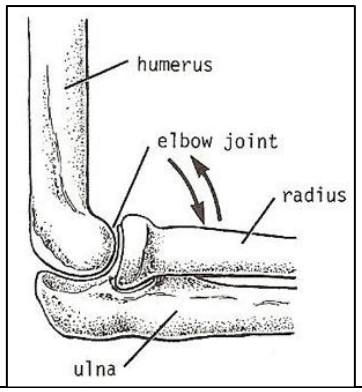
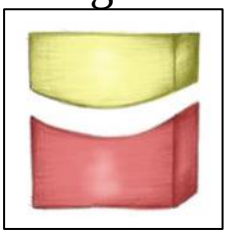
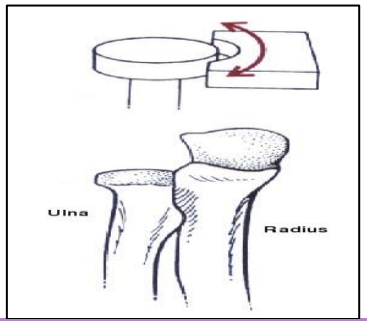
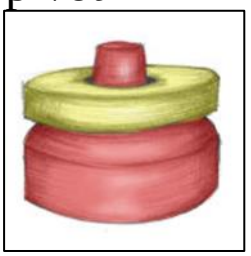




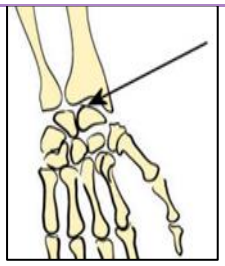
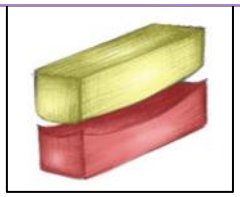
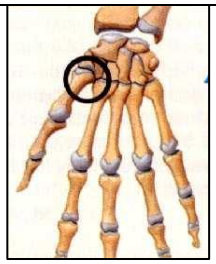
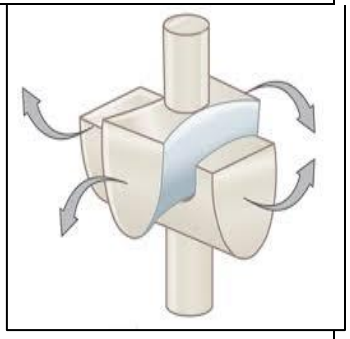
Primary cartilaginous joints

Types of synovial joint

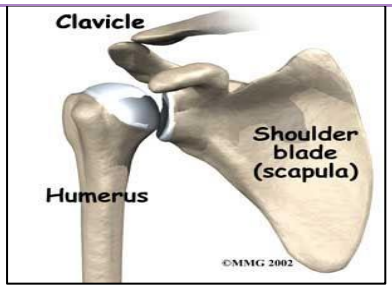
1-uniaxial(The movements are permitted around one axis)

Joint Type	Movement at joint	Examples	Structure
Hinge(Movements are permitted in one plane around a transverse axis)	Flexion/Extension		Hinge 
Pivot(Movements are permitted in one plane around a vertical axis)	Rotation of one bone around another		pivot 

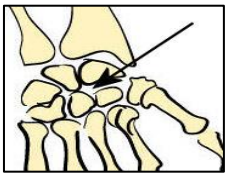
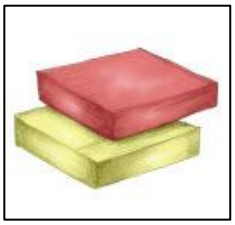
2-Biaxial (The movements are permitted around two axis)

Condyloid(Ellipsoid) (articular surfaces include an oval convex surface fitting into an concave surface.)	Flexion/Extension/Adduction/Abduction	 **Wrist joint**	
Saddle (Articular surfaces are reciprocally concavo-convex. Movements are similar to those permitted by ellipsoid joint with addition of some rotation)	Flexion/Extension/Adduction/Abduction /Circumduction	 first carpometacarpal joint **(joint of the thumb**	

3-multiaxial (The movements are permitted around more than two axis)

Ball and Socket (articular surfaces include a rounded head fitting into a cup shaped socket.)	Flexion/Extension/Adduction/Abduction/Internal & External Rotation	 Shoulder joint	

4- Gliding(non-axial joint)

Plane (Gliding) The opposed surfaces of the bones are flat	Gliding movements(side to side movement)	 **Intercarpal joints**	

3) Movements at joints

1-Flexion and Extension

Flexion: a bending movement that *decreases* the angle between two parts.
Extensionis the opposite of flexion, a straightening movement that *increases* the angle between body parts

2-Abduction and Adduction

Abduction means moving away from the median plane of the body.
Adduction means moving toward the median plane of the body

3- Rotation

referring to rotation towards or away from the center of the body

4-Circumduction

the circular movement combining in sequences the movements of flexion, extension, abduction, and adduction.

4) Factors maintaining Stability of synovial joints

- ✓ **Bones:** The shape of the two articular parts determines the degree of stability.

- ✓ **Ligaments:** preventing any over movement.
- ✓ **Muscles**: The tone of different groups of muscles acting on the joint is the most important factor in maintaining the stability

2 The Muscular System

1) The four basic functions of the muscular System

- ✓ Motion(body movements),
- ✓ movement of substances with the body,
- ✓ stabilizing body position
- ✓ regulating organ volume.

NOTE

The muscular system makes up nearly half the weight of the human body

2) There are three types of muscles

Skeletal muscle	Cardiac muscle	Smooth muscle ((not striated)

Hh

1.Skeletal muscle

- Skeletal muscles are **attached to the skeleton**
- moves bones and other structures
- Skeletal muscles are **voluntary** muscles - in other words we think about what movements we want to make and send messages via our **nervous system** to tell the appropriate muscle(s) to contract.

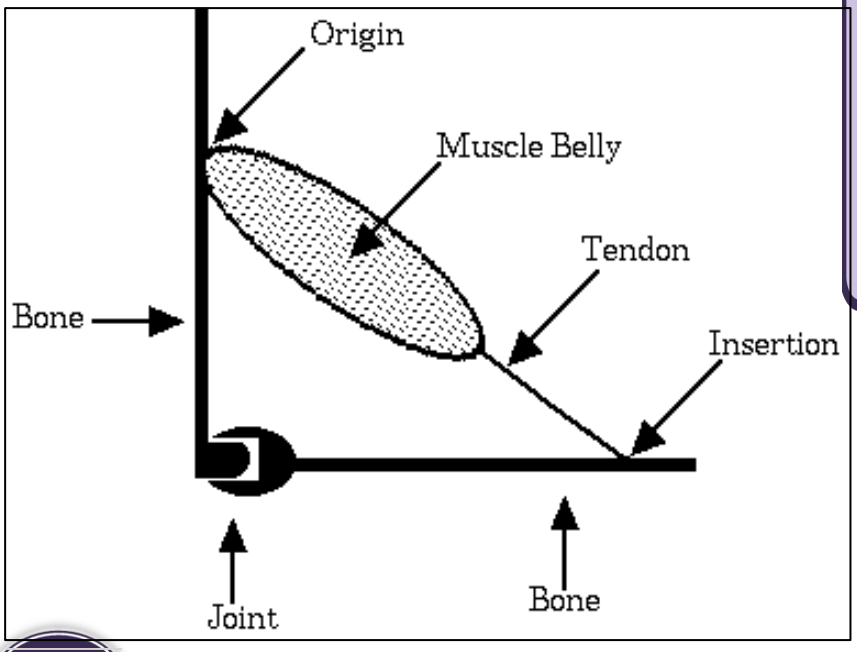
- Skeletal muscles **named striated muscles as they are** formed of fibers that are combined into parallel fibers.

2.Cardiac muscle

- As the name should tell you, cardiac muscle is found only in the heart.
- Like smooth muscle, cardiac muscle is involuntary
- and likeSkeletal muscle cardiac muscle is striated

3.smooth muscle

- ✓ Smooth muscle is found in our **internal organs:** in our digestive system, our blood vessels, our bladder, and our respiratory organs
- ✓ Smooth muscles are **involuntary** muscles - in other words we do **not** have to think about contracting them because they are **controlled automatically** by the **nervous system**.



NOTE

Skeletal muscle Attachments

Origin - The point of attachment of a skeletal muscle to the bone that remains relatively fixed during contraction.

Insertion - The point of attachment of a skeletal muscle to the bone or other body part that it moves (not fixed).

Types of insertion

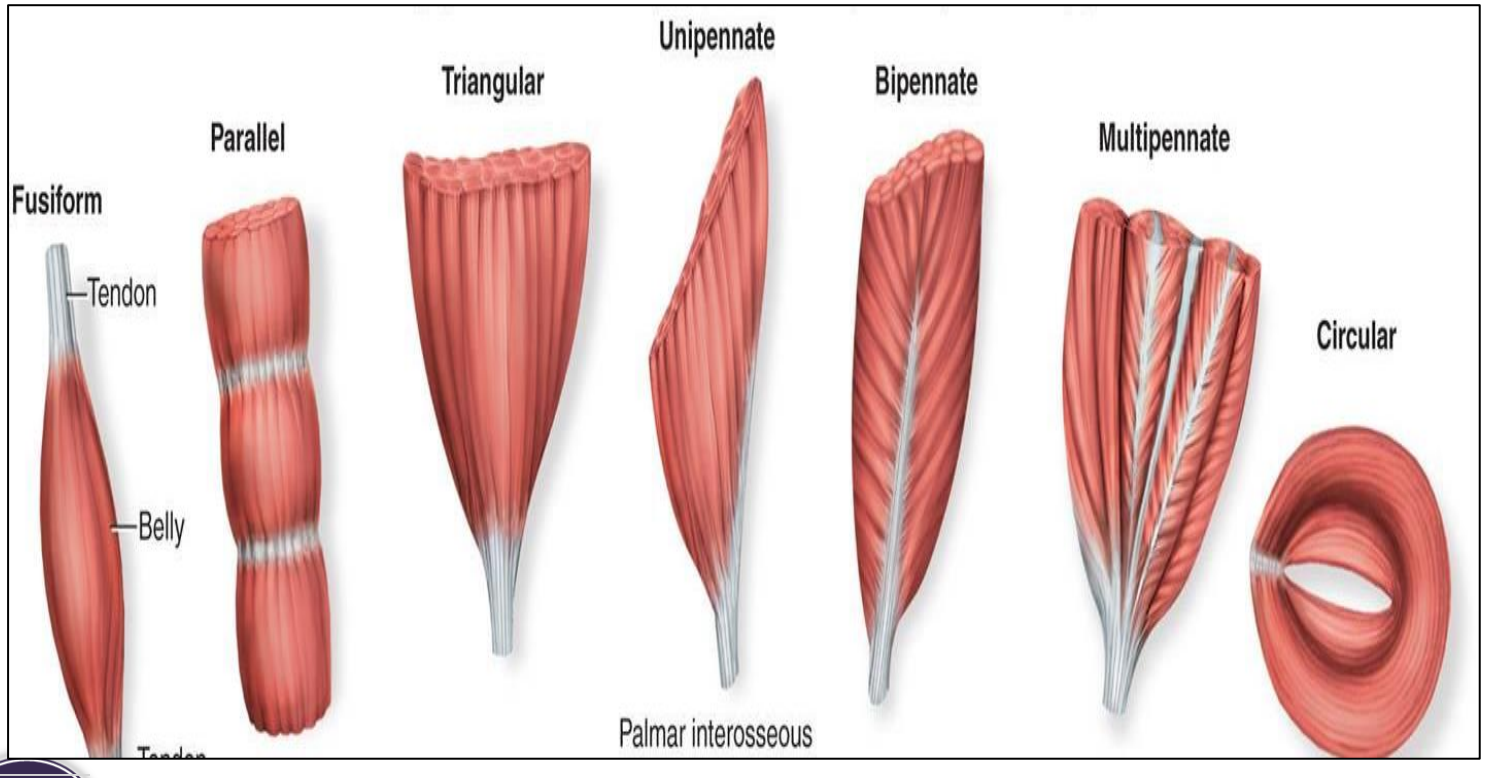
Tendon: When the muscle fibers form a dense cord of connective tissue that enables the attachment of muscle to the periosteum of the bone.

Aponeurosis : When the muscle fibers extend as a broad flat layer

3) ***What are the different shapes of skeletal muscles?***

There are 5 different muscle shapes within the human body:

- **Circular**
- **Convergent**
- **Parallel**
- **Pennate**
- **Fusiform**



1.Circular Muscles: These muscles appear circular in shape which surround an opening such as the mouth and eyes surroundings

2.Convergent (triangular) Muscles: *These are muscles where the origin is wider than the point of insertion.*

3.Parallel Muscles: Parallel muscles have fibers which, as the name suggests, run parallel to each other and are sometimes called strap muscles.

4.Fusiform Muscles: These muscles are more **spindle shaped**, with the muscle belly being wider than the origin and insertion.

5.Pennate Muscles: Pennate muscles are feather-like in the arrangement of their fascicles, they can be divided into:

Unipennate: These muscles have their fibres arranged to insert in a diagonal direction onto the tendon

Multipennate: have multiple rows of diagonal fibers, with a central tendon which branches into two or more tendons

Bipennate: Bipennate muscles have two rows of muscle fibres, facing in opposite diagonal directions, with a central tendon, like a feather

Chapter 2

Systemic Anatomy

1 Cardiovascular System

- An organ system that permits blood to circulate and transport nutrients to and from the cells in the body
- The main components of the cardiovascular system are the heart, and the blood vessels (arteries-veins).

1) heart

- The heart pumps oxygenated blood to the body and deoxygenated blood to the lungs.

- The blood that is returned to the right atrium is deoxygenated (poor in oxygen) and passed into the right ventricle to be pumped through the pulmonary artery to the lungs for re-oxygenation and removal of carbon dioxide.

- The left atrium receives newly oxygenated blood from the lungs which is passed into the strong left ventricle to be pumped through the aorta to the different organs of the body.

Chambers of the heart

- **Right atrium**

Non –oxygenated blood pass to the right atrium through three vessels.
These are :

- **the superior and inferior venae cava:** returns blood from the body;
- the **coronary sinus**: returns blood from the walls of the heart itself

Blood passes into the right ventricle through the **tricuspid valve**

- **Right ventricle:**

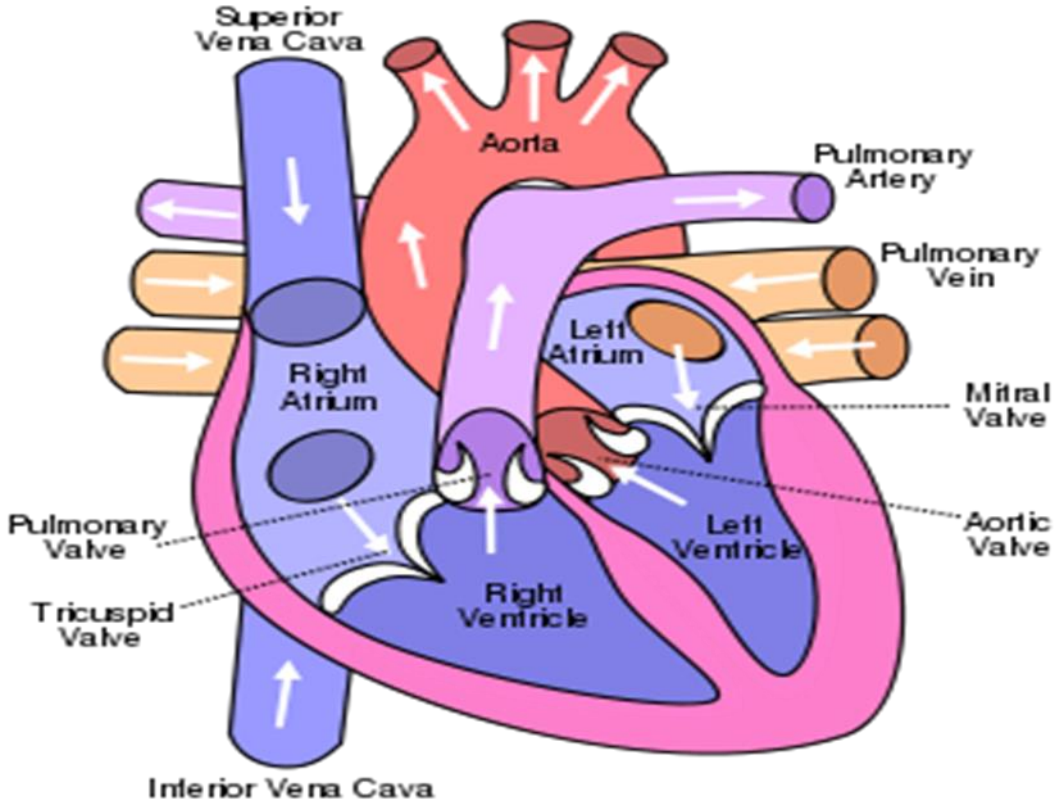
Non –oxygenated blood in the right ventricle pass through the pulmonary artery to the two lungs where oxygenation of this blood occurs.

- **Left atrium:**

Oxygenated blood pass from the lungs to the left atrium through **the four pulmonary veins (two from each lung)** from the left atrium, blood passes into the left ventricle through the **mitral valve**

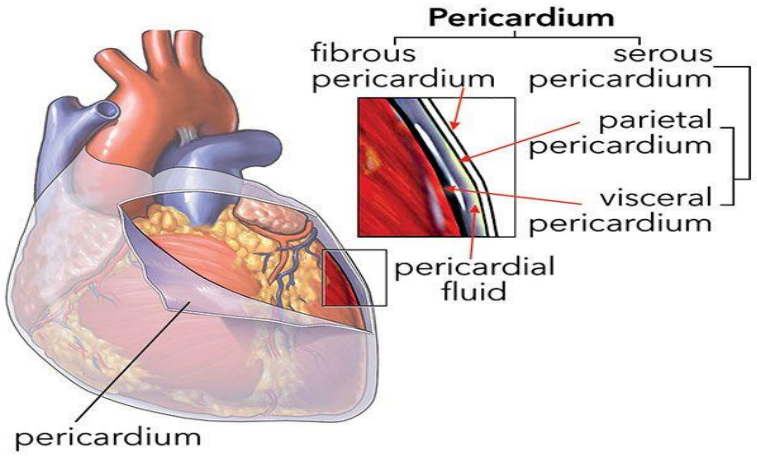
- **Left ventricle**:

Oxygenated blood in the left ventricle pass through the aorta to the different parts of the body.



NOTE

Pericardium Is a double walled fibroserous sac that encloses the heart and the roots of its great vessels. It is formed of two



2) Arteries and veins

Arteries

- Arteries are blood vessels which carry blood from the heart to the body.
- All of which carry oxygenated blood with the exception of the pulmonary artery.
- The most widely known artery within the human body is the Aorta. This is the largest of all blood vessels and transports blood away from the heart where it then branches into smaller arteries.
- Smaller arteries divide further until they are classed as arterioles.
- **Arterioles** continue to branch into smaller and smaller vessels which, once they have decreased in size below 10 micrometers in diameter are known as **capillaries.**

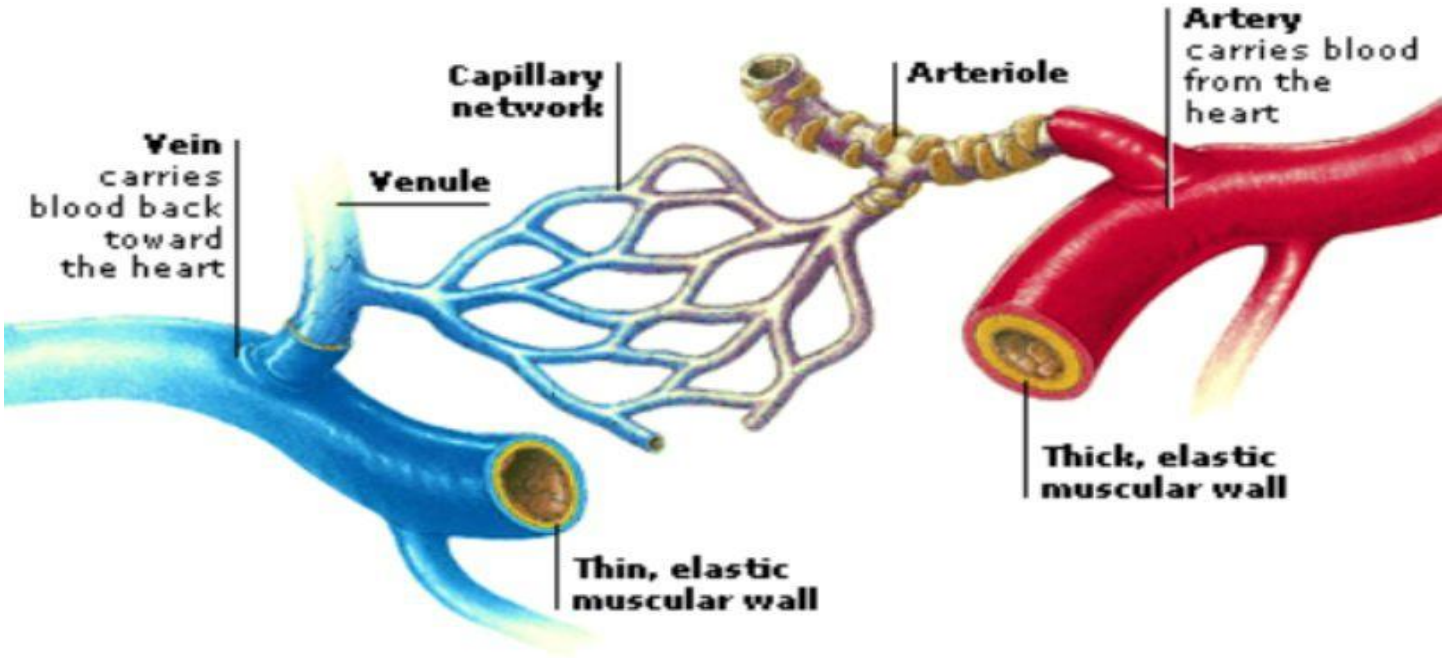
veins

- ✓ Veins are blood vessels which carry deoxygenated blood from the body to the heart.
- ✓ All of which carry deoxygenated blood except the pulmonary veins, which carry oxygenated blood, from the lungs, back to the heart.
- ✓ At tissue level **capillaries** drain blood into **venules**, which are very small veins,

- ✓ As **venules** return to the heart merge into larger veins which drained into either the Superior Vena Cava (it returning from tissues and organs above the heart) or the Inferior Vena Cava (it returning from tissues and organs below to the heart).
- ✓ These two large veins merge and return blood to the right atrium of the heart.
- ✓ Veins also contain **valves** which prevent the back flow of blood.

***** What Are The Differences Between Arteries And Veins?**

ARTERIES	veins
Arteries are the carriers of red oxygenated blood directly away from the heart	veins carry de-oxygenated blood towards the heart
Blood is under great pressure in arteries; hence it flows fast	In veins the blood is not under great pressure, hence it flows more slowly and smoothly.
Arteries have **thick muscular walls**	veins have relatively thin muscular walls.
Arteries have no valves	veins have internal valves along their length to prevent back flow of blood.



***** What Are The Differences Between pulmonary circulation And systemic circulation ?**

pulmonary circulation	systemic circulation
transports deoxygenated blood through the pulmonary artery to the lungs for re-oxygenation and removal of carbon dioxide, newly oxygenated blood from the lungs pass through the pulmonary veins to heart	which transports oxygenated blood away from the heart, to the rest of the body, and returns deoxygenated blood back to the heart

NOTE

Anastomosis: connection between two blood vessels, such as between arteries (arterio-arterial anastomosis), between veins (veno-venous anastomosis) or between an artery and a vein (arterio-venous anastomosis)

2 Respiratory System

1) Air conducting portion

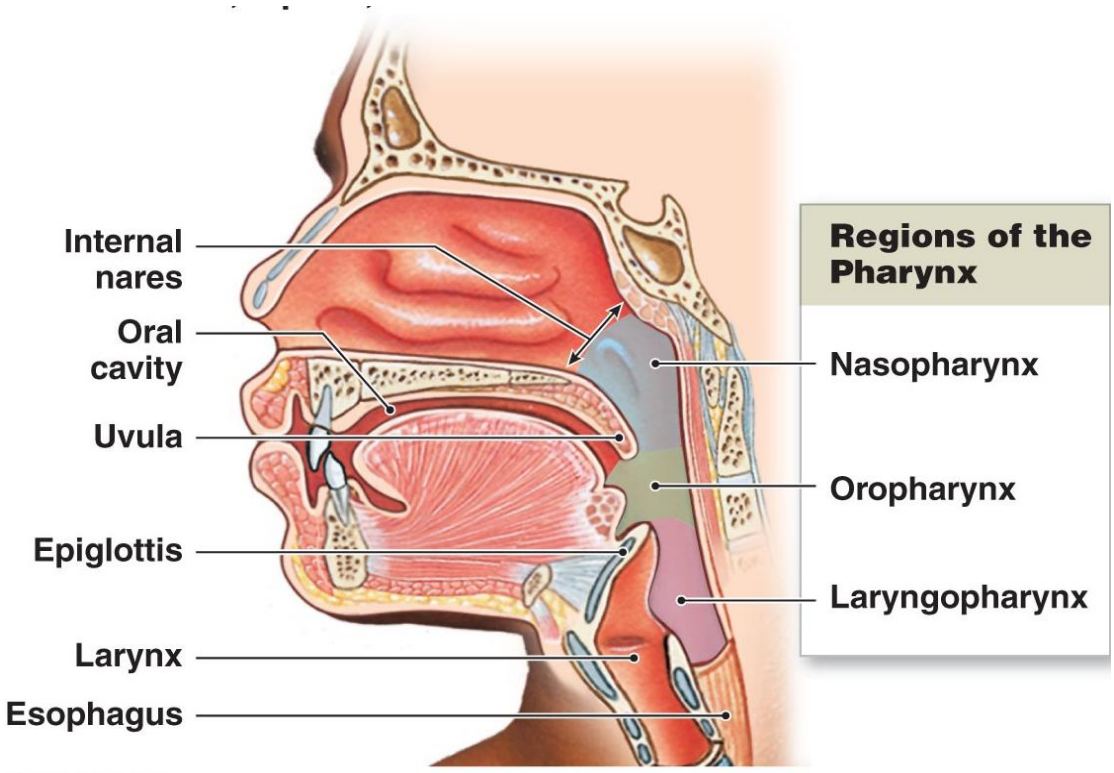
1-Nose:

- ✓ The nose is the upper part of the respiratory tract and contains the peripheral organ(mucosa) of smell .It is divided into right and left nasal cavities by the nasal septum.

2-Pharynx:

- ✓ It is a muscular tube

- ✓ Extends from the base of the skull to the level of sixth cervical vertebrae end where it is continuous with the oesophagus.
- ✓ divided into three sections: nasopharynx, oropharynx and laryngopharynx

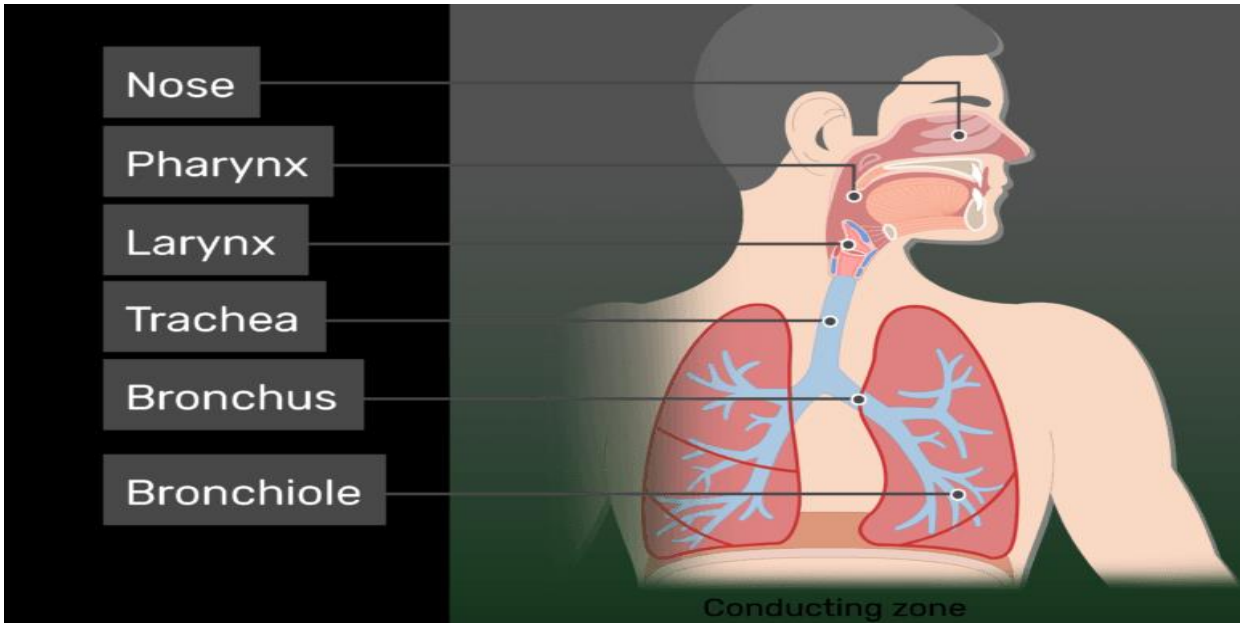


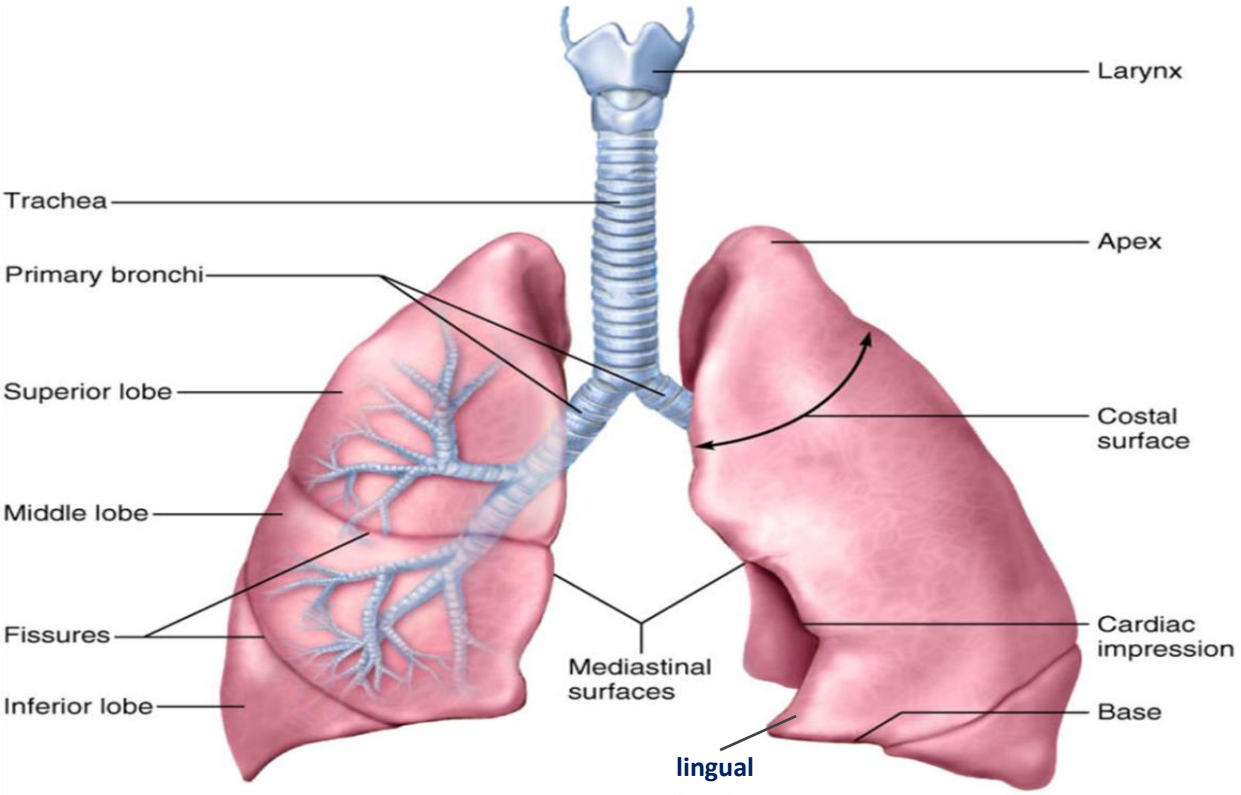
3-Larynx:

- The larynx is a specialized organ consists of cartilages connected by muscles and ligaments.
- **Function:**
  - an open organ for air passage which closed during swallowing (by epiglottis)
  -contains vocal cords which are responsible for the production of voice.
- **Begin:** at the level of third cervical vertebrae.
- **End:** At the level of sixth cervical vertebrae.

4-Trachea (12-14 cm):

- The trachea is a fibrocartilaginous tube formed by incomplete cartilaginous rings (which keep the trachea patent), cartilages are deficient posteriorly where the trachea is adjacent to the esophagus**.**
- The trachea extends from the inferior end of the larynx at the level of C6 to lower border of T4. It ends by dividing into right and left main bronchi





2) the lung

Each lung has pyramidal shape with apex and base.

Apex: Directed upwards and projects into the root of the neck.

Base: Concave and directed downwards where it is related to the diaphragm.

Lobes of the Lung:

Right lung: divided into 3 lobes (upper, middle and lower)

Left lung: divided into 2 lobes only (upper and lower)

Pleura: It is a closed serous sac (one on each side) surrounds each lung

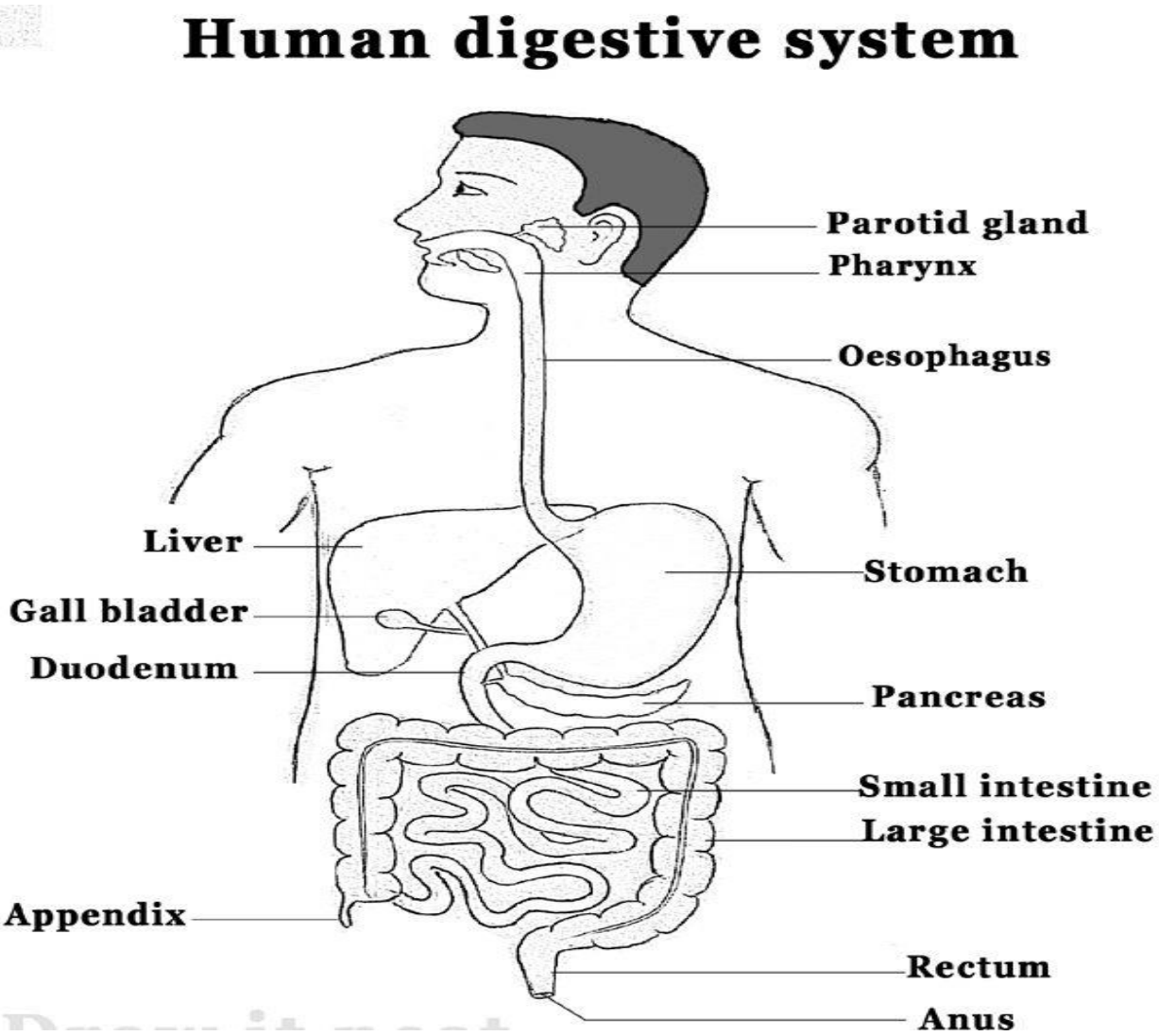
****Differences between right and left lungs:

Right lung	left lung
The right lung has 3 lobes	the left lung has only 2 lobes.
The right lung is short and broad (presence of the liver pushes the diaphragm upwards on the right side).	The left lung is long and narrow (the heart lies on the left side).
	left lung has a cardiac notch on its anterior border and a lingual just below the notch.

3 Digestive system

The digestive system include

- **The gastrointestinal tract** (GI) which starts at the mouth and ends at the anus.
- **Accessory digestive glands**: the liver, gall bladder and pancreas.



1) Gastrointetinal tract (GI)

1-The mouth (oral cavity):

Bounded laterally and in front by the alveolar process (containing the teeth) the mouth is the first portion of the **GI** that receives food and saliva continue into t**he pharynx** which ends at the oesophagus.

2-The oesophagus:

- consists of a fibromuscular tube through which food passes, aided by peristaltic contractions, from the pharynx to the stomach. It is about 25 cm long.

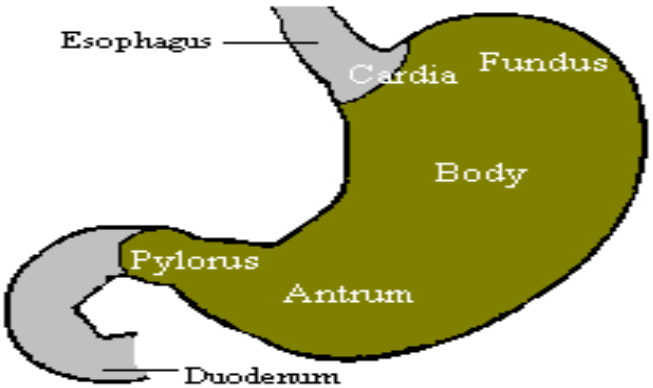
3-Stomach

Position: The stomach occupies left hypochondriac region.

Shape: J-shaped stomach

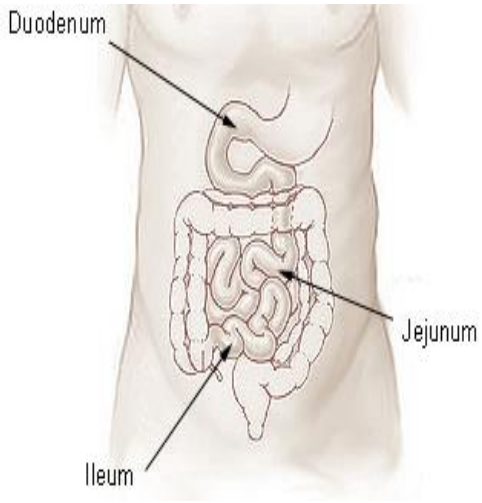
Orifices of the stomach:

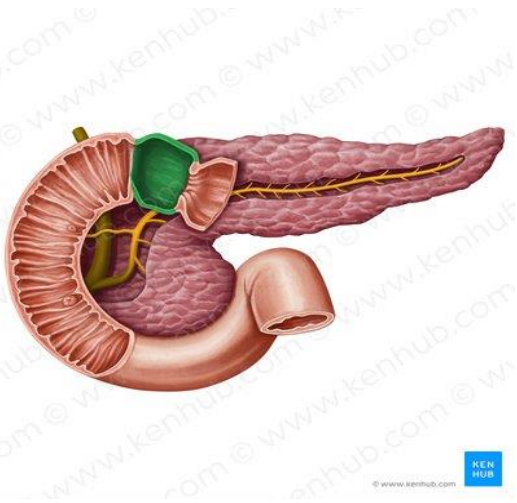
- Cardiac orifice: **(cardia)**: It is the opening by which the oesophagus communicates with the stomach.
- Pyloric orifice: **(pylorus)**: It is the opening by which the stomach communicates with the duodenum.



4-Small intestine

- The small intestine (measures 6 meters) in length and has a smaller diameter than the large intestine. It is divided into 3 parts: **duodenum** (fixed part), **jejunum** and **ileum** (free part).





- **Duodenum**: is the proximal 10 inches, it is C- shaped curvature curved around head of pancreas
- **Jejunum**: follows the duodenum and forms the proximal 2/5 of the free part.
- **Ileum:** is the distal 3/5 of free part of the small intestine and joins the caecum at the ileocaecal junction.

5-Large intestine

☒ The large intestine measures 1.5 meters and its diameter is greater than that of the small intestine
☒ It consists of the caecum, appendix, colon, rectum and anal canal.

The caecum:

✓ The most dilated part of the large intestine.
☒ It forms the beginning of the large intestine and lies in the right side
☒ The caecum is about 10 cm length.
☒ It contains the vermiform appendix: The appendix length is from 2 to 20 cm (average 10 cm)

• Colon:

Consists of 4 parts: ascending, transverse, descending and sigmoid.

Ascending colon:

1. It is 15 cm long, and begins just above the ileocaecal junction as the upward continuation of the caecum. It ends just below the liver by forming the right colic flexure

Transverse colon:

It is 50 cm long

- Begins as the continuation of the right colic flexure.
- Ends at the left colic flexure.
-

Descending colon:

It is 25 cm long.

- It extends from the left colic flexure down to enter the pelvis.

Sigmoid colon(pelvic colon):

It is 40 cm long.

- ✓ It is S-shaped.
- ✓ It begins at the left side of the inlet of the pelvis and it is continuous with the rectum.

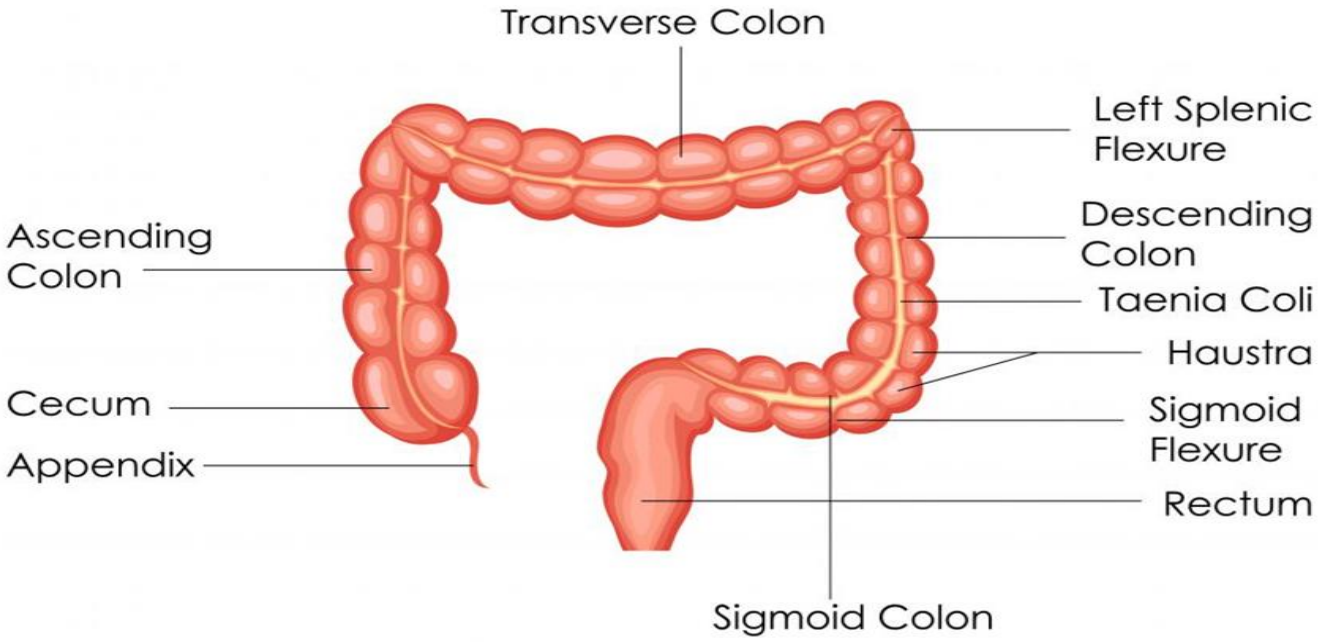
• Rectum

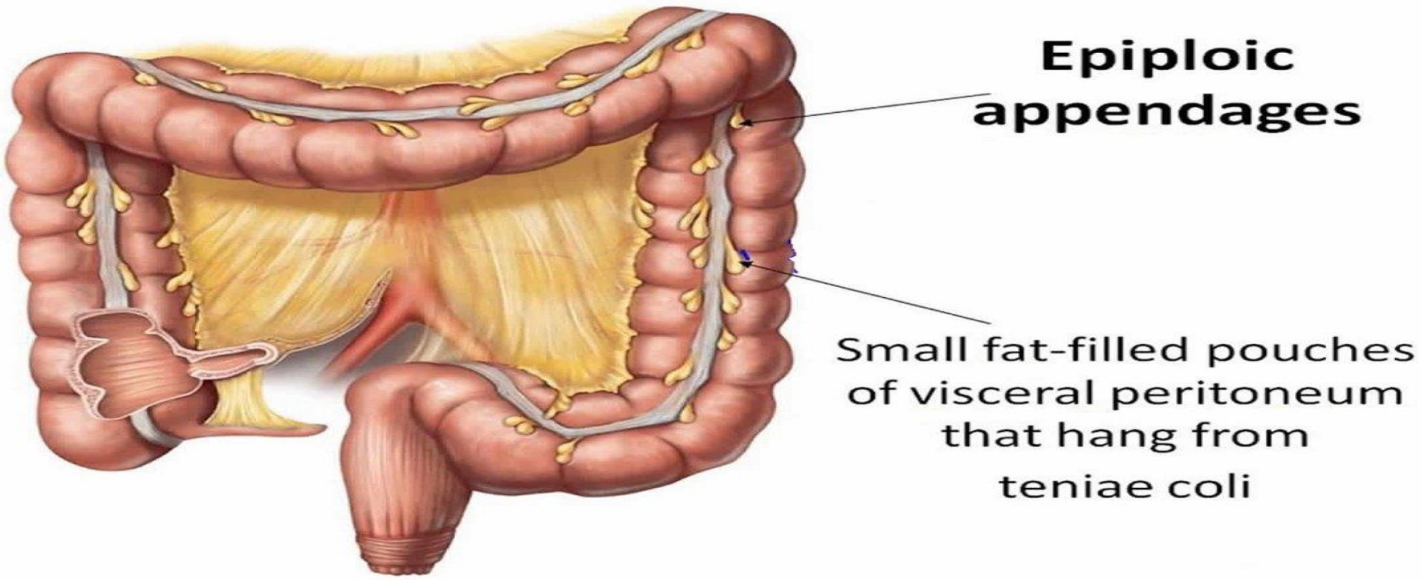
- Five inches long, and its lower part is dilated to form the rectal ampulla.
- **Continue below as the anal canal (4cm) which ends as at the anus**.

The large intestine differs from the small intestine by the presence of the following main features:

- **Taeniae coli**: these are 3 longitudinal muscular bands in the wall of the colon
- **Sacculations (haustrations):** these are sac-like dilatations in the wall of the large intestine especially in the colon.

- **Appendices epiploicae:** these are small projections of fat scattered over the surface of the colon.





2) Accessory digestive glands

1- There are four accessory glands of the digestive system and they are the salivary glands, the liver and the pancreas

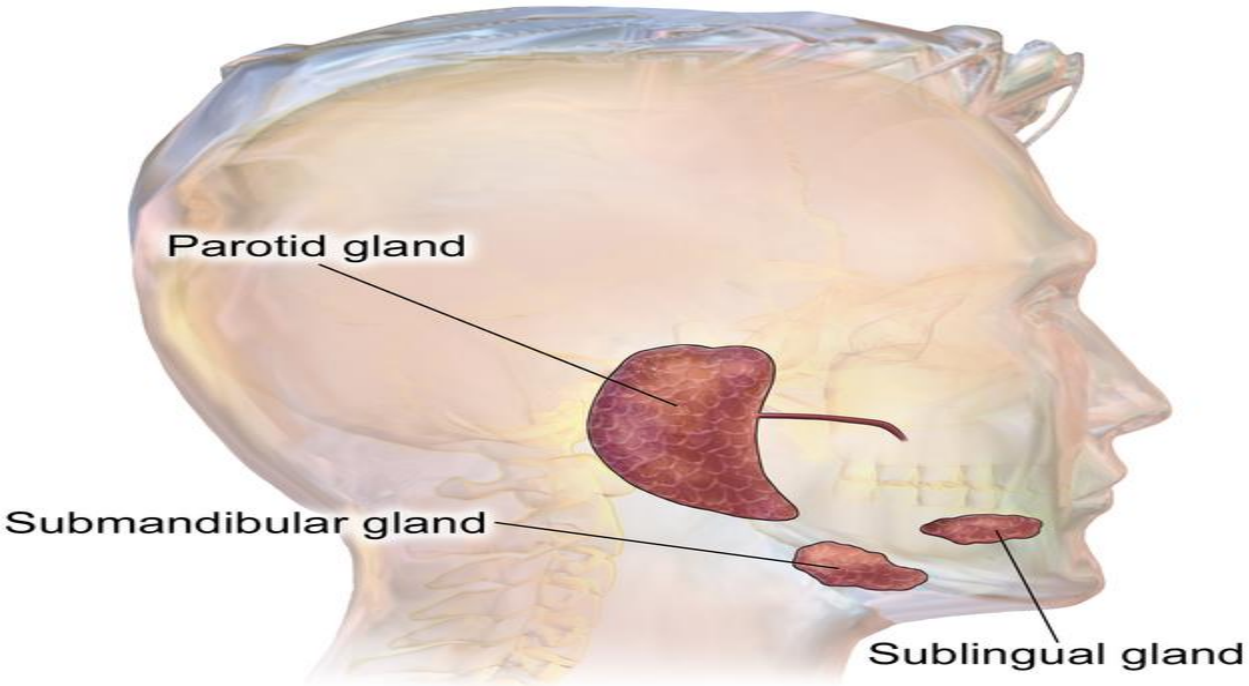
The salivary glands

- ✓ are exocrine glands that produce saliva.
- ✓ They also secrete amylase,an enzyme that breaks down starch into glucose
- ✓ **Site :** The glands are found in and around your mouth and pharynx.
- ✓ We call the major salivary glands the parotid, submandibular, and sublingual glands.

- **The parotid glands** are a pair of major salivary glands wrapped around the mandibular ramus. It is one of a pair being the largest of the salivary glands

2. **The submandibular glands** are a pair of major salivary glands located beneath the lower jaws.

3**. The sublingual glands** are a pair of major salivary glands located inferior to the tongue, anterior to the submandibular glands



Salivary Glands

The liver

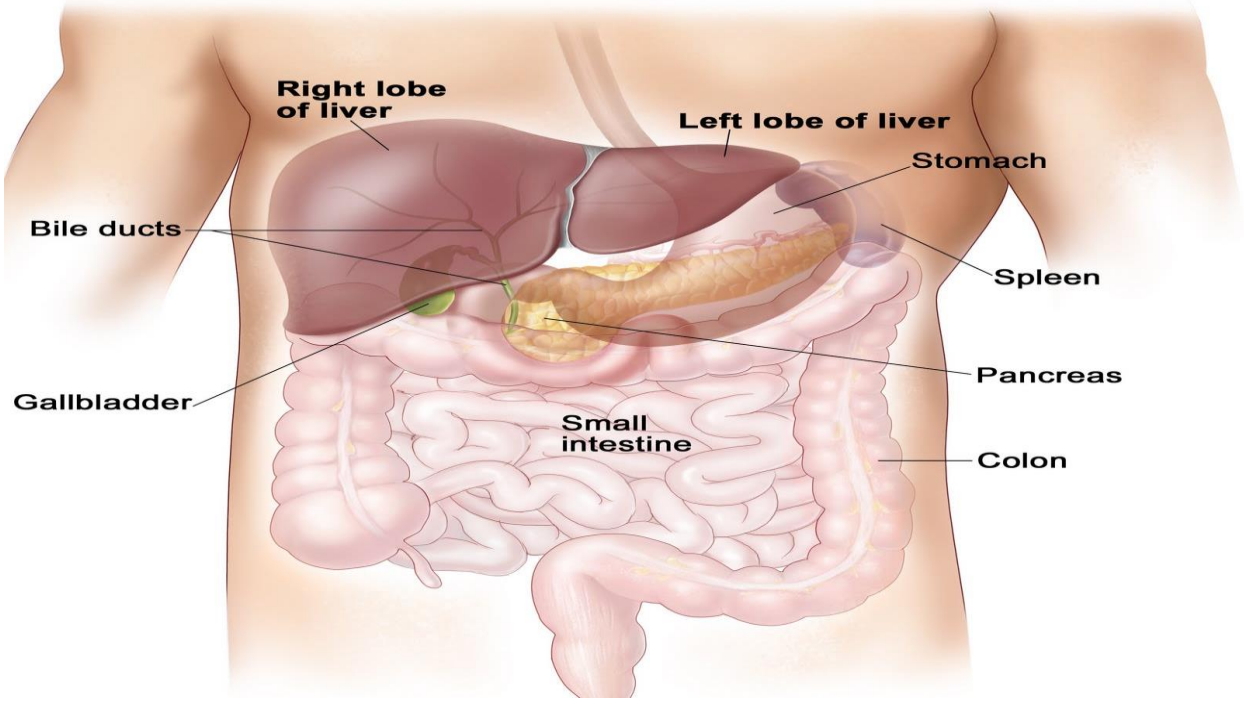
is a wedge-shaped and reddish brown in color because of its great vascularity.

It lies mainly in the right side(right hypochondriac region)

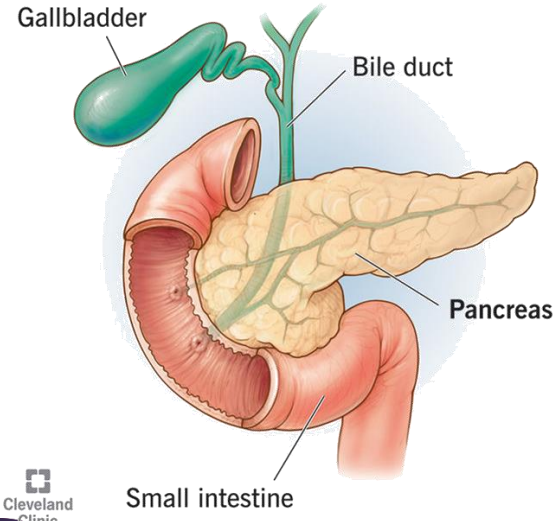
- The liver plays a major role in metabolism and has a number of functions in the body, including

1. **glycogen storage**,
2. **decomposition of red blood cells**,
3. **plasma protein synthesis**,
4. **detoxification** and
5. **produces bile** which aids the digestion of lipids in the small intestine.

Lobes of the liver: The liver consists of a large right lobe and a smaller left lobe.



Pancreas



Pancreas

-The pancreas lies in contact with the upper part of the posterior .abdominal wall
The pancreas extending transversely from the duodenum to the -.spleen
-The pancreas is both an endocrine gland (producing insulin), as well as an exocrine gland, secreting pancreatic juice containing digestive enzymes that pass to the small intestine (Digests Starch)

The pancreas composed of 4 parts: head, neck, body and tail

1- Head: occupies the concavity of the duodenum.

2- Neck: is the slightly constricted area between head and body.

1- Body: extends to the left crossing over the lumbar vertebrae

1- Tail: is the narrow left end of the pancreas

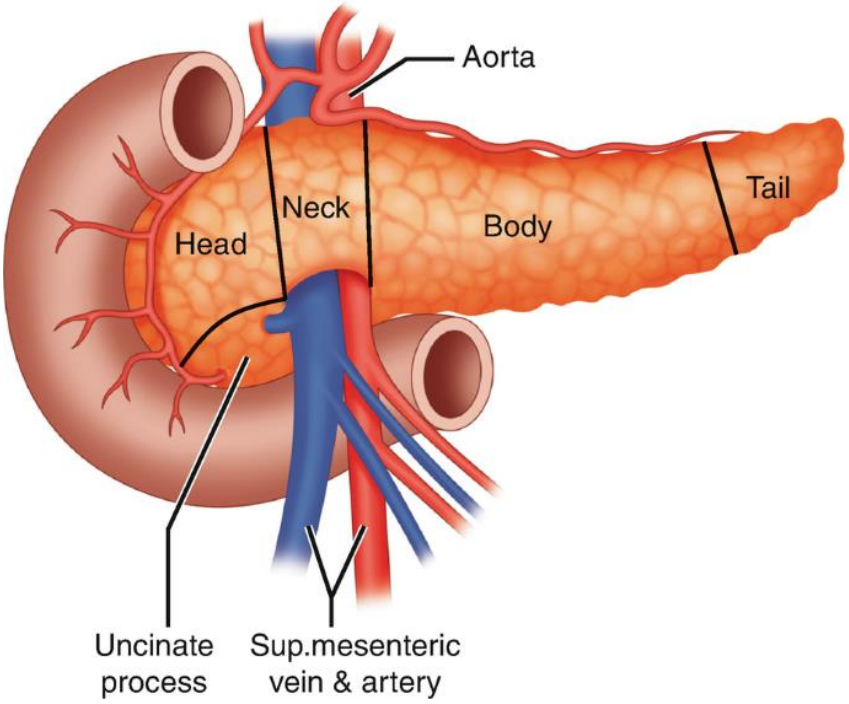


Fig. Parts of pancreas

Lymphatic system

Main functions of the lymphatic system

- To collect and return interstitial fluid (surrounds the tissue cells), including plasma protein to the blood, and thus help maintain fluid balance (this means it collects liquid lost from the circulatory system-returns proteins to circulation)
- To absorb lipids from the intestine and transport them to the blood. (this means it transports fats to circulation)
- To defend the body against disease by producing lymphocytes and carries bacteria to the lymph nodes for destruction

The lymphatic system consists of :

1) Lymphatic vessels (lymphatics)

a nearly body wide network of **thin-walled vessels with abundant valves**. Lymphatic vessels occur almost everywhere blood capillaries are found, except teeth, bone, bone marrow, and the entire central nervous system. It function is to collect interstitial fluid.

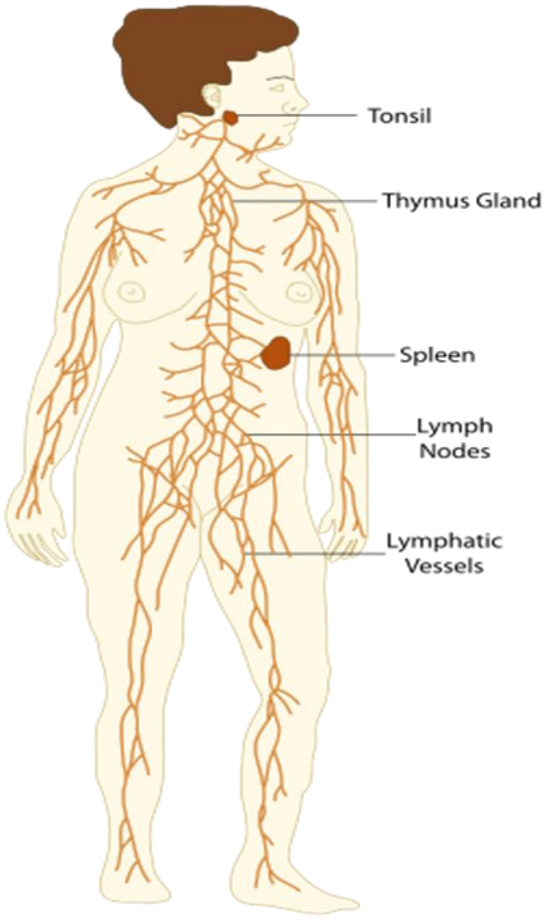


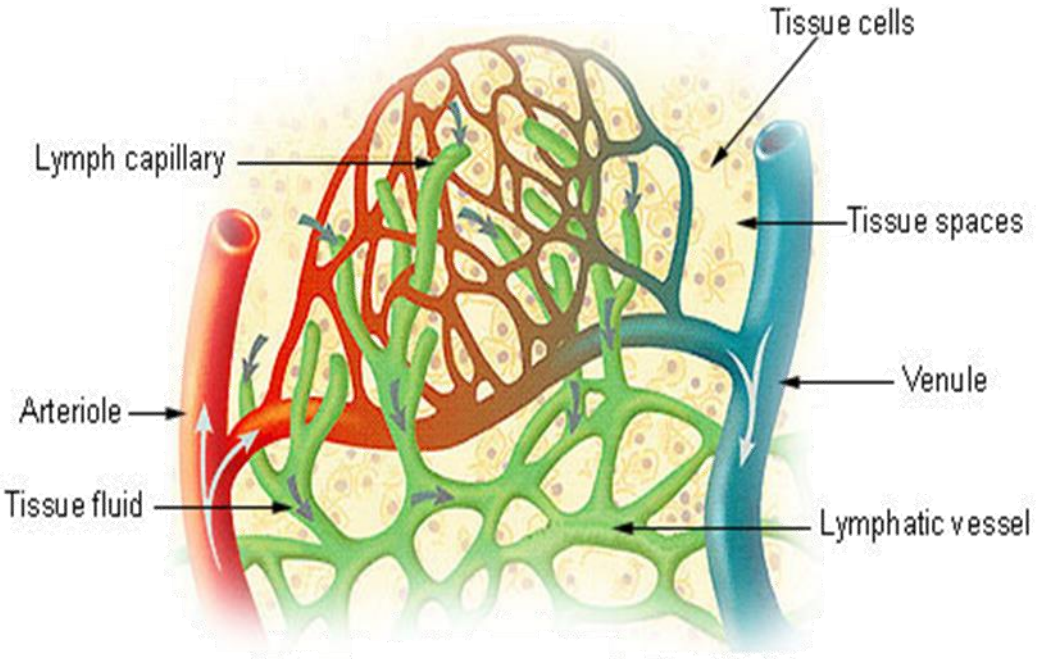
Fig. Lymphatic system

2) Lymph nodes

small masses of lymphatic tissue through which lymph is filtered on its way to the venous system.

3) Lymphoid tissue

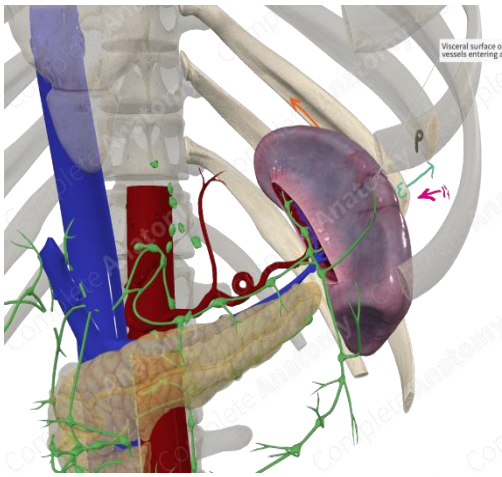
sites that produce **lymphocytes**, such as that found in the walls of the digestive tract; in the spleen, tonsils, thymus, and lymph nodes; and in myeloid tissue in red bone marrow.



(Lymph is the name for fluid inside the lymph vessels)

Spleen

It is soft, very vascular and very friable organ. It lies in the left side between the fundus of the stomach and the diaphragm

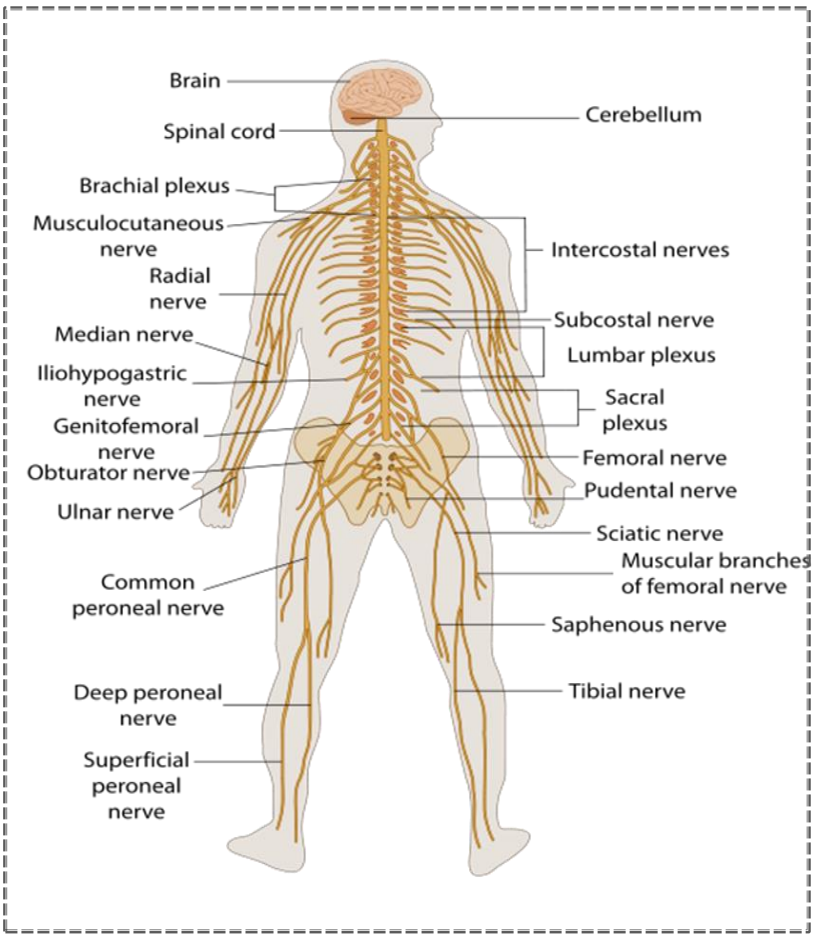


Function of spleen

- ☒ It mainly acts as a filter for purifying the blood
- ☒ Removing microbes and worn out or damaged red blood cells.
- ☒ It is also an important organ in the immune system, producing the white blood cells that fight infection and synthesize antibodies.

The nervous system

- ☒ Consists of highly specialized cells (neurons) that transmit information rapidly between various parts of body.
- ☒ Coordinates body voluntary and involuntary actions
- ☒ Enables the body to react to continuous changes in its external and internal environments.
- ☒ It controls various activities of the body, such as circulation and respiration,etc.



☒ The nervous system is divided **Structurally** into follows:

- the **central nervous system (CNS),** made up of the brain and spinal cord
- the **peripheral nervous system (PNS),** consisting of nerve fibers and cell bodies outside the CNS that conduct impulses to or away from the CNS.(12 pairs of cranial nerves from the brain and 31 pairs of spinal nerves from the spinal cord)

Brain

- 1300 to 1400 grams.
- It lies within the skull.
- 12 pairs of nerves extend from it to supply the body .

Spinal Cord

- ✓ Along tube like structure (43-45 cm) which extends from the brain.
- ✓ It lies within the vertebral column.
- ✓ 31 pairs of nerves extend from it to supply the body .

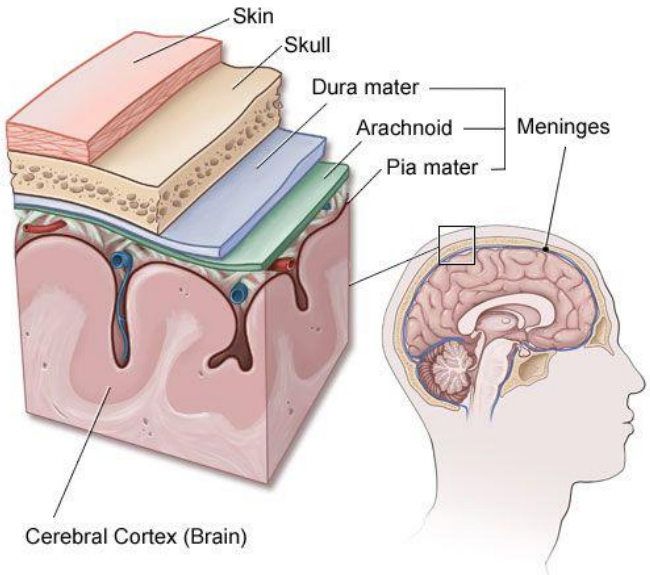
The meninges

The **meninges** are three layers or membranes that cover the brain and the spinal cord

- The outermost layer is the **dura mater**.
- The middle layer is the **arachnoid mater.**
- The innermost layer is the **pia mater**.

The meninges offer protection to the brain and the spinal cord

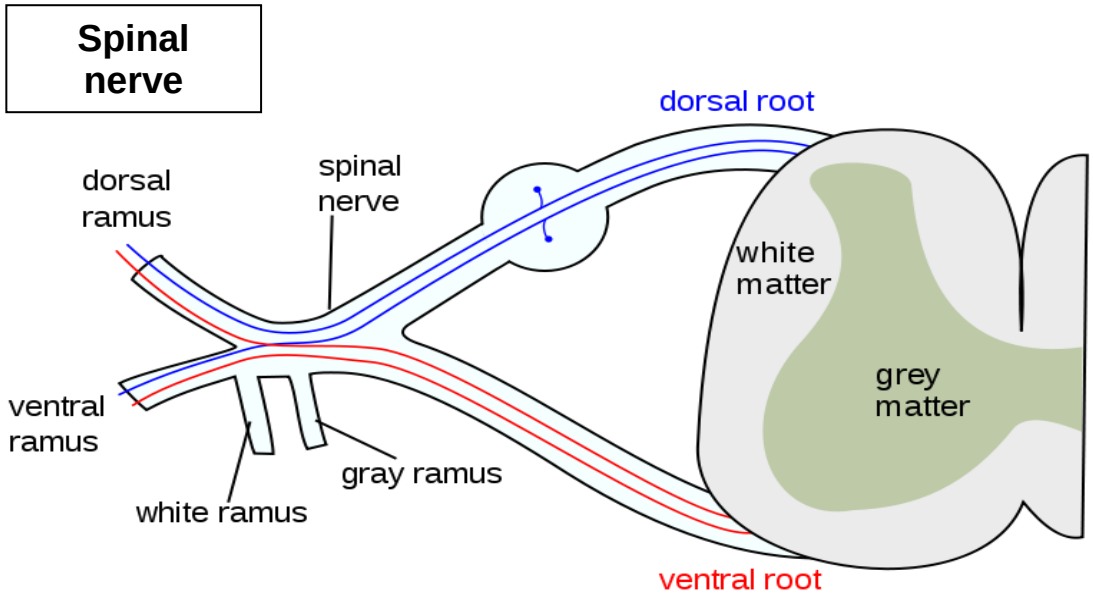
The Cerebrospinal Fluid (CSF) circulates around the brain and spinal cord (between arachnoid and pia matter). It **protects and nourishes the brain and spinal cord .**



The Peripheral nervous system

- **Cranial nerves (12 pairs)** connect the brain to different parts head, neck, and trunk. Each nerve has a corresponding roman numeral between I and XII
- **spinal nerves (31 pairs)**

The term **spinal nerve** generally refer to a mixed spinal nerve, which carries motor(ventral root), sensory(dorsal root), and autonomic signals between the spinal cord and the body. Humans have 31 pairs of spinal nerves, each roughly corresponding to a segment of the **vertebral column**: **8** cervical spinal nerve pairs (C1-C8), **12** thoracic pairs (T1-T12), **5** lumbar pairs (L1-L5), and **5** sacral pairs (S1-S5) and **1** coccygeal pair.



The Peripheral nervous system is made up (functionally) of two parts:

- **Somatic** nervous system**(SNS**); the voluntary nervous system (controls functions that are under conscious), which carries sensation (e.g., pain) from the skin and joints and supplies skeletal muscle

- **Autonomic** nervous system **(ANS),** the involuntary/visceral nervous system, which supplies smooth muscle, glands and viscera (internal organs) in the body cavities.

Within the ANS, there are two subsystems that have mostly opposing effects - the sympathetic and parasympathetic. These systems generally but not always work in opposition to each other; the sympathetic system activates and prepares the body for vigorous muscular activity, stress, and emergencies. While the parasympathetic system lowers activity, operates during normal situations, permits digestion, and conservation of energy.

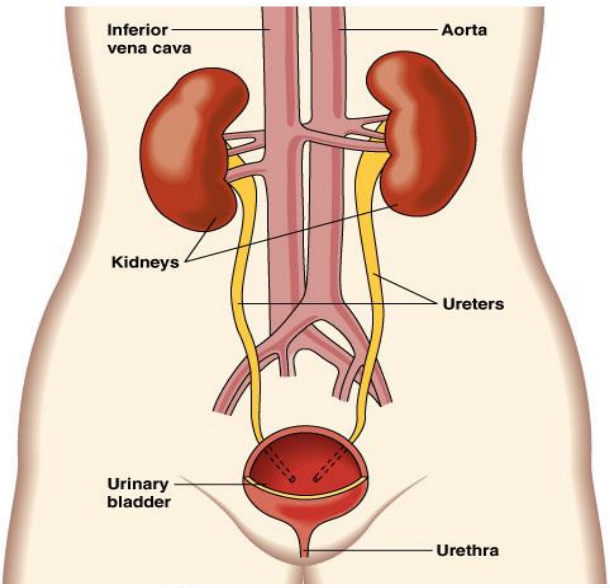
Urinary system

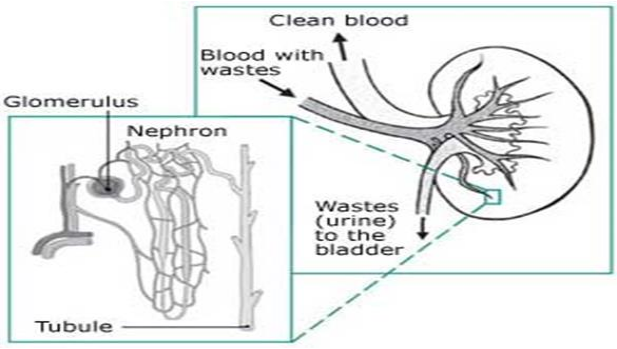
Consists of the 2 kidneys, 2 ureters, the bladder, and the urethra.

The purpose of the renal system is to eliminate wastes from the body, regulate blood volume and pressure, and control levels of electrolytes and metabolites

Kidneys

- Lie on the posterior abdominal wall, one on each side of the vertebral column.
- Each kidney extends from level of 12th thoracic to level of the 3rd lumbar vertebrae, with its long axis directed downwards and laterally.
- On the upper end of each kidney the adrenal gland rests
- Each kidney consists of millions of functional units called nephrons
- Nephrons lies mainly in the cortex except for the collecting tubules which lies in the medulla
- Collecting tubules collect to form minor calyces➔major calyces➔pelvis of the kidney➔ureter





Ureters

10 inches (25 cm) long, with its upper 1/2 situated in the abdomen and its lower 1/2 in the pelvis.

Urinary bladder

- ☒ Lies in the pelvic cavity directly behind the pubic bones.
- ☒ An empty bladder is pyramidal in shape with a mean capacity of (250 ml**)**
- ☒ **The neck of the bladder:** It is the lowest and most fixed part of the bladder, and is continuous with the internal urethral orifice

Male urethra

- ☒ S-shaped
- ☒ 20 cm Length.
- ☒ It extends from the internal urethral orifice at the neck of the urinary bladder to the external urethral orifice at the tip of the penis.
- ☒ Division: It is divided into 3 parts: prostatic, membranous and spongy.

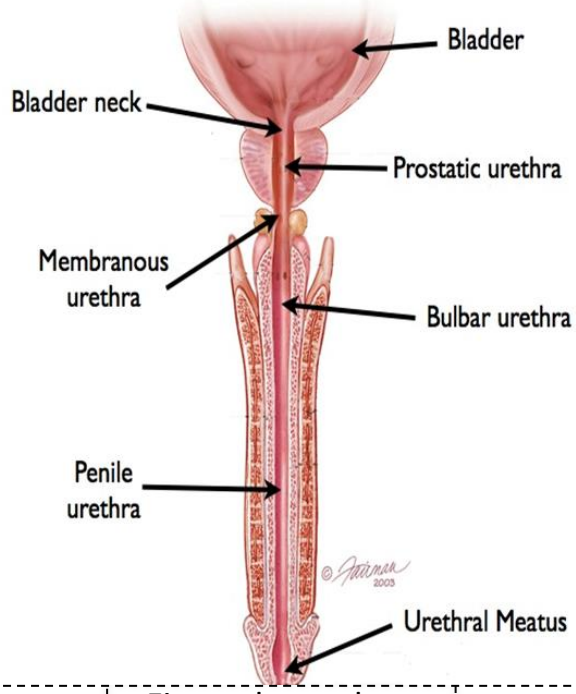


Fig. male urethera

Female urethra

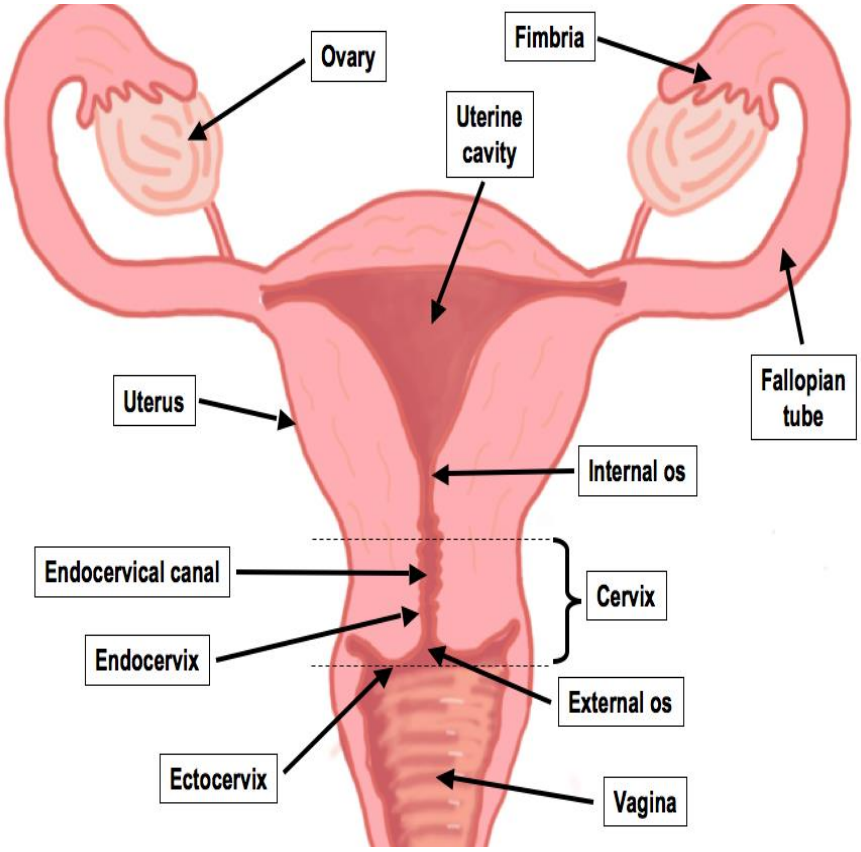
- ☒ It is 4 cm long.
- ☒ It begins at the internal urethral orifice and opens into the vestibule of the vagina by the external urethral orifice which lies directly in front of the vaginal orifice.

Sphincters of the urethra :

- ☒ Internal urethral sphincter (involuntary), and surrounds the internal urethral orifice and neck of the bladder.
- ☒ External urethral sphincter (voluntary- skeletal muscles), and surrounds the membranous urethra in male and intermediate part of the urethra in females

The female reproductive system

(or female genital system)



- **Internal:** The female internal reproductive organs are the vagina, uterus, uterine tubes (Fallopian tubes, oviducts) and ovaries.
- **External: external genitalia (skin folds)**

1 Ovaries

☒ a pair of small glands about the size and shape of almonds,
☒ Located on the left and right sides of the pelvic body cavity.
☒ Ovaries produce female sex hormones such as estrogen and progesterone as well as ova (commonly called "eggs").

2 Fallopian Tubes

- ☒ A pair of muscular tubes that extend from the left and right superior corners of the uterus to the edge of the ovaries.
- ☒ The fallopian tubes end in a funnel-shaped structure called the infundibulum, which is covered with small finger-like projections called fimbriae.
- ☒ The fimbriae swipe over the outside of the ovaries to pick up released ova and carry them into the infundibulum for transport to the uterus.
- ☒ The inside of each fallopian tube is covered in cilia that work with the smooth muscle of the tube to carry the ovum to the uterus.

3 Uterus (womb)

- a hollow, muscular, pear-shaped organ
- The lower part known as the cervix
- Connected to the two fallopian tubes on its superior end and to the vagina (via the **cervix**) on its inferior end,
- The inner lining of the uterus, known as the **endometrium**, provides support to the embryo during early development When fertilization does not occurs endometrium breakdown in the form of **menstruation** at the end of the cycle
- . The visceral muscles of the uterus contract during childbirth to push the fetus through the birth canal.

4 Vagina

- An elastic, muscular tube that connects the **cervix of the uterus** to the exterior of the body.
- Its outer opening lies posterior to the urethra and closed in virgins by a thin membrane (hymen).

Male Reproductive (genital) System

1) Scrotum

- ✓ A sac-like organ made of skin and muscles that houses the testes.
- ✓ The scrotum is made up of 2 side-by-side pouches with a testis located in each pouch.

2) Testis (plural testis)

The 2 **testes** are the male gonads responsible for the production of sperm and testosterone.

- The testes are ellipsoid glandular organs around 1.5 to 2 inches long and an inch in diameter.
- Each testis is found inside its own pouch on one side of the scrotum.

NOTE

NB: The scrotum is located outside of the body to keep the temperature of the testis below the average body temperature, which is an essential factor for sperm

3) Epididymis (within scrotum)

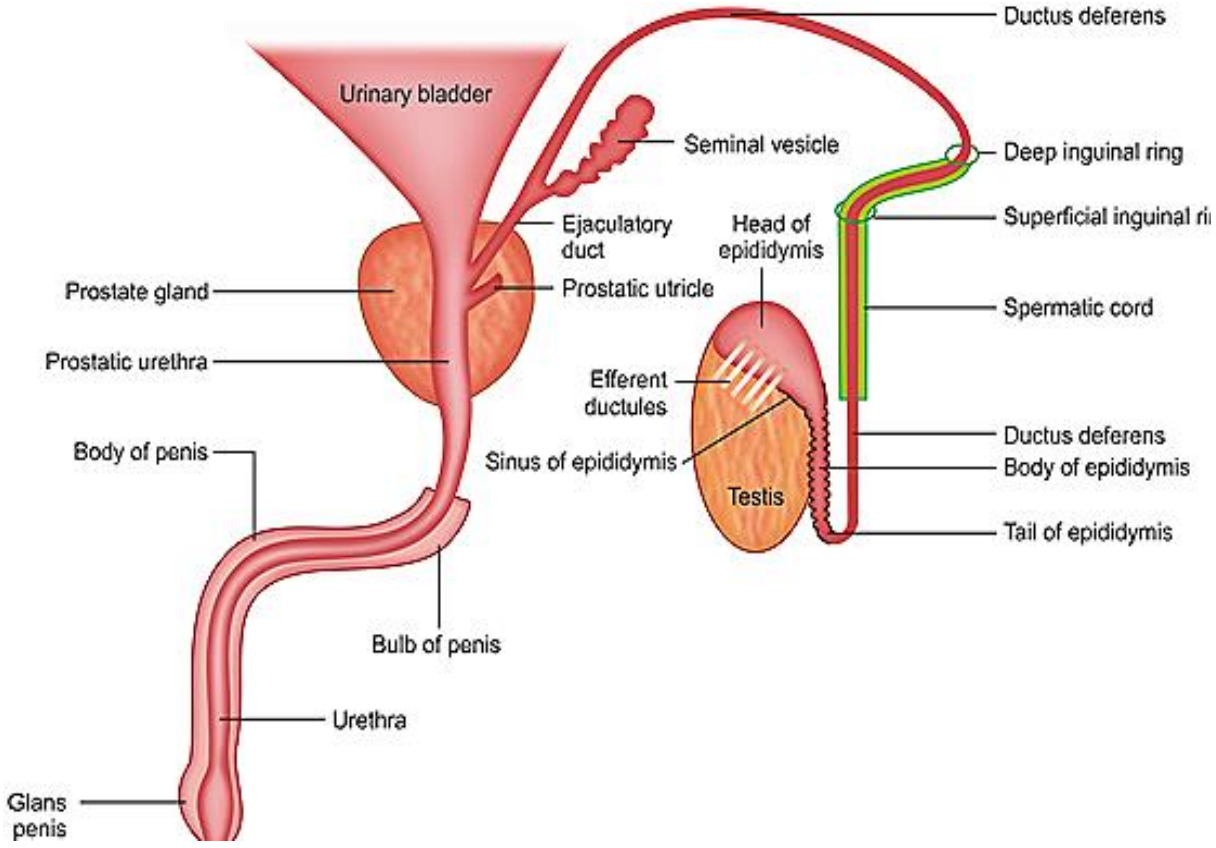
- The **epididymis** is a **sperm storage** area that wraps around the superior and posterior edge of the testes.
- The epididymis is thin tubules that are tightly coiled into a small mass.
- Sperm produced in the testes moves into the epididymis to **mature** before being passed on through the **male reproductive organs**.
- **Consists of head (above), body and tail below.**

4) The ductus deferens

- Known as the vas deferens
- 30-45 cm
- A muscular tube that carries sperm superiorly from the epididymis tail into the abdominal cavity to the ejaculatory duct.

5) Ejaculatory duct (2cm)

✓
✓ The union of the ductus deferens with the ducts from the seminal gland (seminal vesicle).
✓ During ejaculation, the ejaculatory duct opens and expels sperm and the secretions from the seminal vesicles into the urethra.
✓ The urethra in males is a common pathway for urine and semen.

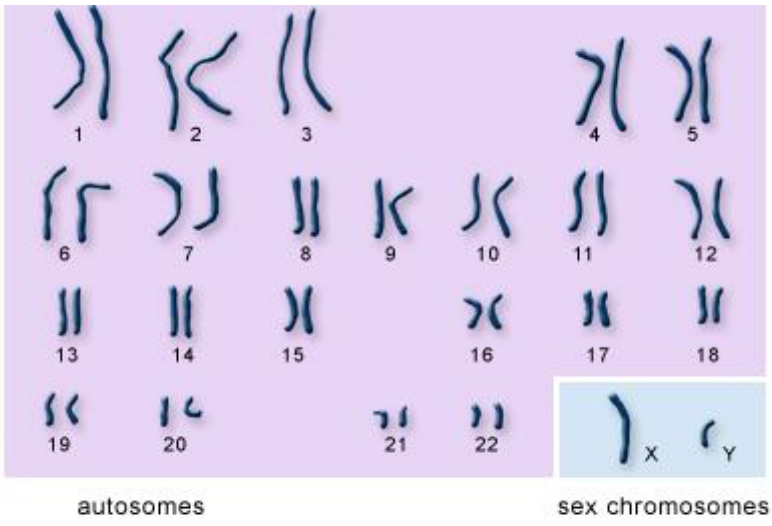


Chapter 3

General Embryology

- **How many chromosomes do people have?**
  - In humans, each cell normally contains 23 pairs of chromosomes, for a total of 46.
  - Twenty-two of these pairs, called **autosomes**, look the same in both males and females.
  - The 23rd pair, the **sex chromosomes**, differ between males and females. Females have two copies of the X chromosome, while males have one X and one Y chromosome.
  - The 22 autosomes are numbered by size. The other two chromosomes, X and Y, are the sex chromosomes.

This picture of chromosomes is called a



the human lined up in pairs karyotype.

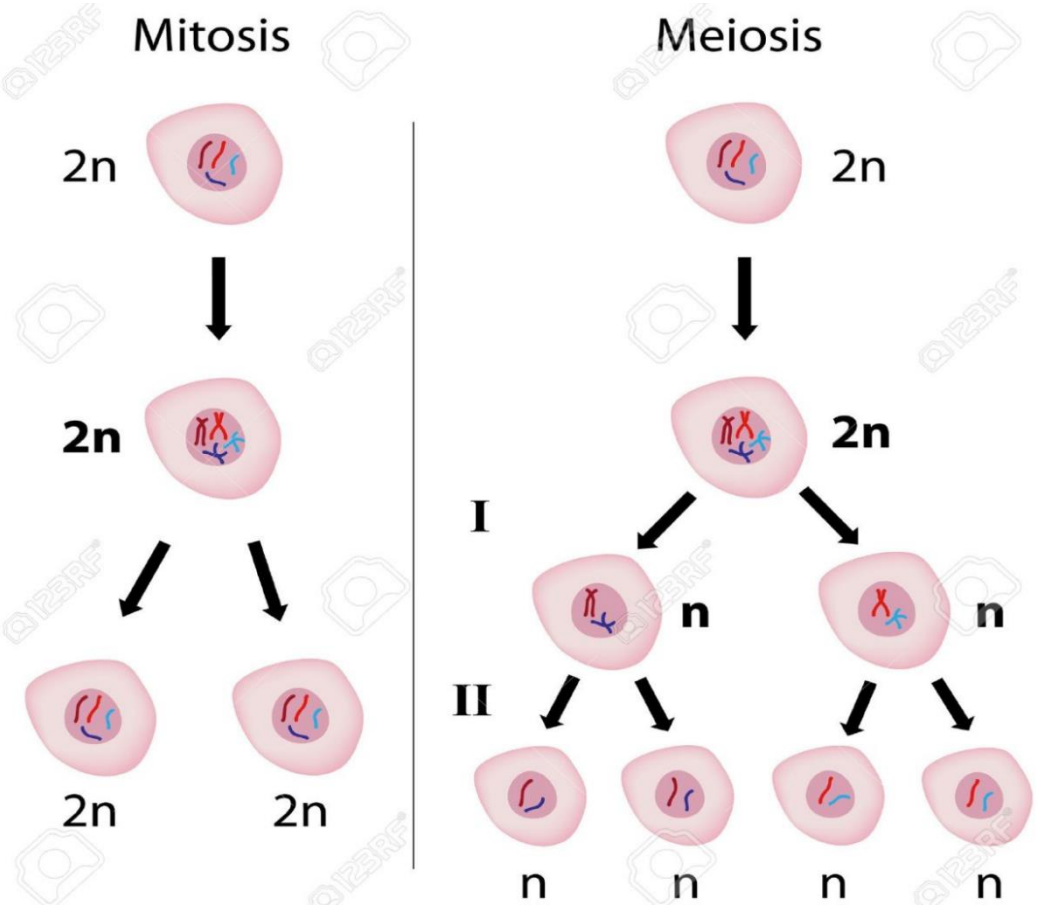
Definition of some Terms

- **A karyotype:**
  - is an individual's collection of chromosomes.
  - The term also refers to a laboratory technique that produces an image of an individual's chromosomes.
  - The karyotype is used to look for abnormal numbers or structures of chromosomes.
- **Chromosomes:**
  - thread-like structures located inside the nucleus of animal and plant cells.
  - The term chromosome comes from the Greek words for color (chroma) and body (soma). Scientists gave this name to chromosomes because they are cell structures, or bodies, that are strongly stained by some colorful dyes used in research.
  - Each chromosome is made of protein and a single molecule of deoxyribonucleic acid (DNA).
  - Passed from parents to offspring, DNA contains the specific instructions that make each type of living creature unique.

- Chromosomes are divided into two chromatids with a constriction point called a centromere in the middle.

- **Mitosis:**
  - Mitosis is a process where a single cell divides into two identical daughter cells (cell division).
  - During mitosis **one** cell divides once to form **two** identical cells.
  - The major purpose of mitosis is for growth and to replace worn out cells.

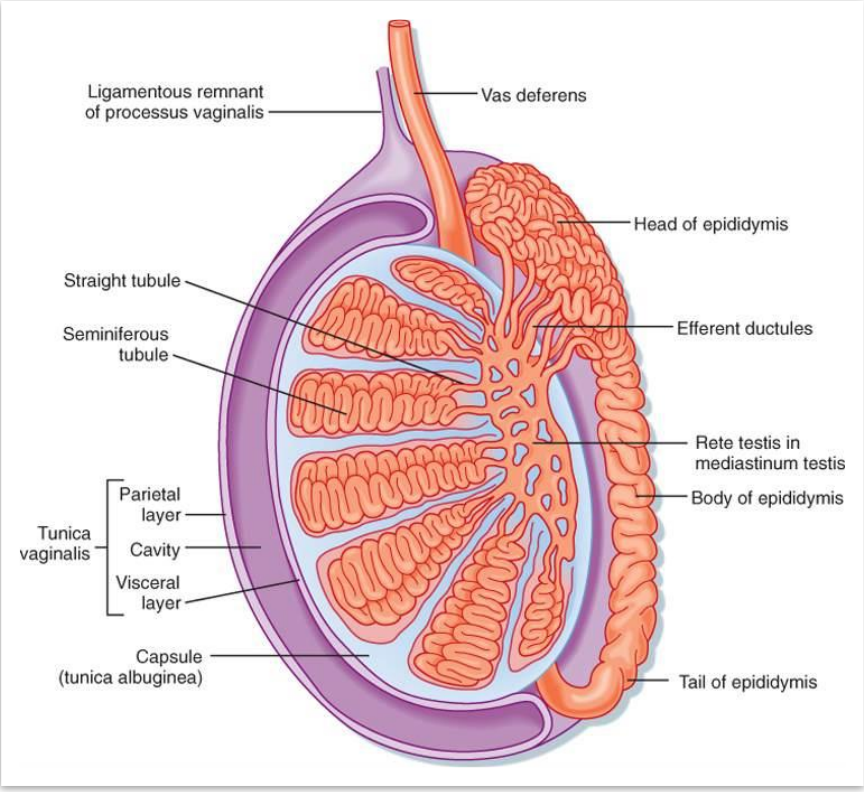
- **Miosis:**
  - Meiosis is a process where a single cell divides twice to produce four cells containing half the original amount of genetic information. These cells are our sex cells – sperm in males, eggs in females.
  - During meiosis one cell Divides twice to form four daughter cells.
  - These four daughter cells only have half the number of chromosomes? of the parent cell – they are haploid.
  - Meiosis produces our sex cells or gametes (eggs in females and sperm in males).
  - Meiosis occurs through 2 stages (meiosis I) followed by (meiosis II):In meiosis I, homologous chromosomes separate, while in meiosis II, sister chromatids separate.



Primary Sex organs

1) Structure of the testis

- The testis consists of about 250 compartments each of which lodges 2-3 highly convoluted tubules called seminiferous tubules.
- Each seminiferous tubule has a blind end and a straight open end which joins a network of tubules called the rete testis.
- This rete testis communicates with the epididymis by efferent ductules called vasa efferentia.
- In Brief: Testis >>> 250 compartments >>> 2-3 seminiferous tubules >>> Rete testis >>> Vasa efferentia >>> epididymis

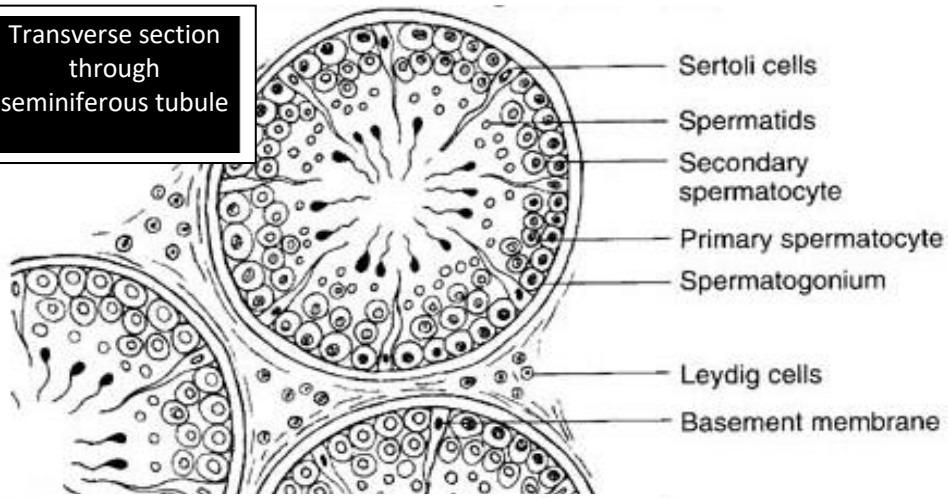


1) Scrotum

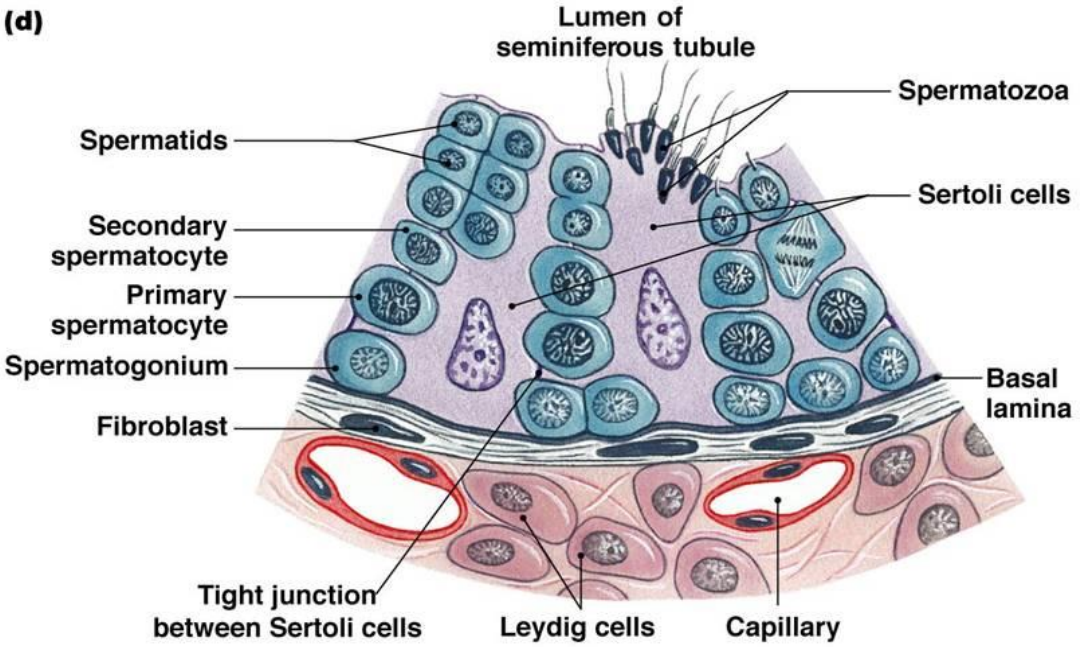
The seminiferous tubule

- The wall of the seminiferous tubule consists of several layers of germ cells. The cells lying on the basement membrane are the most primitive and called **spermatogonia** (form sperms).
- Scattered among the germ cells are special type of supporting cells called **Sertoli cells** (supporting cells) which assist in maturation of the sperms.
- In between the seminiferous tubule scattered cells called the **leydig** cells which secrete hormones **(testosterone).**

Transverse section through seminiferous tubule

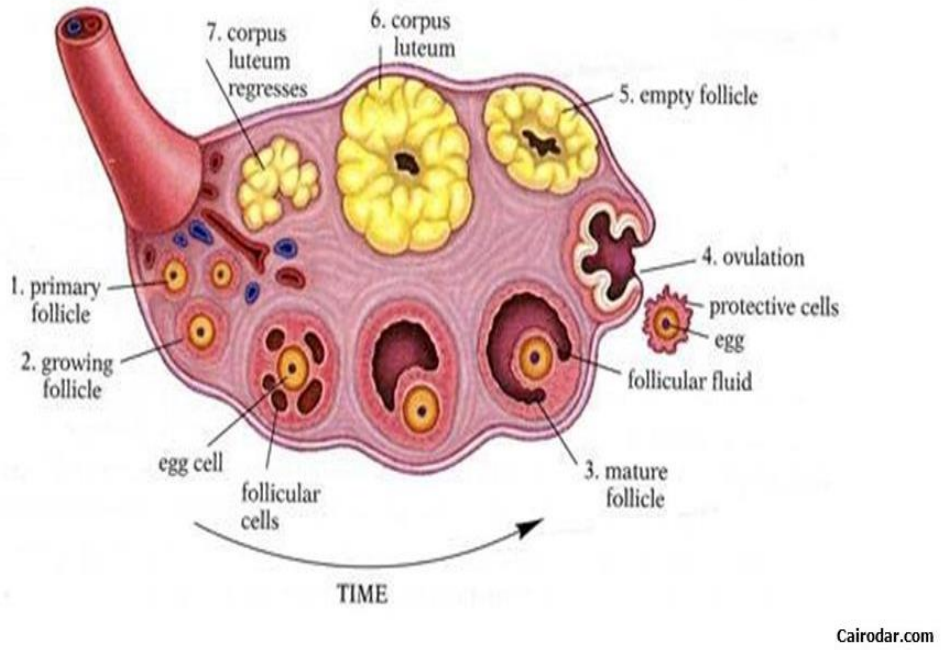


Magnified part of the semineferous tubule



2) Structure of the ovary

- The ovary consists of an outer cortex and an inner medulla. The cortex lodges the ovarian follicles which undergo maturation after puberty to produce ova(ovum).



Gametogenesis

- The process by which mature gametes (sperms or ova) are formed from primitive sex cells (genesis = formation).
- In the male, gametogenesis is called **spermatogenesis** and results in the formation of sperms, while in the female, it is called **oogenesis** and results in the formation of mature ova
- Spermatogenesis is a continuous process extending from puberty to old age, while oogenesis extends only from puberty to menopause (menopause is stoppage of menstruation which occurs at the age of 45-50 years).

1) Spermatogenesis

- The process, by which fully formed sperms (spermatozoa) are produced from primitive sex cells called Spermatogonia.

Steps of spermatogenesis:

1. Spermatogonia:

- Spermatogonia (singular: spermatogonium) are the most primitive sex cells which lie directly on the basement membrane of the seminiferous tubule.
- Each spermatogonium divides by **mitosis** (no reduction of chromosomes) into 2 daughter cells called "type A" and "typeB" Spermatogonia (each has 46 chromosomes).

NOTE

The "type A" cell remains in contact with the basement membrane (for further mitotic division), while "type B"cell continues maturation to form a primary spermatocyte.

2. Spermatocytes:

The "type B" spermatogonium enlarges in size and becomes transformed into a **primary spermatocyte** (still having 46chromosomes=diploid).
The primary spermatocyte undergoes the **1st meiotic division**(with reduction of the chromosomes to their half number=haploid).
This result in 2 daughter cells called **secondary spermatocytes** each of which contains 23 chromosomes.
The secondary spermatocyte is carrying either Y-chromosome (form male) or X-chromosome (form female).

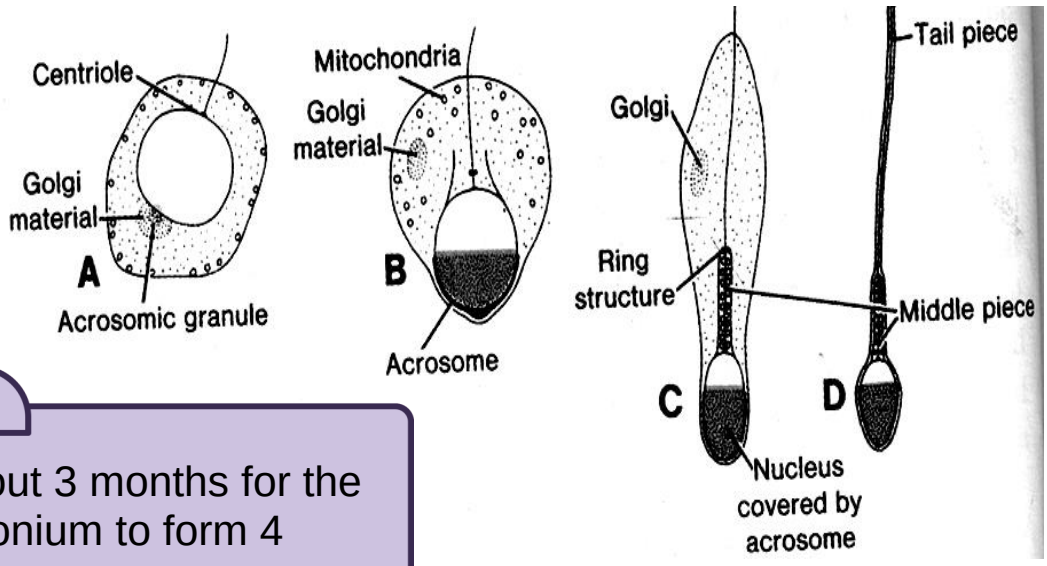
3. Spermatids:

Each secondary spermatocyte undergoes the **2nd meiotic division**to form2 daughter cells called **spermatids** each of which still has 23 chromosomes.

4. Spermiogenesis: This is the process by which the spermatids are transformed from rounded cells to complete sperms by the support of Sertoli cells. This transformation

involves a number of morphological changes in the cytoplasm of the spermatid as follows:

- Formation of head cap: the Golgi apparatus is transformed into the head or acrosomal cap which covers the anterior ½ of the head of the sperm. This cap is active enzymatically and thus facilitates penetration of the sperm into the ovum.
- Formation of axial filament: derived from one of the centrioles which elongates and forms the middle piece as well as tail of the sperm.
- Formation of mitochondrial sheath: derived from the mitochondria which condense and form sheath around the middle piece.
- Formation of the head of the sperm: it is actually the nucleus of the spermatid.

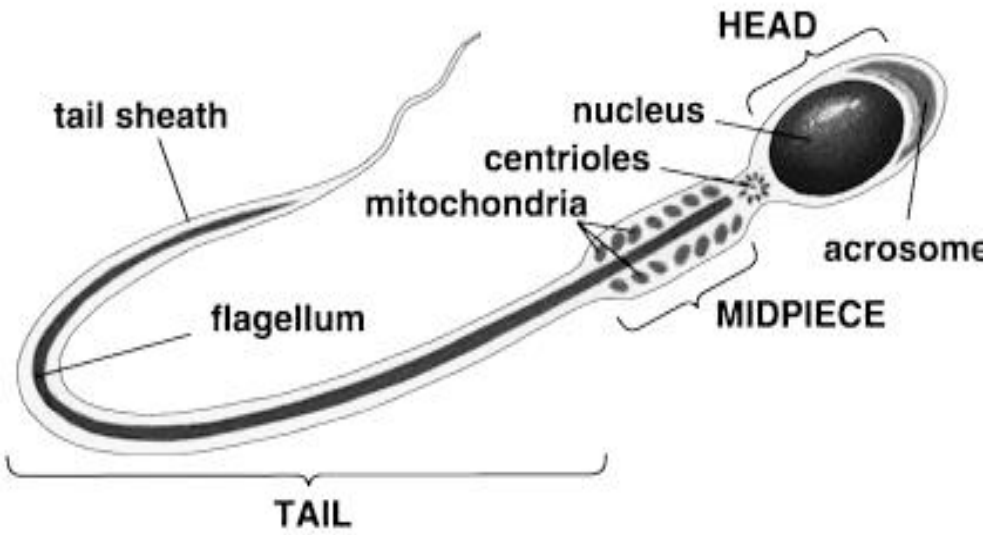


NOTE

It takes about 3 months for the spermatogonium to form 4 complete mature sperms.

Structure of the mature sperm: It consists of head, neck, middle piece and tail.

- ✓ **Head:** consists of the nucleus together with the acrosomal cap covering its anterior half.
- ✓ **Neck:** is the narrow junction between the head and the middle piece.
- ✓ **Middle piece:** consists of the axial filament together with the surrounding mitochondrial sheath.
- ✓ **Tail:** is the terminal part of the axial filament.
- ✓ The length of the whole sperm is about 60 microns



Seminal fluid: composed of 2 components: sperms and secretions.

1. **Sperms:** In the normal ejaculate(3-5 cc) the sperms are present in a concentration of 100 million sperms per cc.
2. **Secretions** are produced by the accessory sex glands; these are: seminal vesicles, prostate and bulbo-urethral glands.

Sperm transport:

The sperms leave the seminiferous tubules and pass through the following ducts: rete testis > vasa efferentia > epididymis > duct deferens > ejaculatory duct and finally the urethra from which they are ejaculated.

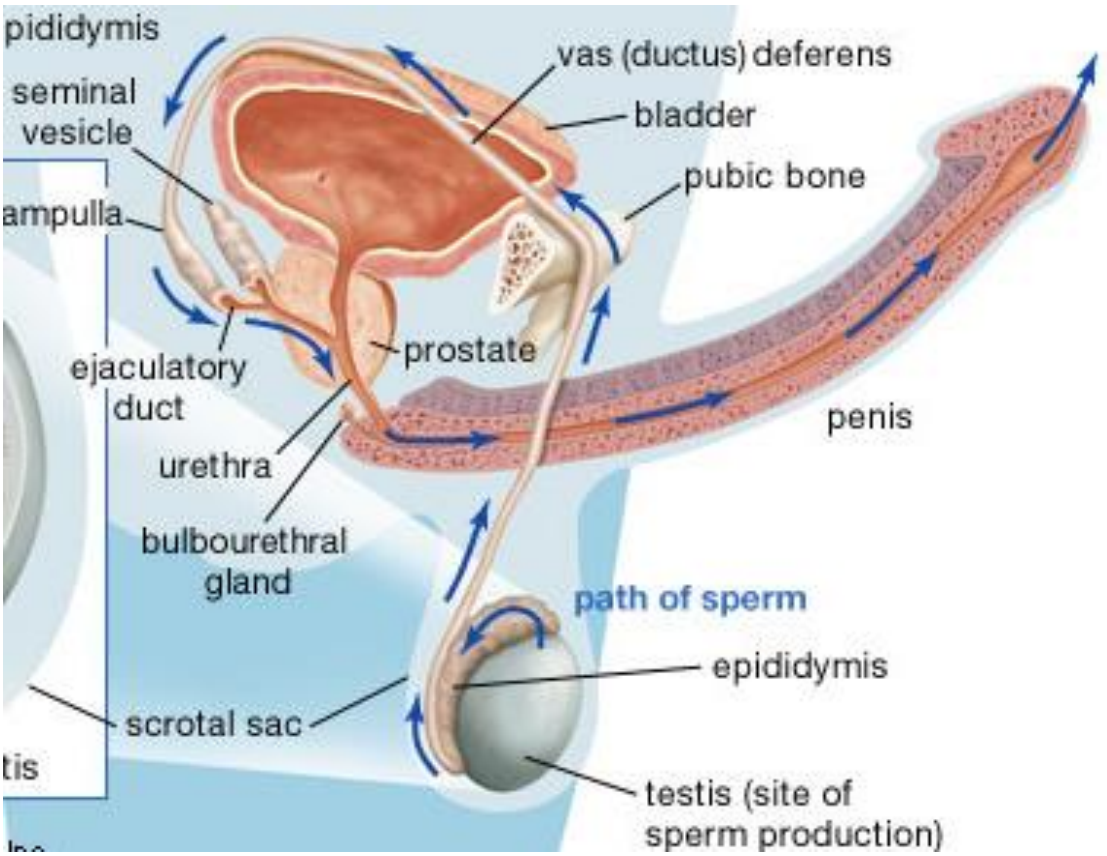
The movement of the sperms is by 3 main mechanisms:

a) **Ciliary movement** of the lining cells of the rete testis and vasa efferentia.

b) **Peristaltic contraction** of the muscle wall of the duct deferens.

c) **Active forward projecting motility** of the sperms

- The accessory sex glands pour their secretions over the sperms to form the seminal fluid. These glands are:
    - seminal vesicles
    - prostate
    - bulbo-urethral glands.


pididymis
seminal vesicle
ampulla
ejaculatory duct
urethra
bulbourethral gland
scrotal sac
tis
vas (ductus) deferens
bladder
pubic bone
prostate
penis
path of sperm
epididymis
testis (site of sperm production)

2) Oogenesis

- ✓ the process by which mature ova (singular ovum) are produced from primitive sex cells called oogonia (singular oogonium).
- ✓ It is a cyclic process, i.e. it occurs once every month. (This contrasts with spermatogenesis which is a continuous process.)

The Intra-uterine stage:

-oogonia:

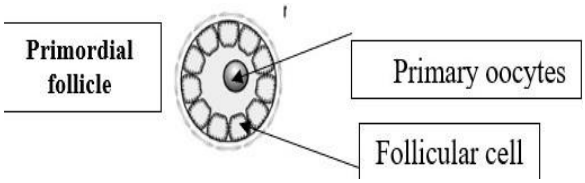
Early during intra-uterine life, the ovary contains only **oogonia** which reach about 5,000,000 by the 5th month. Then, the majority of the oogonia degenerate while the surviving ones become transformed into much larger cells called **primary oocytes**.

-primary oocytes:

These primary oocytes remain arrested in the prophase stage of meiotic division I from the intrauterine life until sexual maturity.

-primordial follicle:

Each primary oocyte is surrounded by a layer of flat follicular cell to form the **primordial follicle**. The primordial follicles remain dormant until puberty when they start maturation



NOTE

-The total number of primary oocytes diminishes from about 1,000,000 at birth to about 40,000 at puberty.
-Oogenesis takes place in the cortex of the ovary, where only one ovarian follicle completes maturation every month under the effect of the follicle stimulating hormone (F.S.H) secreted by the anterior lobe of pituitary gland.

Steps of oogenesis (Occurs every month after puberty)

I. Maturation of primordial follicle:

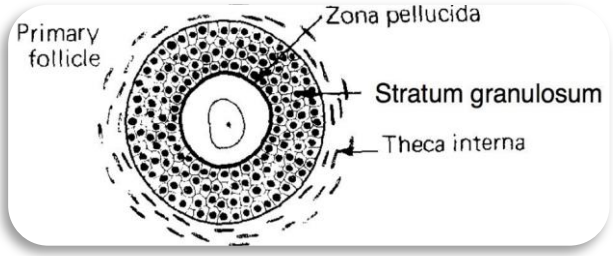
The **primordial follicle** consists of main components: a primary oocyte and follicular cells surrounding the oocyte.

1. Maturation of the primary oocyte:

-The primary oocyte (contains 46 chromosomes) undergoes the **1st meiotic division** giving rise *secondary oocyte* (23 chromosomes) and a small mass of nuclear material called ***1st*** *polar body* (23 chromosomes) which is situated at the pole of the cell (hence its name polar) and contains no cytoplasm.

-The secondary oocyte, then, starts the **2nd meiotic division**; this division is not completed only at the time of fertilization when the ovum is penetrated by the sperm.

2. Maturation of the follicular cells:



✓ **formation of the primary follicle**:

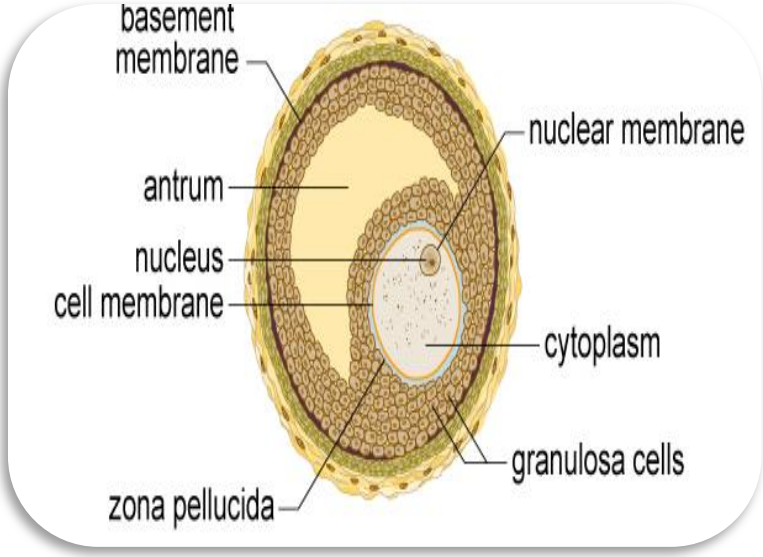
-The follicular cells of the primordial follicle divide repeatedly by mitosis to form several layers around the oocyte. These cells accumulate granules in their cytoplasm, and thus are called granulosa cells

- The granulosa cells become separated from the oocyte by a thick hyaline membrane called zona pellucida (pellucid = transparent).

✓ **Formation of the secondary follicle**:

-Small cavities filled with fluid appear between the granulosa cells. These cavities then form a single cavity called **follicular antrum**.

-The follicular antrum separates the granulosa cells into 2 layers (outer and inner):



2. **Outer layer**: forms called stratum granulosum (stratum = layer)
3. **Inner layer**: surrounds the secondary oocyte and bulges into the antrum as an elevation called cumulus oophores (cumulus = elevation; oophores= bearing the ovum).

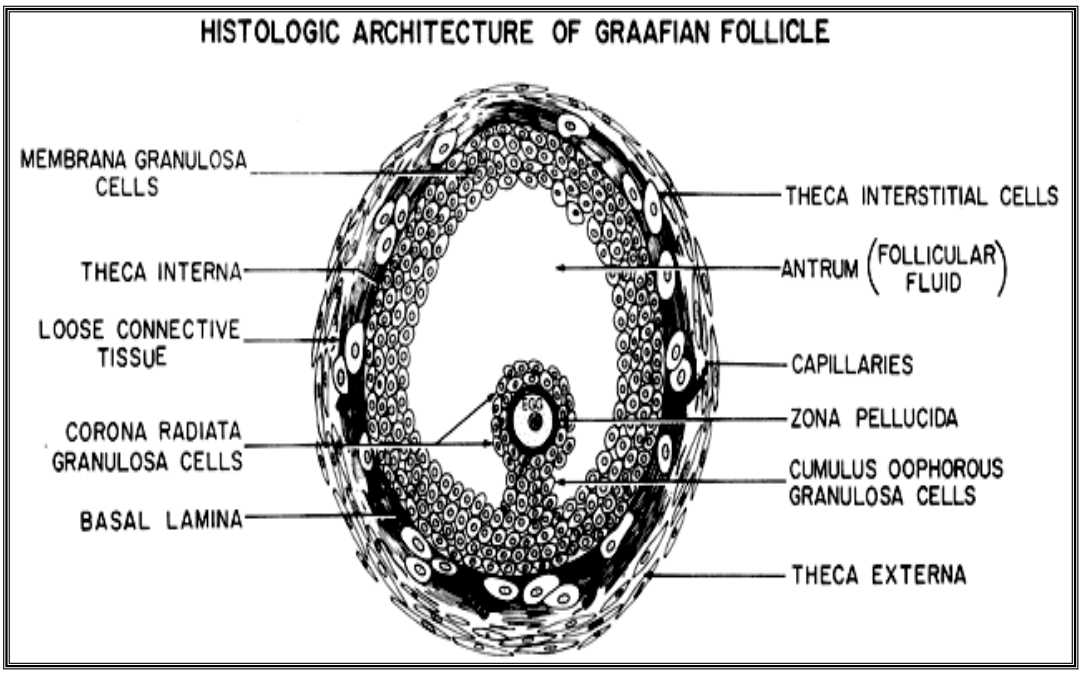
✓ **Formation of mature (Graafian) follicle**

Pressure increases inside the antrum

To the outside of the stratum granulosum, a sheath of stroma cells (cells of the ovary itself) is formed around the whole follicle and called the theca follicle (theca = sheath). The theca is divided into 2 layers (external and internal):

1-Theca externa: outer layer and consists of fibrous tissue and big blood vessels

2-Theca interna: inner layer and consists of cells and fine blood vessels (**These cells are the main source of estrogen hormone).**



II)Rupture of graafian follicle (OVULATION):

-Once the Graafian follicle becomes fully mature, it comes to lie close to the surface of the ovary.

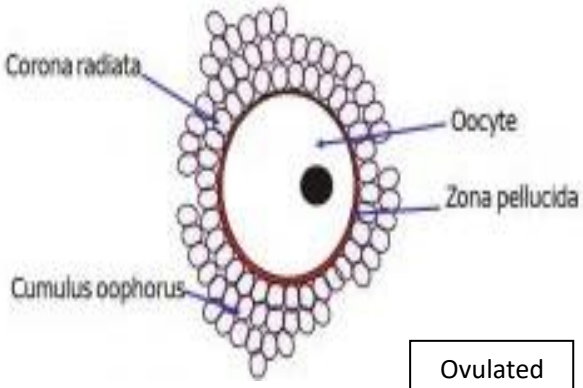
-It then ruptures through overlying cortex of the ovary which is thin and ischemic at the site of rupture.

-Ovulation takes place 14 days before the beginning of the following menstrual cycle.

-As a result of rupture, the secondary oocyte together with surrounding zona pellucida and corona radiata is discharged from the ovary to enter the uterine tube

The following factors leading to the rupture of the follicle

1. Increase in size of the follicle.
2. Thinning of the overlying cortex of the ovary.
3. Rise of the pressure of the fluid inside the follicular antrum



Ovulated oocyte

NOTE

-The fimbriae of the uterine tube covering this ovary guide the ovulated oocyte to enter the uterine tube
- The corona radiata is formed by the granulosa cells which remain attached to the zona pellucida and stay around it after ovulation.

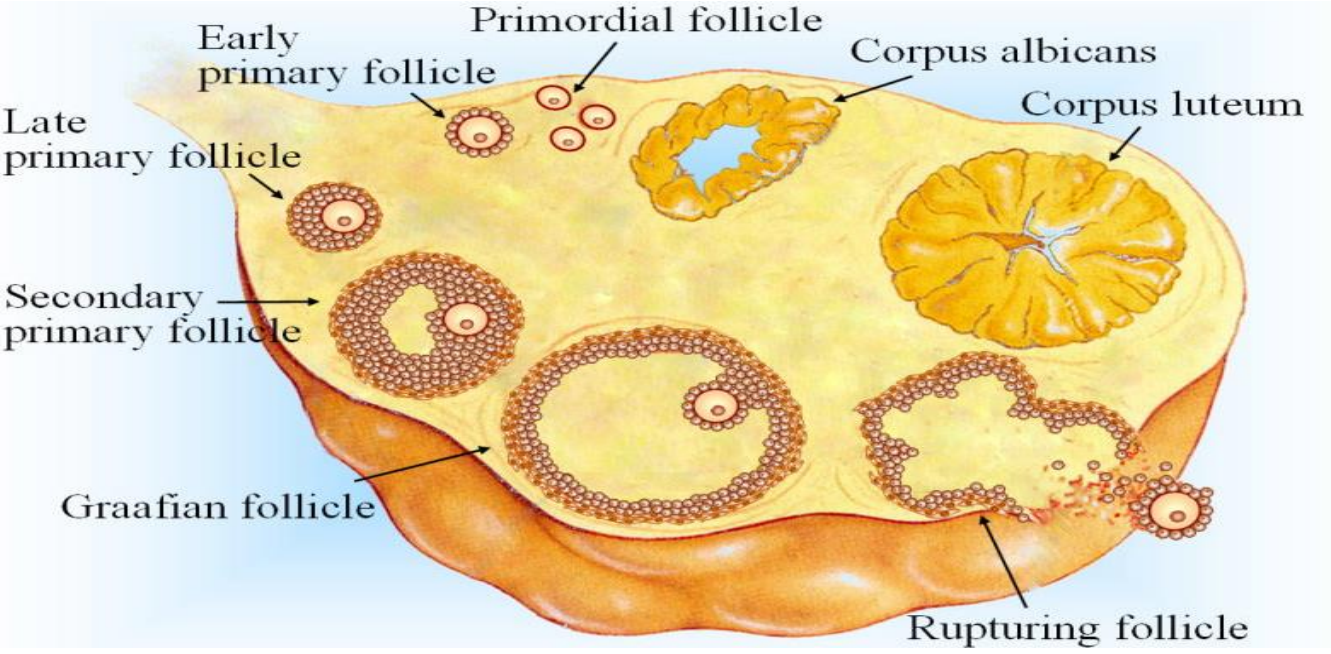
III. Formation of corpus luteum

-Following ovulation, the wall of the Graafian follicle collapses and rapidly undergoes luteinization (appearance of yellow pigment in the wall of the follicle).
-This luteinization takes place under effect of the luteinizing hormone (L.H.) secreted by the pituitary gland.
-As a result, the Graafian follicle becomes transformed into a yellow body called corpus luteum (corpus = body; luteus = yellow).
-These luteinized cells secrete progesterone hormone, but few of the theca interna cells continue to secrete estrogen hormone➔the progesterone hormone increases in the blood while the estrogen hormone decreases.

IV. Fate of corpus luteum:

It degenerates after some time and transforms into a white fibrous mass called corpus albicans (albus = white).

There are 2 types of corpuses luteum:
1. In absence of fertilization: the corpus luteum degenerates within 10 days of ovulation and is called corpus luteum of menstruation.
2. In case of fertilization:
the corpus luteum remains active for a longer period (4-5 months) until the placenta becomes fully functioning and then the corpus luteum degenerates. It is called corpus luteum of pregnancy.



OVARIAN CYCLE (28 days)

A series of changes which take place in the ovary every month, under the effect of the follicle stimulating hormone (F.S.H.) and the luteinizing hormone (L.H.). These hormones are secreted by the pituitary gland.

PHASES:

✓ **Follicular phase** **(**Begin from the 1st day of menstrual flow**)**

- Starts by secretion of F.S.H. which initiates maturation of the primordial follicle to become the mature Graafian follicle
- During maturation, the estrogen hormone is secreted by the cainterna cells of the Graafian follicle.

2) Ovulation (at day 14):

The Graafian follicle ruptures and the ovum (in the stage of secondary oocyte) is extruded to enter the uterine tube. The wall of the Graafian follicle is then transformed into the corpus luteum.

3) Luteal phase: **(**stage of formation of corpus luteum under effect of the luteinizing hormone (L.H.) secreted by the pituitary gland**)**

The progesterone hormone increases in the blood while the estrogen hormone decreases.

The estrogen and progesterone hormones act in the following ways

a) They suppress secretion of F.S.H. of the pituitary gland, this preventing maturation of further primordial follicles during the same cycle.

b) They pass to the endometrium of the uterus (through the blood) which responds by proliferation as well as by increasing activity of its glands. At the end of the ovarian cycle, the corpus luteum degenerates **(corpus luteum of menstruation)** and becomes the corpus albicans leading to cessation of secretion of the estrogen and progesterone hormones. As a result, the F.S.H. released again from the pituitary gland and a new ovarian cycle starts in the ovary.

If fertilization takes place, the corpus luteum continues functioning for a longer period (**corpus luteum of pregnancy**), and the ovarian cycles will stop until after childbirth, when they start again.

MENSTRUAL CYCLE

A series of changes which occur in the endometrium of the uterus every month under the effect of the ovarian hormones (estrogen and progesterone).

It lasts 28 days, i.e. one lunar month, and its beginning is marked by the passage of menstrual blood flow.

Phases (stages): The menstrual cycle is divided into 3phases:

A) Phase of menstrual flow:

In this phase, the menstrual blood flows for about 3-5 days. This blood (about 50 cc) consists of the following:

1. Shedded epithelium of the endometrium. However, the basal layer of the endometrium is left intact for further regeneration.
2. Secretions of the degenerated uterine glands.
3. Venous and arterial blood: from the degenerated vessels.

B) Proliferative phase (also called follicular phase):

It lasts for about 10 days. It is under the influence of estrogen hormone (only) and ends by the time of ovulation.

The basal part of the endometrium (left after shedding) shows proliferation of its 3 main constituents: glands, vessels and stroma cells, as follows:

1. Glands: grow in size, increase in number and become tubularin shape
2. Blood vessels: grow in size and increase in number (but not yet spiral).
3. Stroma cells: divide and thus increase in number (but not in size).

This phase is also called estrogenic phase because it is under the influence of estrogens secreted by the Graafian follicle**.**

C) Secretory phase (pregestational or luteal phase):

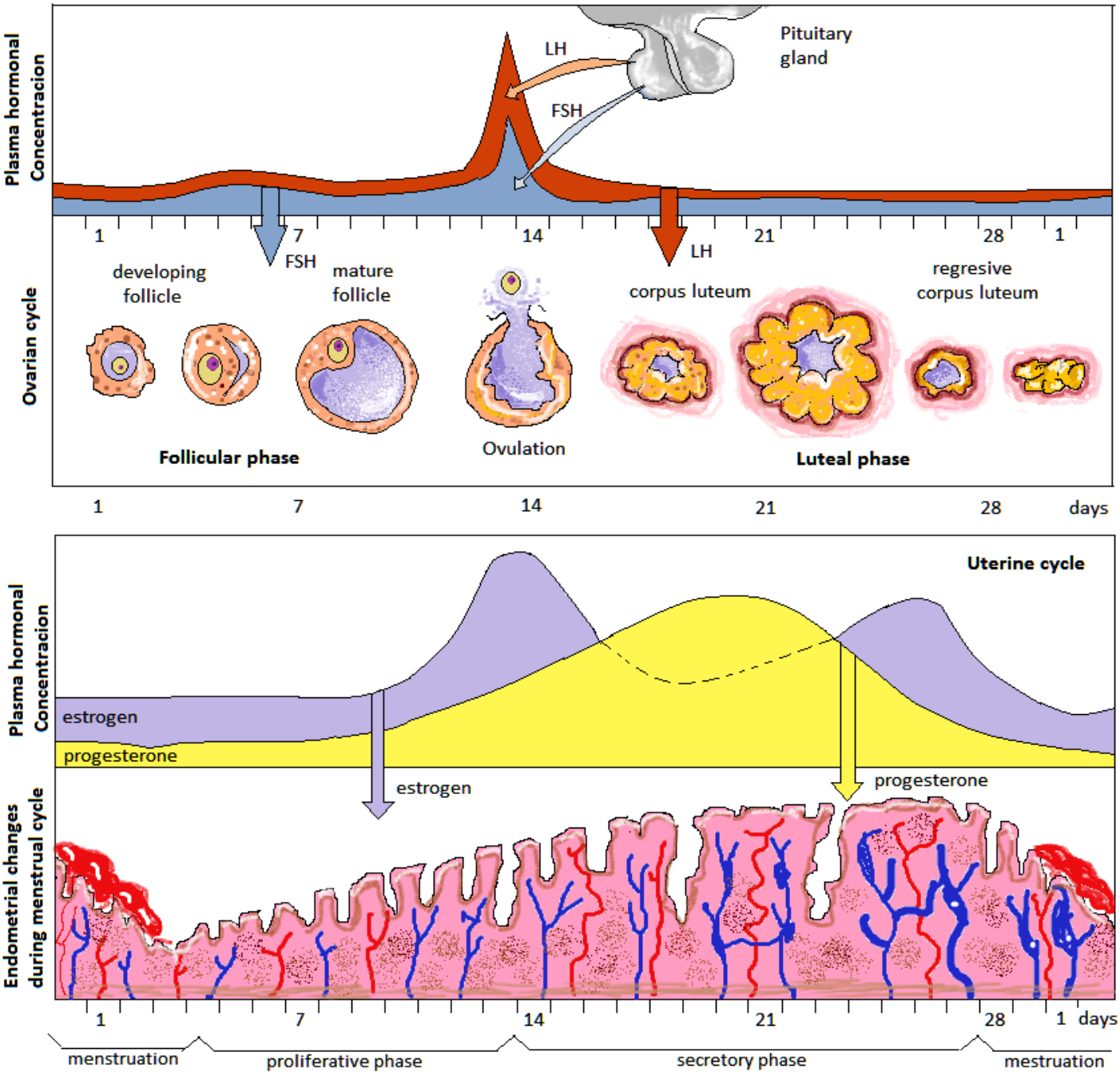
It follows the proliferative phase, i.e. just after ovulation. It I under the influence of the progesterone hormone secreted by the corpus luteum It lasts for about 10 days after the time of ovulation.

The endometrium as a whole becomes thicker, edematous and its epithelium becomes tall columnar and undergo the following changes:

1. Uterine glands: become dilated and distended with secretion and rich in glycogen and mucin.
2. Blood vessels: become dilated and spiral and show many arterio-venous anastomoses.
3. Stroma cells: become large in size due to accumulation of glycogen and lipids.

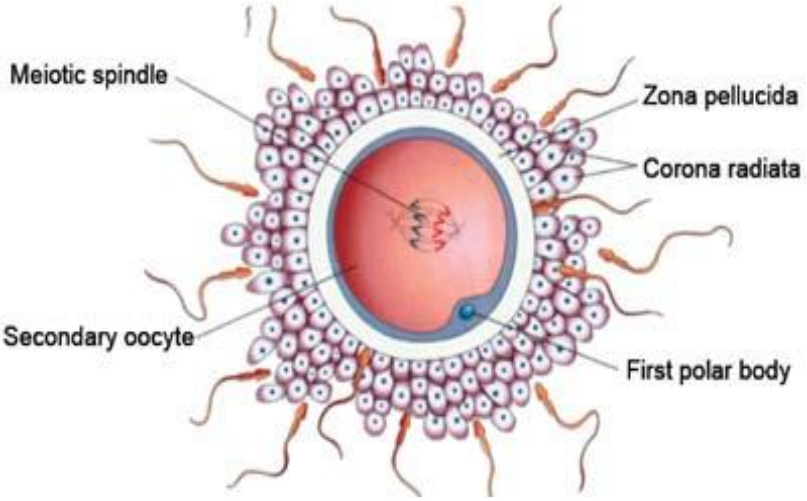
NOTE

This phase ends by cessation of secretion of progesterone and estrogen (because of degeneration of the corpus luteum) which results in degeneration of the surface epithelium and shedding of the endometrium, thus marking the beginning of the menstrual flow of the next cycle.



1 1st week of development

- Fertilization (day 1)
- Mitotic cleavage of the blastomere
- Morula formation
- Blastocyst formation and Implantation

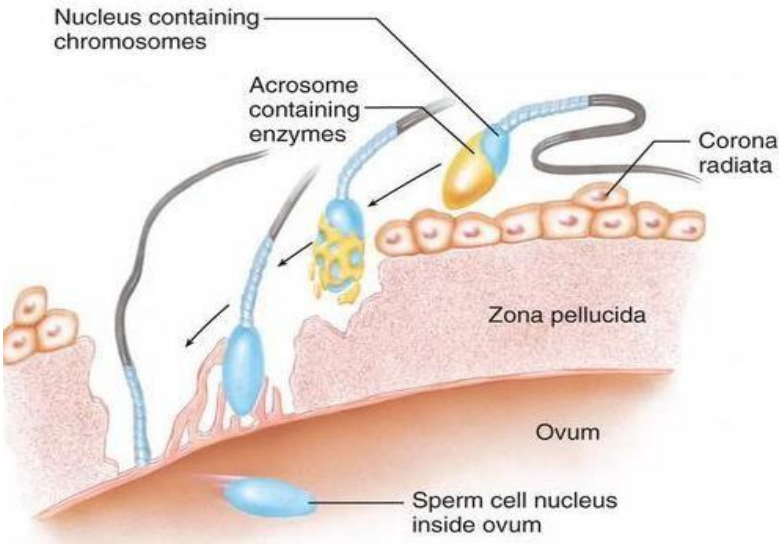


Fertilization

Fertilization is the process by which the male and female gametes fuse to form the zygote. It takes place in the ampullary dilated lateral part of the uterine tube.

STEPS

The sperm enters the ovum, which is in the stage of secondary oocyte, by penetrating the following 3 coverings of the ovum:



- **Penetration of Corona radiata**.

Penetration of the sperm is facilitated by the activity of the enzyme hyaluronidase which comes out from the head cap of the sperm.

Capacitation: The process by which the sperm becomes capable of fertilizing an egg

NOTE

Function of Zona pellucida
-surrounds the oocyte in the ovary
-protects oocyte in the uterine tube

- **Penetration of Zona pellucida**.

Immediately after penetration the zona pellucida undergoes changes by which it acts as a barrier preventing any further penetration *so only one sperm is capable of penetrating the ovum.*

- **Penetration of Cell membrane of the ovum (Vitelline membrane).**

Immediately following penetration, the secondary oocyte completes the 2nd meiotic division to become a mature ovum, and the 2nd polar body is formed.

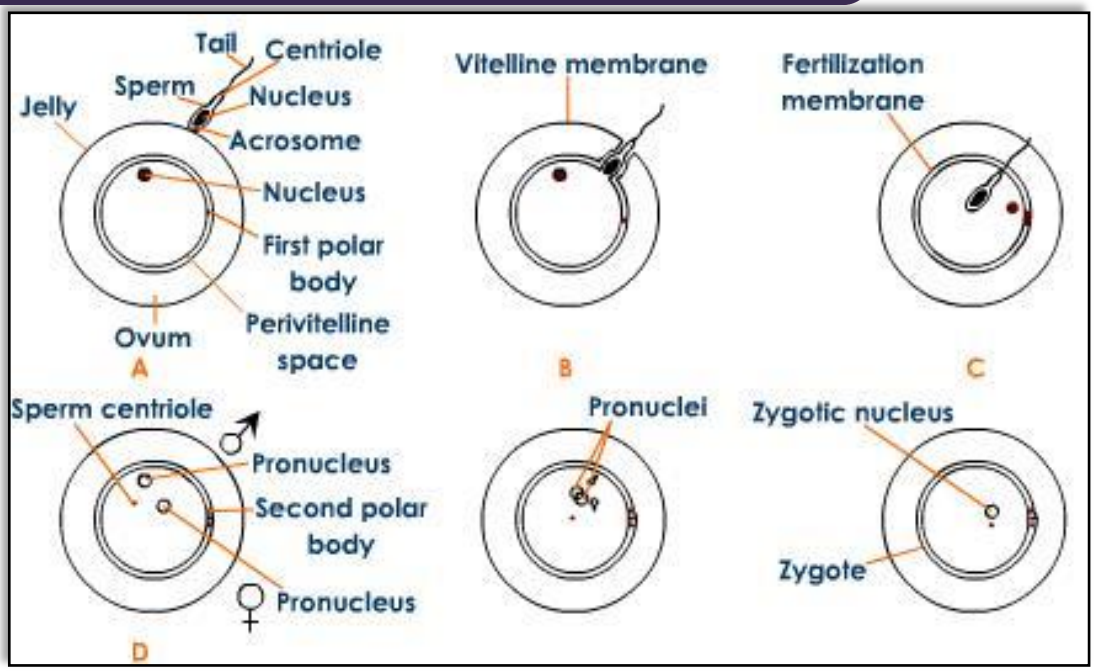
- **Fusion of male and female pronucleus**

It is only the head of the sperm which enters the ovum to form the male pronucleus, it comes to the side of the female pronucleus (of the ovum) and fuse together to form the nucleus of the zygote.

The formed zygote is still surrounded by the zona pellucida

NOTE

- ✓ **Pronuclei:** The transitional male and female nuclei formed in the egg after fertilization.
- ✓ **pronucleus differs from the nucleus in having only 23 chromosomes (n=haploid number).**
- ✓ **Pronuclei** fuse to form the diploid(n=46) zygote nucleus

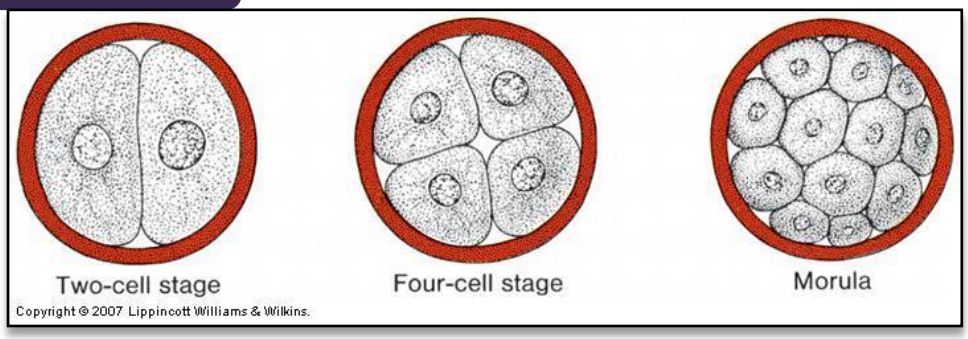


The ovum, **if not fertilized**, loses its viability and degenerates about **24 hours after ovulation**.
Similarly, *the sperms* lose their power of penetration and fertilization after **48 hours of ejaculation**.

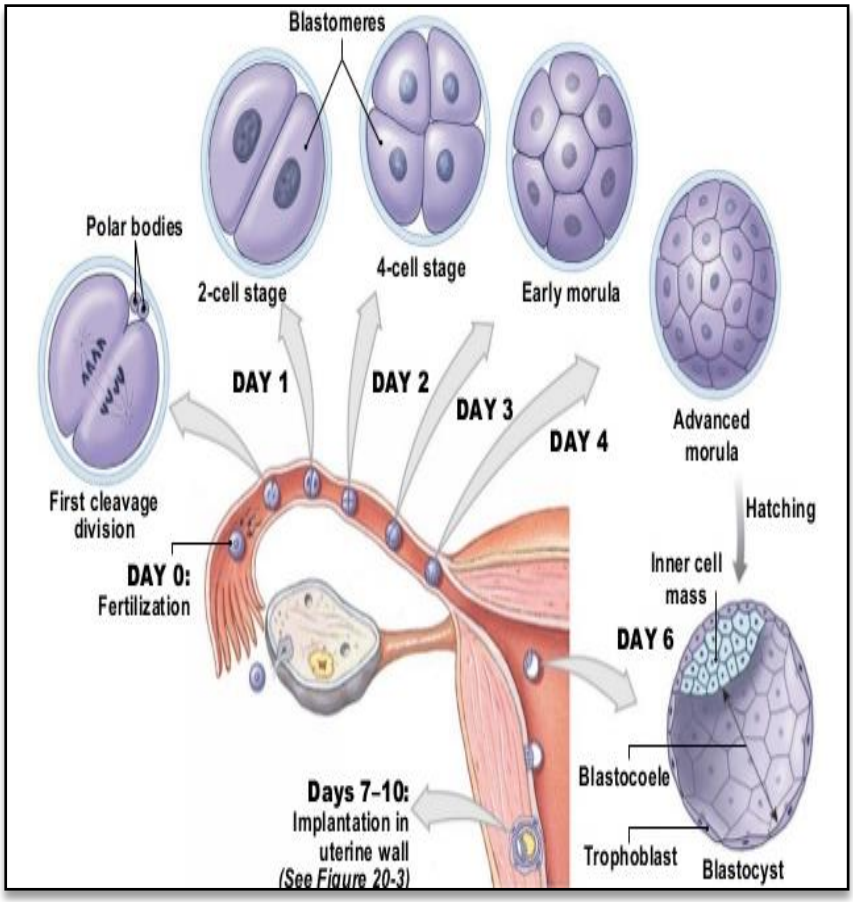
RESULTS OF FERTILIZATION

1. The **diploid number** of chromosomes (46) is restored *(23 from the sperm and 23 from the ovum)*.
2. The **sex of the zygote** is determined, being either a male (XY) or a female (XX).
3. **Cleavage (segmentation**) is stimulated to start.

Cleavage and Morula formation



- Immediately after completion of fertilization, the zygote divides repeatedly by mitosis; a process called cleavage or segmentation.



✓ During cleavage the zygote moves in the uterine tube to enteruterus, by the activity of the cilia of the cells lining the tube.

✓ The cells formed by cleavage are called **blastomeres**. These are at 1st: **2 cells**, then 4, 8, 16 and finally **32** when the zygote becomes transformed into a **morula**.

✓ The morula is still surrounded by the **zonapellucida**, and it **reaches the uterine cavity about *3 to 4 days* after fertilization**. As it enters the uterus the zonapellucida disintegrates and the morula becomes transformed into **a blastocyst**.

What is the function of zonapellucida?

1-Once the egg is fertilized, the zona pellucida **keeps other sperm out**(allow only one sperm to enter the ovum)

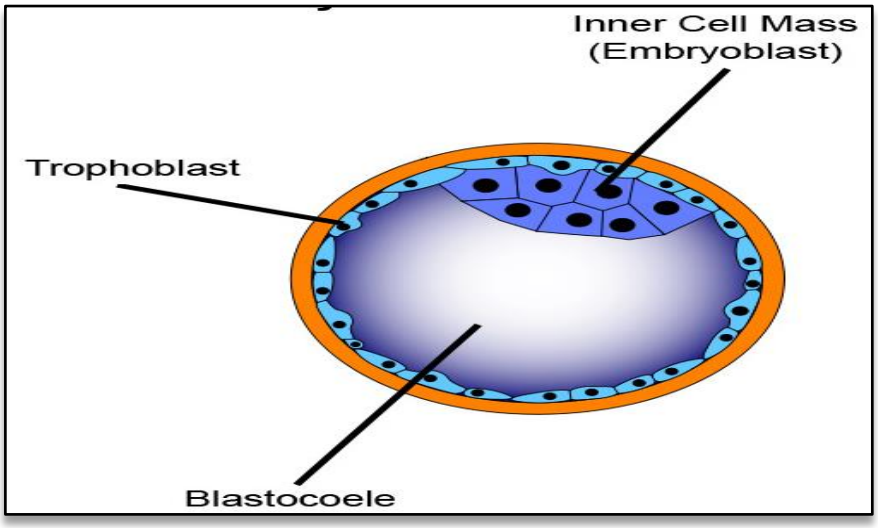
2-protects the cell as it travels **from the ovary to the uterus**

Blastocyst formation

The blastocyst is formed in about **the 6th day of fertilization**, having a cavity called **blastocoele**. Its cells are arranged into **2 groups** : outer cell mass and inner cell mass, as follows:

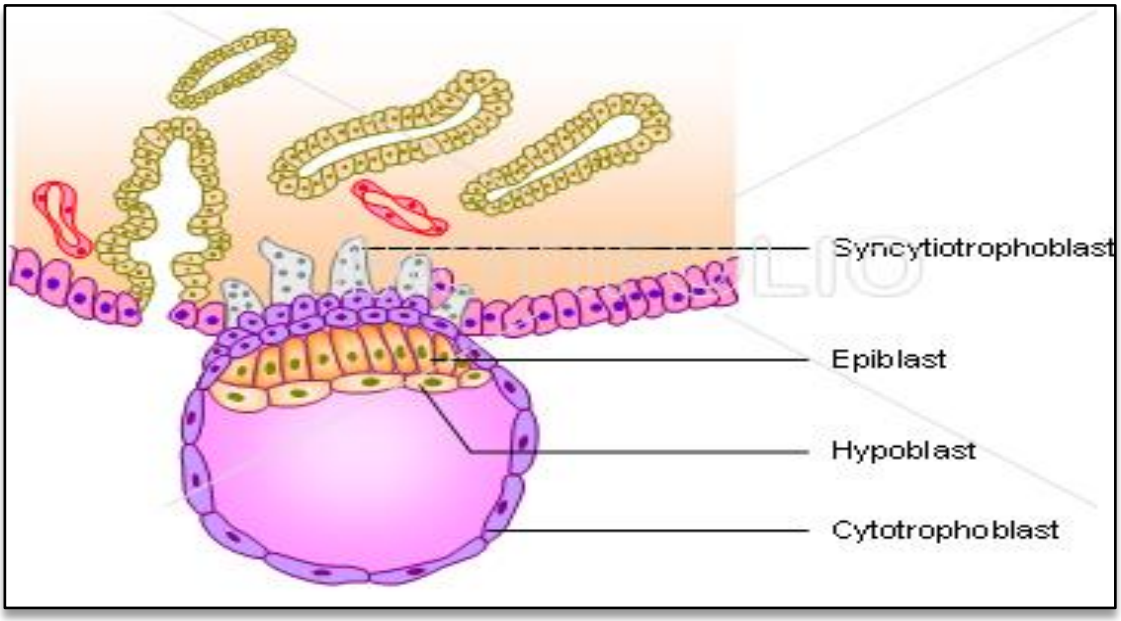
1.***Outer cell mass***: forms the wall of the vesicle, and will develop into the trophoblast which has nutritive and protective functions (tropho = related to nutrition).

2**. Inner cell mass**: projects in the form of a mass into the cavity of the blastocyst and will develop into the tissues of the embryo (**embryoblast**).



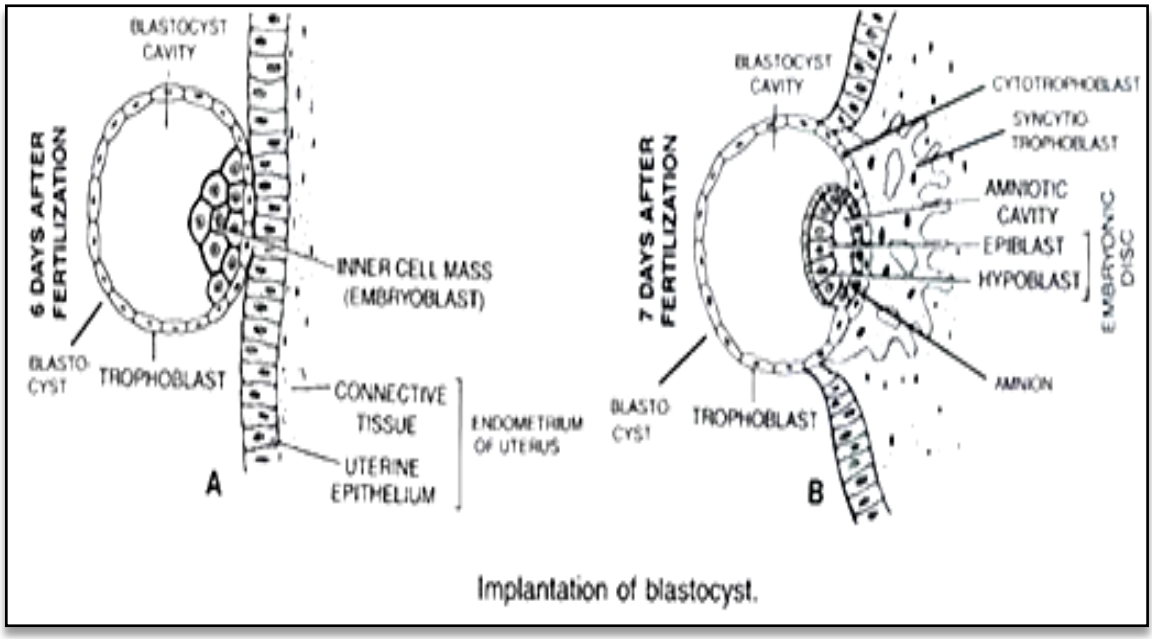
The blastocyst, then, gets implanted into the endometrium. During the process of implantation the blastocyst undergoes **the following changes**:

1. **The trophoblast** becomes differentiated into 2 layers:
   a) **Outer syncytiotrophoblast** (multinucleated mass) and is formed of a mass in which the nuclei are randomly dispersed with no cell walls between the nuclei.
   b**) Inner cytotrophoblast** formed of cells with well-defined walls.
2. The inner cell mass is transformed into a flat bilaminar embryonic disc consisting of 2 layers (dorsal and ventral):
   a) **Dorsal layer**: consists of columnar (**Epiblast**).
   b) **Ventral layer**: consists of cuboidal cells (**Hypoblast**).



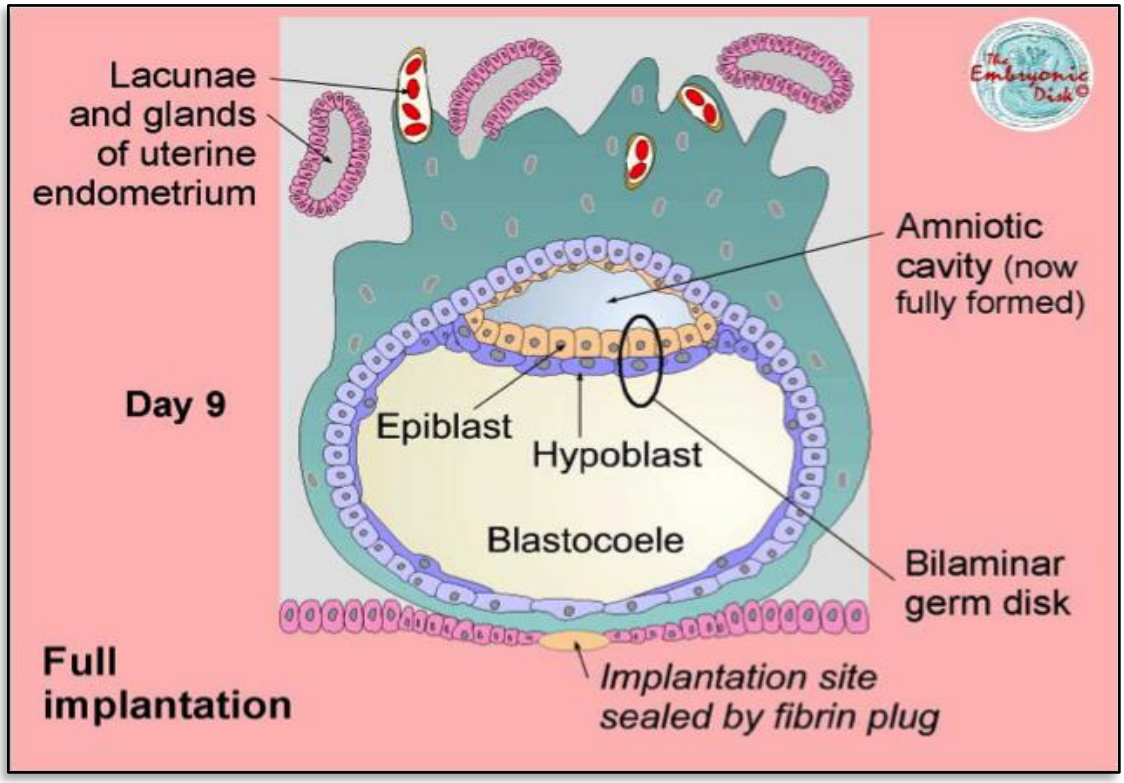
Implantation

Adhesion that the blastocyst to the endometrium of the uterus



Implantation of blastocyst.

- By the end of the 1st week after fertilization the blastocyst starts implantation into the endometrium of **the upper part of the posterior wall of the uterus (commonest site)**.
- It invades the endometrium by the embryonic pole which overlies the inner cell mass, the opposite pole is called the **ab-embryonic pole**, and it is **the last point of the blastocyst to enter the endometrium**.
- The trophoblast erodes the endometrium by producing **proteolytic enzymes**, thus forming an implantation cavity into which the blastocyst sinks. As invasion is completed, the blastocyst is embedded inside the endometrium and the penetration site closes behind.
- ***Embedding is completed about 9th day after fertilization.***



- It should be noted that at this stage the trophoblast secrete hormone called **chorionic gonadotrophin** which prolongs the life of the corpus luteum to continue secreting progesterone and oestrogen hormones.

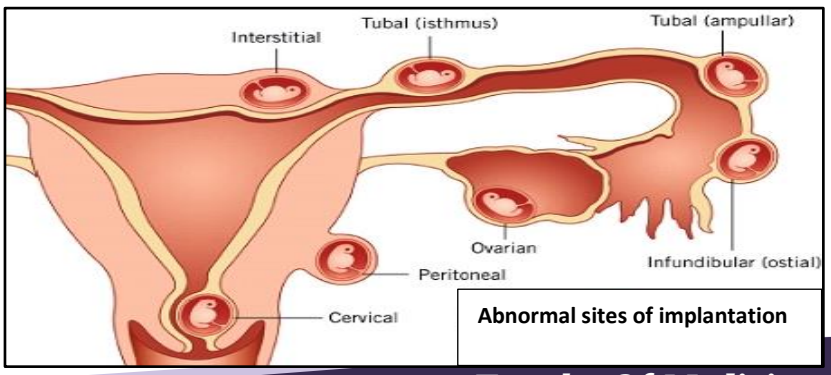
Abnormal sites of implantation

The abnormal site may be **inside the uterus** very close to the cervix where condition is called **placenta previa**, or **outside the uterus** where the condition is called **ectopic pregnancy**.

1) Ectopic pregnancy:

The blastocyst is implanted into any of the following sites, **outside the uterus** (ectopic = away from the normal site):

a) In the uterine tube: is called **tubal pregnancy** (*commonest site*).
b) In the abdominal cavity is called **abdominal pregnancy**.
c) On the surface of the ovary is called **ovarian pregnancy**.



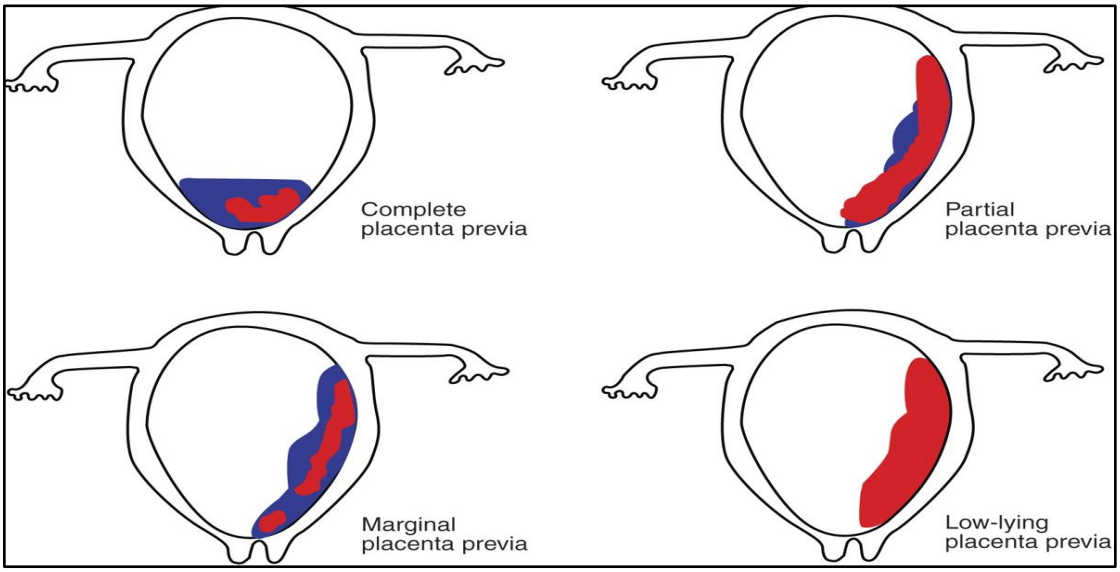
Abnormal sites of implantation

2) Placenta previa:

In this anomaly, the blastocyst implants very close to the cervix, and as a result:
the placenta will be placed below the fetus, a situation which is different from the normal implantation where **placenta lies above the fetus**(previa = which comes in front of).

There are **4 types of placenta previa**:

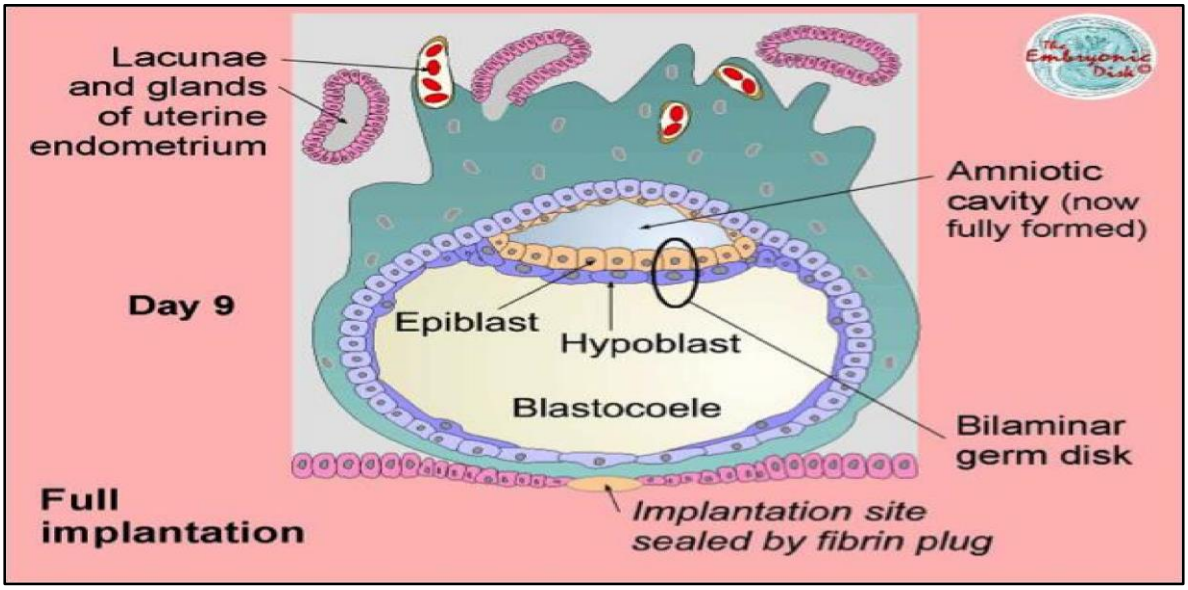
- ☒ **Type I**: Placenta previa centralis (complete): the placenta completely overlies the internal os of the cervix.
- ☒ **Type II**: Partial placenta previa: the placenta partially overlies internal os.
- ☒ **Type III**: Placenta previa marginalis: the placenta lies just at the margin of the internal os without overlying it.
- ☒ **Type IV**: Low-lying placenta previa

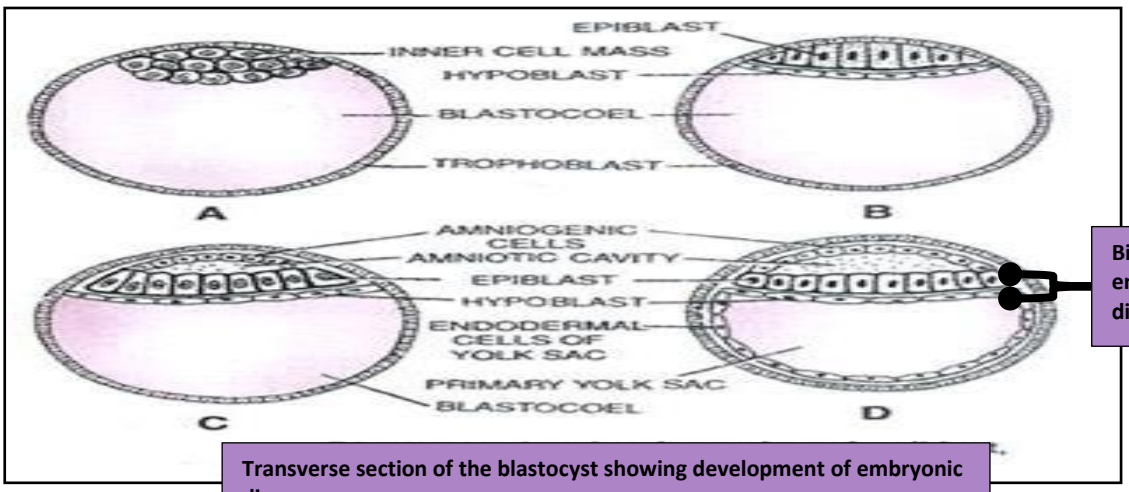


2 The 2 ND Week of development (week of twos)

- It's the week when the **embryoblast** (inner cell mass), **extraembryonic mesoderm** and **trophoblast** (outer cell mass) each separate into **two distinct layers**.
- Additionally, there are **two cavities (amniotic +yolk sac)** that develop within the embryonic unit at this time as well.
- formation of **extra-embryonic mesoderm**
- the blastocyst is transformed into a **chorionic vesicle**

A-Embryonic Disc





Transverse section of the blastocyst showing development of embryonic disc

The embryonic (germ) disc is developed from the inner cell mass of the blastocyst, where its cells become arranged in the form of a flat bilaminar plate intervening between the amnion (dorsally) and the yolk sac (ventrally).

It is at 1st bilaminar, consisting of **2 layers (dorsal and ventral):**

1 .Dorsal layer (epiblast): is formed of columnar ectodermal cells and lies in the floor of the amniotic cavity.

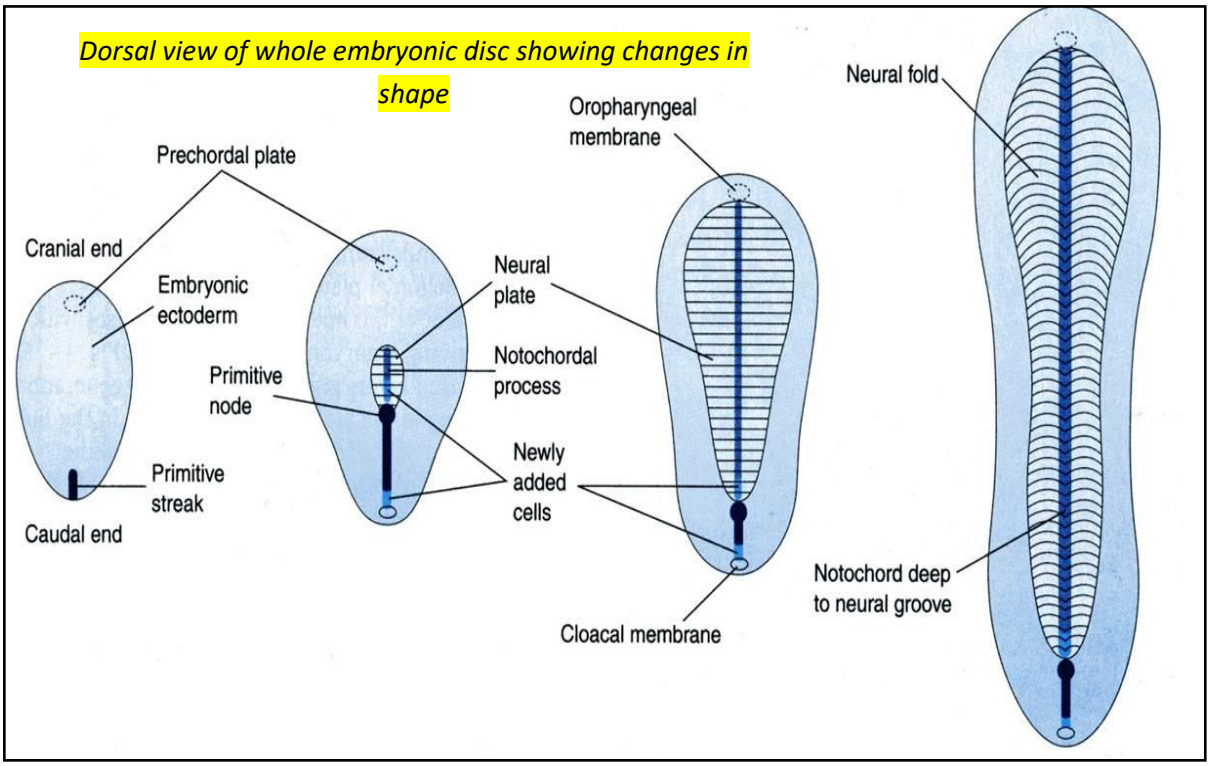
2. Ventral layer (hypoblast):is formed of cuboidal endodermal cells and lies in the roof of the yolk sac.

NOTE

The 2 layers of the disc are fused together only at 2 areas: one situated cranially called buccopharyngeal membrane and one situated causally called cloacal membrane

Shape of the disc:

The embryonic disc is **at first circular** in outline, but as it grows it **becomes oval** and then **pear-shaped** with a narrow caudal end, and a broader cranial end.



B-Trophoblast

Become differentiated into **2 layers**:

1-Outer synciteotrophoblast: formed by multinucleated mass which randomly dispersed **with no cell walls in-between.**

2-Inner cytotrophoblast: formed of cells **with apparent walls between nuclei**.

c-Two cavities

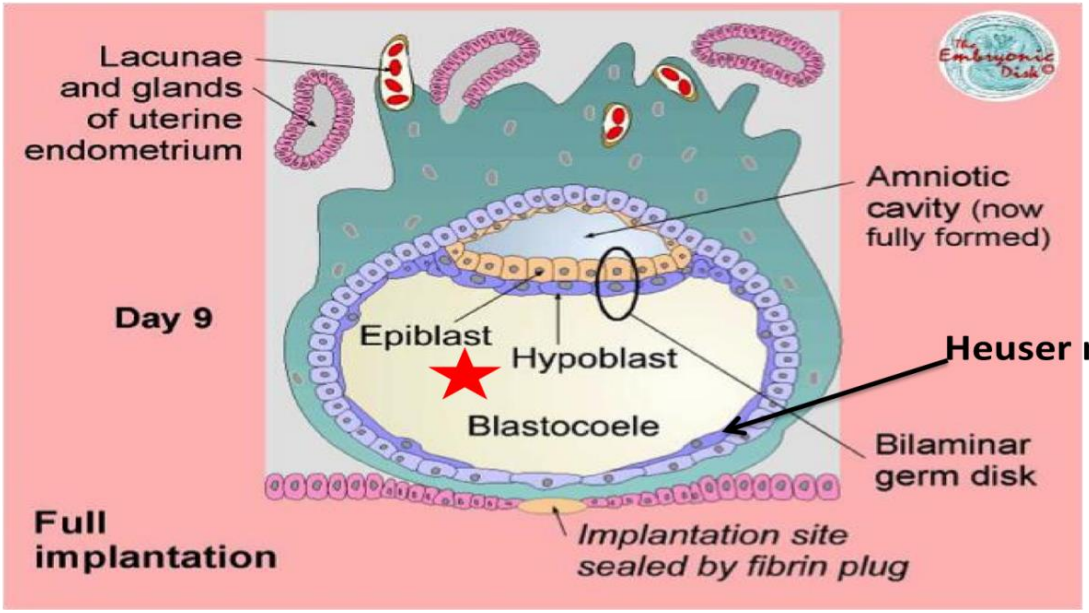
1-Amniotic cavity:

- A small space develops relative to the epiblast; it is the precursor of the amniotic cavity.
- **Epiblastic cells** forming the floor of the cavity subsequently separate to form amnioblasts that will form the **amnion** (surrounding the amniotic cavity).
- The sac and cavity will eventually become filled with amniotic fluid later on in the pregnancy.

2-yolk sac:

Primitive yolk sac: (★)

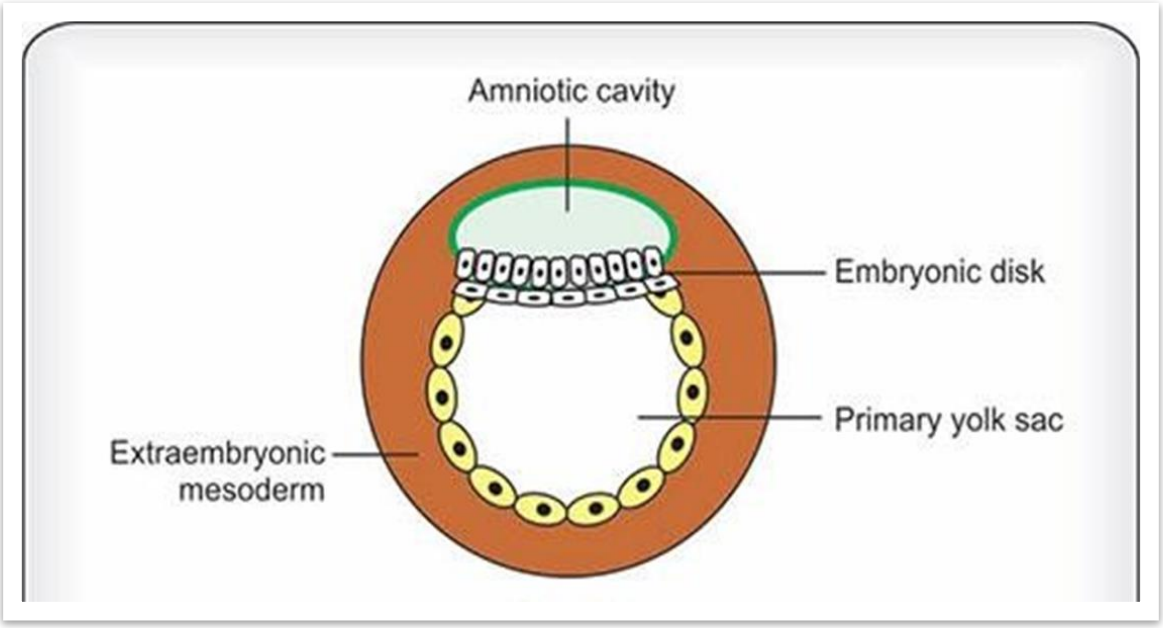
- Its roof is formed by the hypoblast layer of the embryonic disc.
- while the rest of its wall is lined with hypoblast cells in the form of a membrane called Heuser's membrane.



D-The extra-embryonic mesoderm

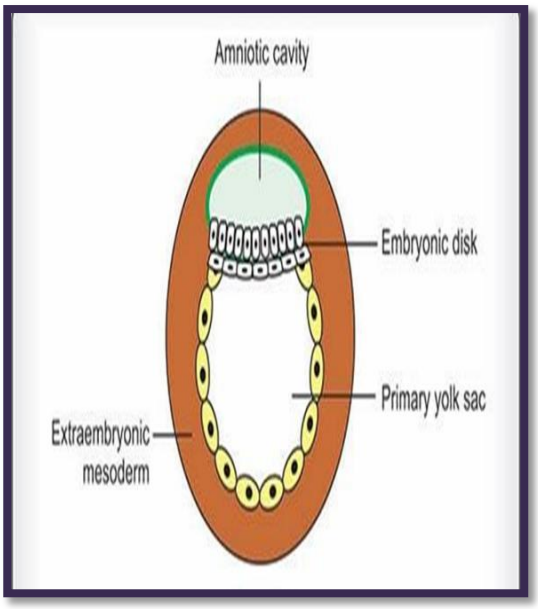
(called **primary mesoderm** as it appears before the intra-embryonic mesoderm)

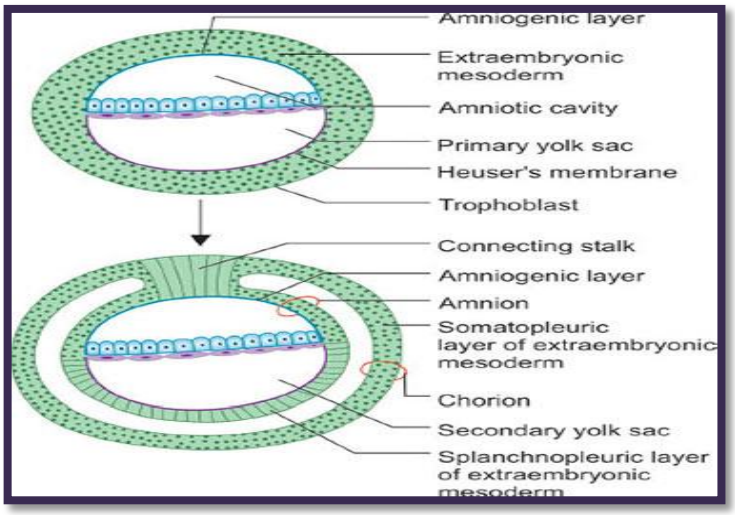
- Several layers of cells
- It surrounds the amniotic cavity as well as the yolk sac.
- Develops from the inner aspect of the cytotrophoblast (hypoblastic cells may share also).

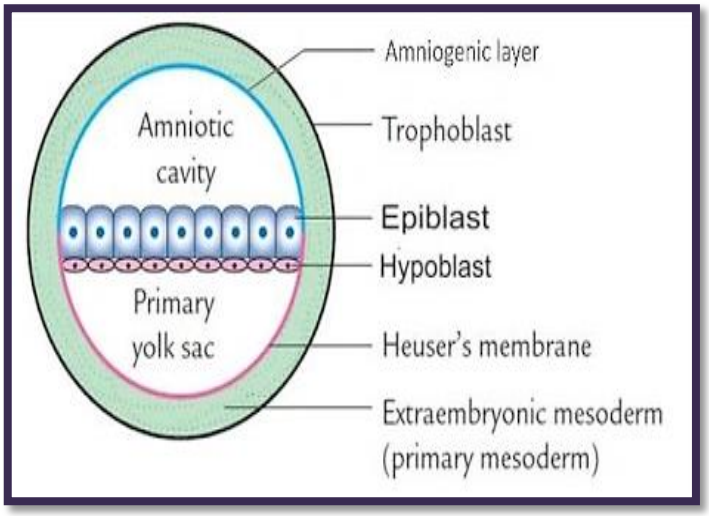


D-The extra-embryonic mesoderm:
(primary mesoderm)

- It is called **Primary mesoderm** as it appears before the intra-embryonic mesoderm.
- Consist of Several layers of cells.
- It surrounds the amniotic cavity as well as the yolk sac.
- Develops from the inner aspect of the cytotrophoblast (hypoblastic cells may share also)
- **This mesoderm** develops multiple small cavities which coalesce together to form a single cavity called extra-embryonic coelom.



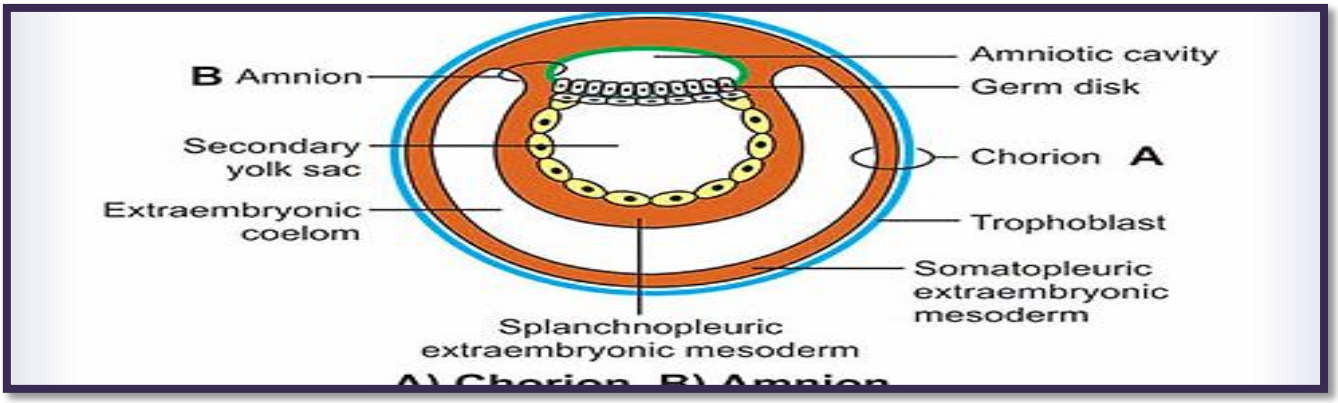




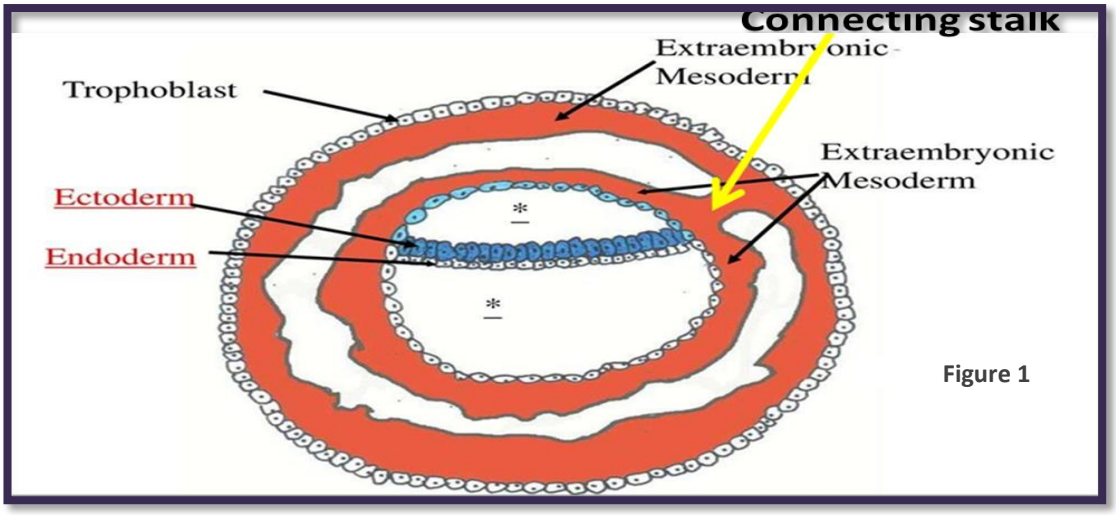
extra-embryonic coelom splits the mesoderm into two layers:

1. **Inner layer (visceral) or splanchnopleuric extra-embryonic mesoderm:**
   - surrounds the amniotic cavity as well as the yolk sac.

2- **Outer layer (Somatic) or somatopleuric extra-embryonic mesoderm:**
   - in contact with the trophoblast.



- The splitting of the extraembryonic mesoderm by **the extra-embryonic coelom** is not complete throughout the whole circumference of the vesicle, but a mass of mesoderm persists non splitted at the caudal end of the bilaminar disc called **the connecting stalk (future umbilical cord)**. **(Figure1)**
- This stalk connects the embryonic disc to the wall of the chorionic vesicle and forms



the basis of the future umbilical cord.

- **The trophoblast** forming the wall of the blastocyst becomes transformed into the chorion by the appearance of the extra-embryonic mesoderm on the inner aspect of the trophoblast. Consequently, the blastocyst is transformed into a chorionic vesicle (At the end of 2nd week).

- **The chorion** forms the wall of the chorionic vesicle. It consists of 3 layers as follows, from outside to inside (Figure 2):
  1. Syncytiotrophoblast.
  2. Cytotrophoblast.
  3. Extra-embryonic mesoderm.

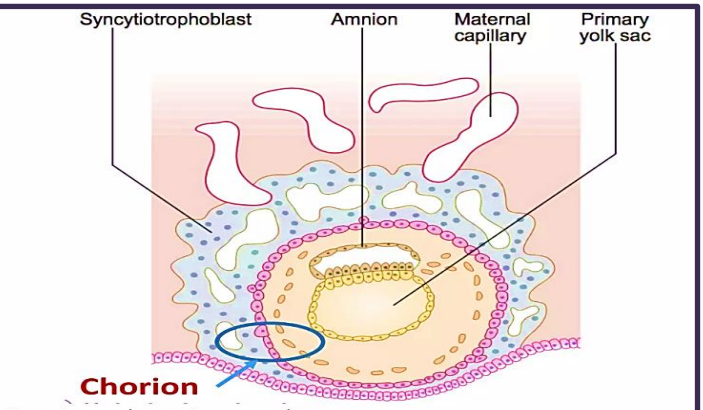
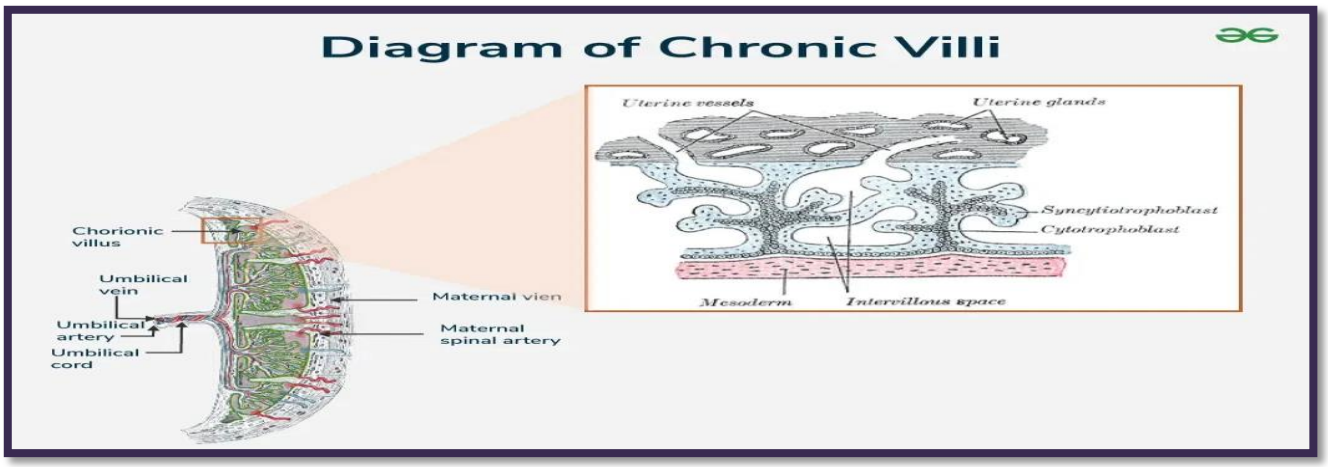


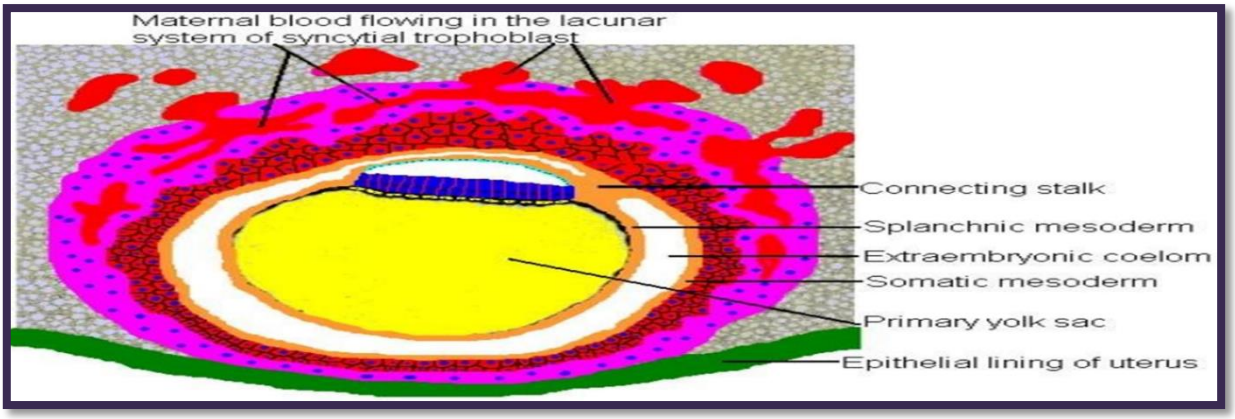
Figure 2

FEATURES OF THE CHORIONIC VESICLE

The chorionic vesicle is characterized by the following features:

1. The wall of the chorionic vesicle is called **chorion**. (The wall of the blastocyst is called trophoblast.)
2. The wall of the chorionic vesicle is covered by chorionic villi. (The wall of the blastocyst is not covered by villi.)
3. The chorionic vesicle shows the appearance of extra-embryonic mesoderm. (There is no extra embryonic mesoderm in the blastocyst.)
4. The inner cell mass is transformed into a bilaminar embryonic disc formed of 2 layers (ectodermal and endodermal).
5. The chorionic vesicle has 2 cavities called amniotic cavity (dorsal to the disc) and yolk sac (ventral to the disc), in addition to the extra-embryonic coelom.
6. The chorionic vesicle shows blood vessels at the following sites:
   - In the chorionic villi: they form part of the placenta.
   - In the connecting stalk: they form the umbilical vessels.
   - In the wall of yolk sac: they form the vitelline vessels.

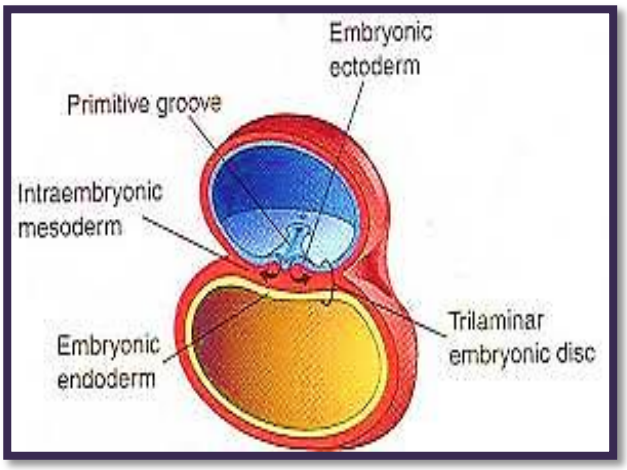

Diagram of Chronic Villi
Chorionic villus
Umbilical vein
Umbilical artery
Umbilical cord
Maternal vien
Maternal spinal artery
Uterine vessels
Uterine glands
Syncytiotrophoblast
Cytotrophoblast
Mesoderm
Intervillous space


Maternal blood flowing in the lacunar system of syncytial trophoblast
Connecting stalk
Splanchnic mesoderm
Extraembryonic coelom
Somatic mesoderm
Primary yolk sac
Epithelial lining of uterus

3 Third week of development

(trilaminar germ disc stage)

Gastrulation

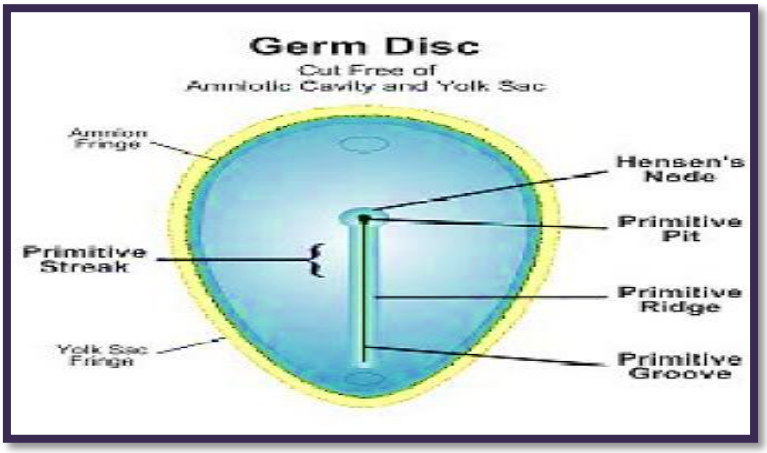


Definition: is the process by which the bilaminar embryonic disc becomes a trilaminar embryo composed of ectoderm, mesoderm and endoderm (**the three primary germ layers**).

- The process begins around day 15.
- Gastrulation begins with the formation of the primitive streak.

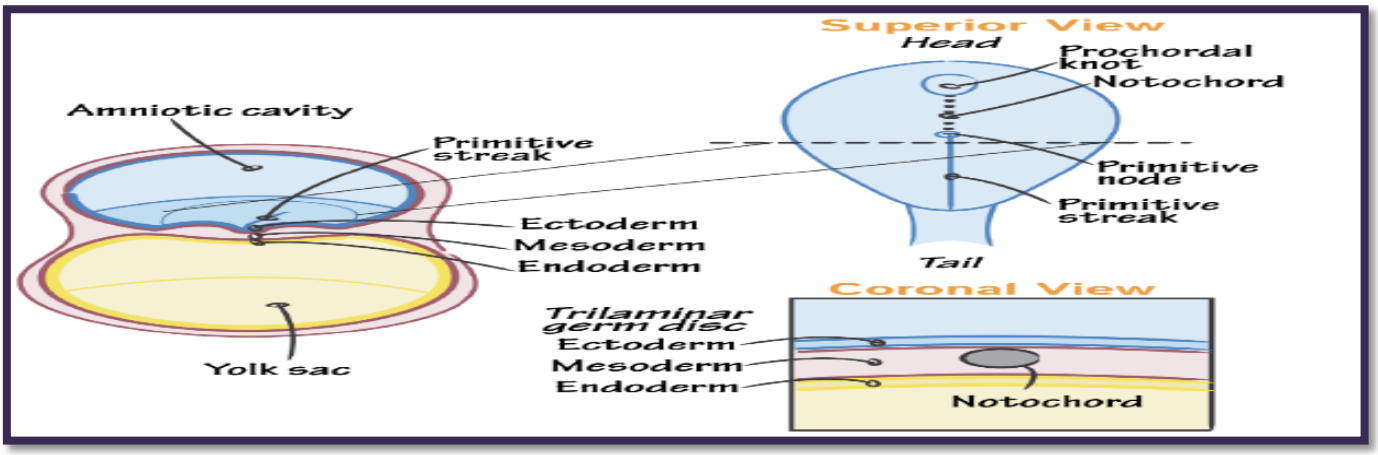
The primitive streak

- A transient structure whose formation, on day 15 of human development, marks the start of gastrulation.
- It appears as an elongating groove (primitive groove) on the dorsal surface of the epiblast (at midline), first appears at the caudal end of the embryo and grows cranially.
- At the cranial end its cells proliferate to form the primitive knot (primitive node).



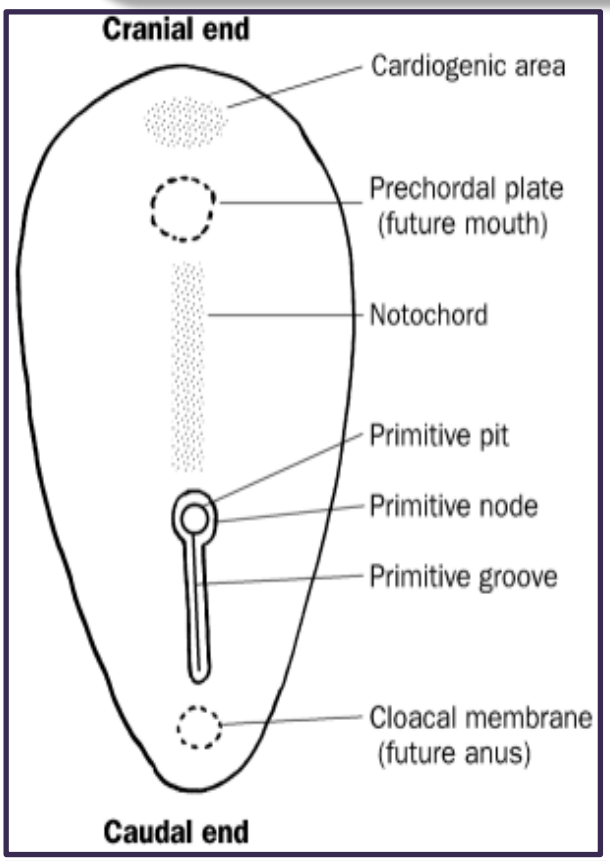
- The primitive node, or **Hensen's node**, contains a depression called the primitive pit. The primitive pit is continuous with the primitive groove.
- Cells (mesoblast cells) proliferate and migrate from the primitive pit ventrally and take place between the epiblast and hypoblast forming a third layer called **the mesoderm**

- **After gastrulation, the epiblast is renamed ectoderm and hypoblast is renamed endoderm.**

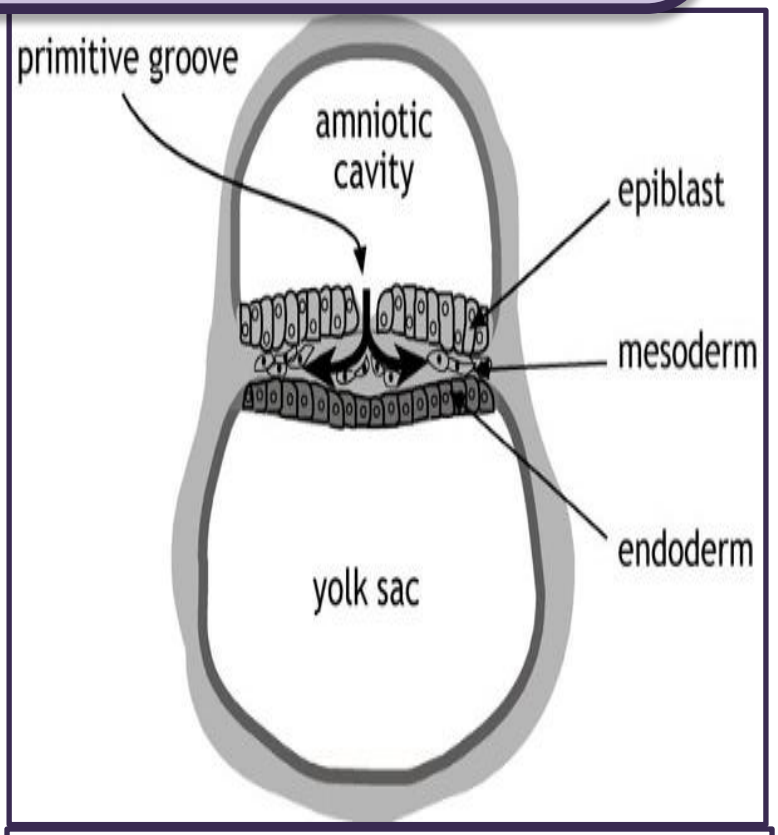


Note:

- **2 sites with no mesoderm The 2 layers of the disc (ectoderm + endoderm) are fused together only at 2 areas in the midline:**
  1. **one situated cranially called buccopharyngeal membrane and**

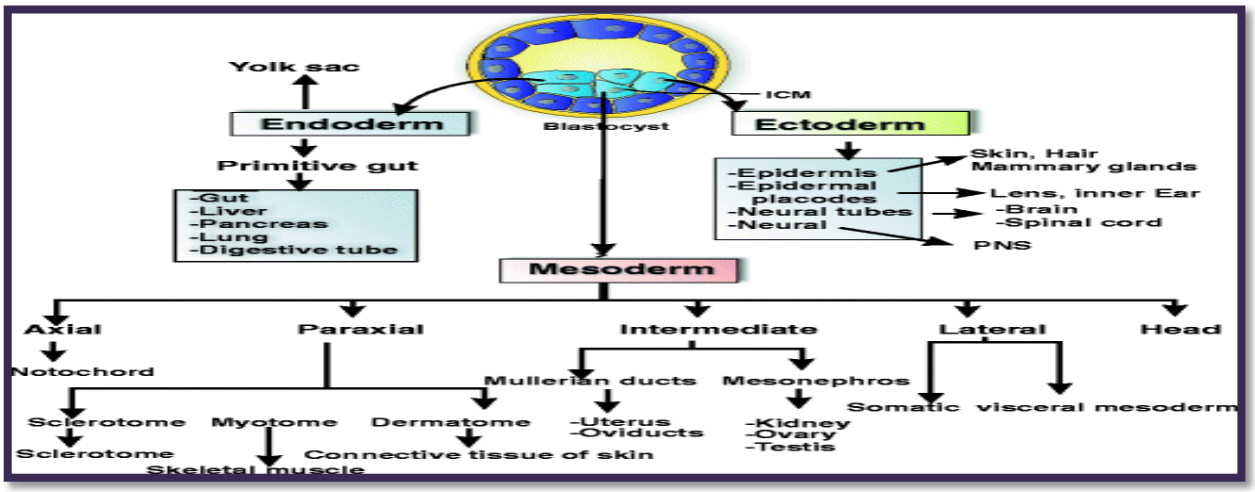


Longitudinal section of embryonic disc (dorsal view)



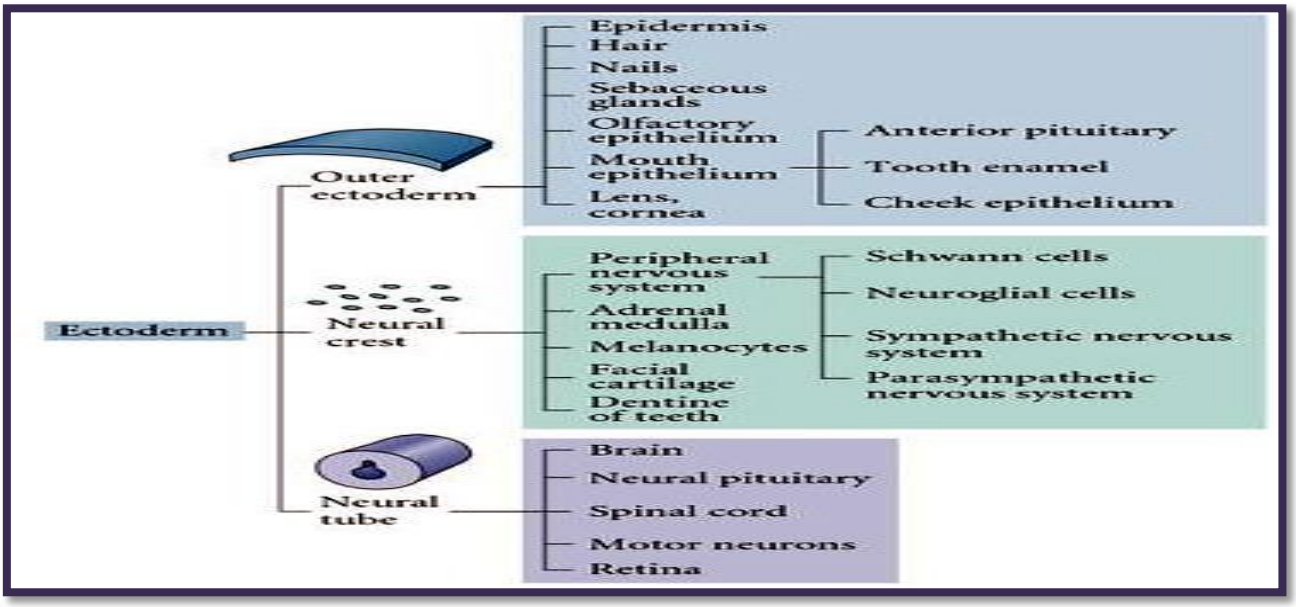
Transverse section through the embryonic disc showing the migrating cells from the primitive streak that forms the mesoderm (the third layer)

Germ layers and their derivatives



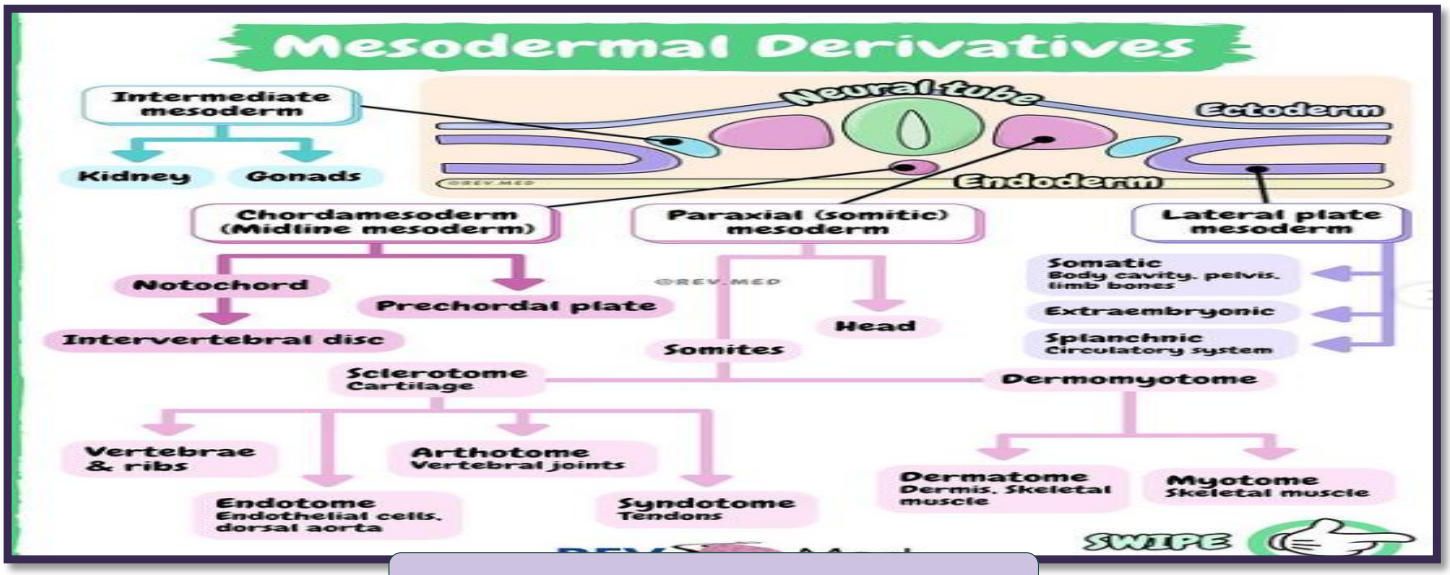
Derivatives of Ectoderm

1. Epidermis of the skin, hair and nails.
2. The brain and spinal cord.
3. Pia, and arachnoid mater (not the dura).
4. Pituitary gland and medulla of the suprarenal gland (not its cortex).
5. Lining of part of oral cavity and lower 1/2 of anal canal.
6. Epithelium of the nasal cavity.



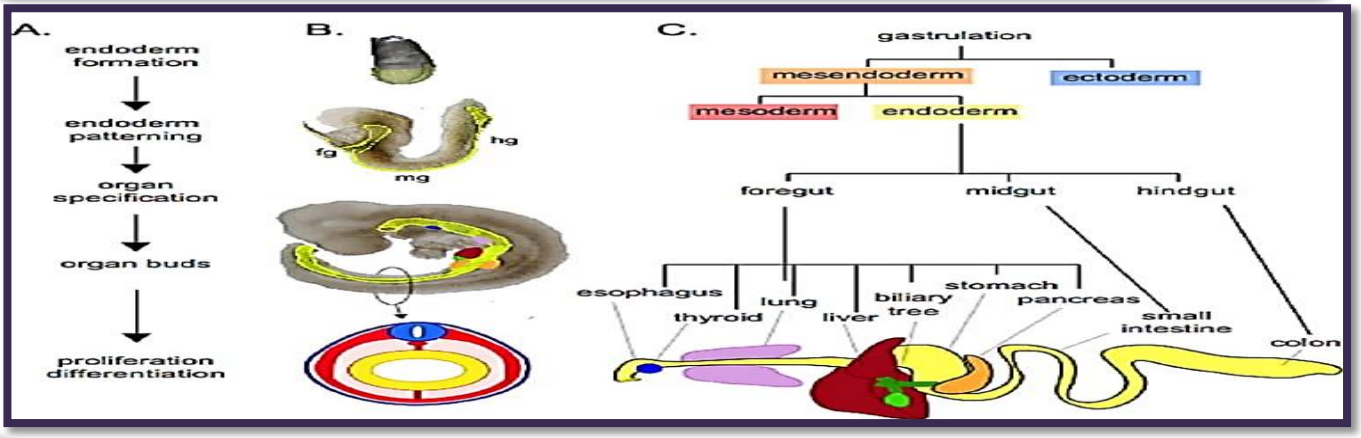
Derivatives of Mesoderm

1. A11 types of connective tissue, cartilage and bone.
2. All types of muscles.
3. Lymph vessels and lymphatic tissues.
4. The whole cardiovascular system: heart, blood vessels and blood cells.
5. Lining of the cavities of the body: pericardium, pleura and peritoneum.
6. Most of the urinary and genital organs and ducts (except parts of the bladder and urethra).
7. Cortex of the suprarenal gland.



Derivatives of Endoderm

1. Most of the lining of the gut (except inlet and outlet).
2. Pancreas and lining of the biliary passages.
3. Lining of the respiratory tract: larynx, trachea and bronchi.
4. Epithelium of the thyroid, parathyroid and thymus glands.
5. Most lining of the urinary bladder and most of the urethra.
6. Lining of the tympanic cavity (middle ear) and the auditory tube.



Notochord

Definition: A cellular rod and forms the first longitudinal midline axis of the embryo around which the vertebral bodies are organized.

The notochord develops from the ectodermal cells of the primitive node according to the following steps:

1. Formation of notochordal process:

- The cells of the primitive node proliferate to form a solid cord called the notochordal process extends in the midline between the ectoderms and endodermal layers.

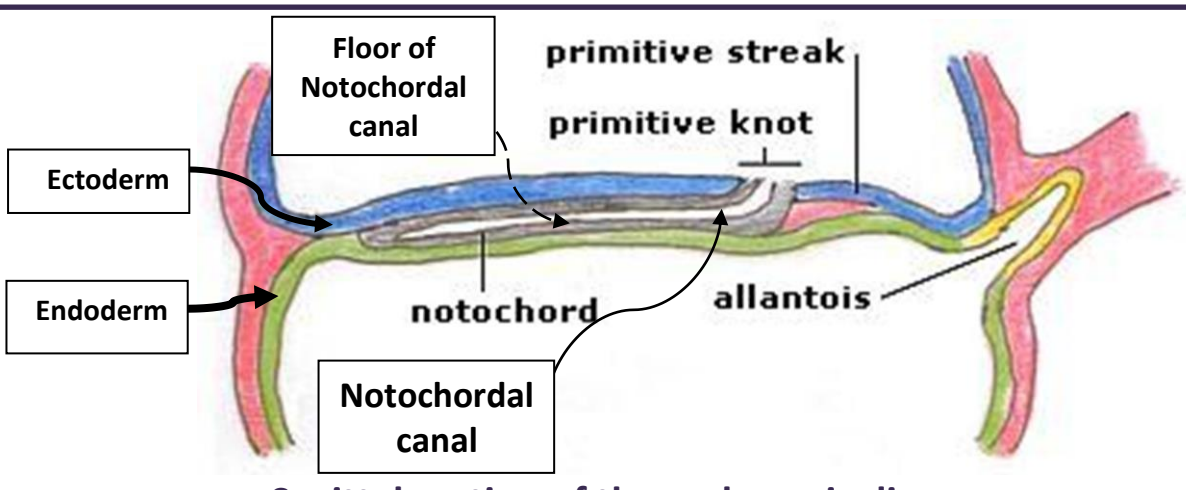
2. Formation of notochordal canal:

- A depression appears on the surface of the primitive node called the primitive pit. It deepens progressively inside the notochordal process, thus transforming it into the notochordal canal.
- The roof of this canal lies in contact with the ectodermal layer, while its floor lies in contact with the endodermal layer.

N.B.

The notochordal process stops at 2 sites (the area where the ectoderm is attached directly to the endoderm without intervening mesoderm):

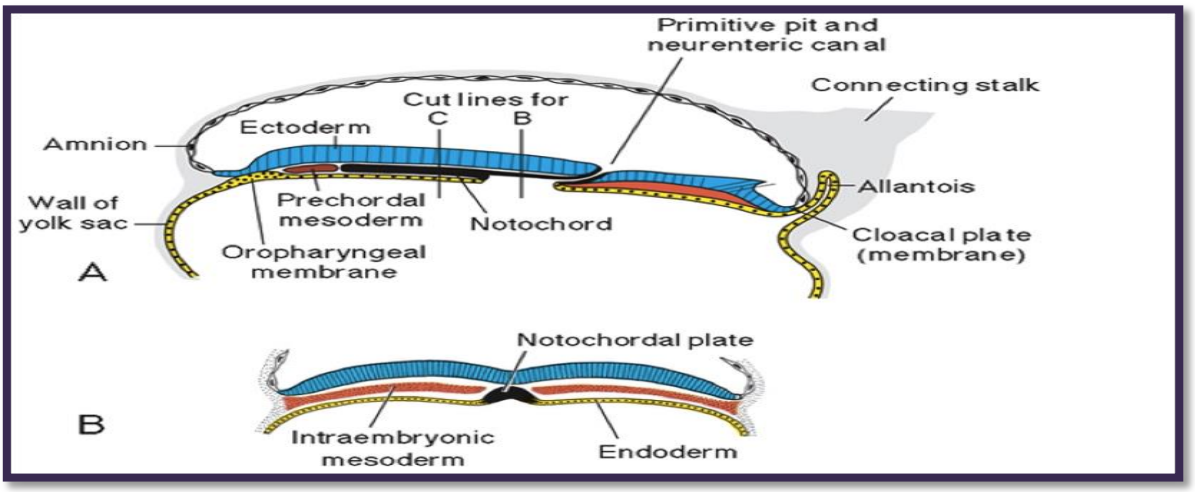
- The buccopharyngeal membrane which will break down to become the mouth.
- The cloacal membrane (proctodeum), or primordial anus.

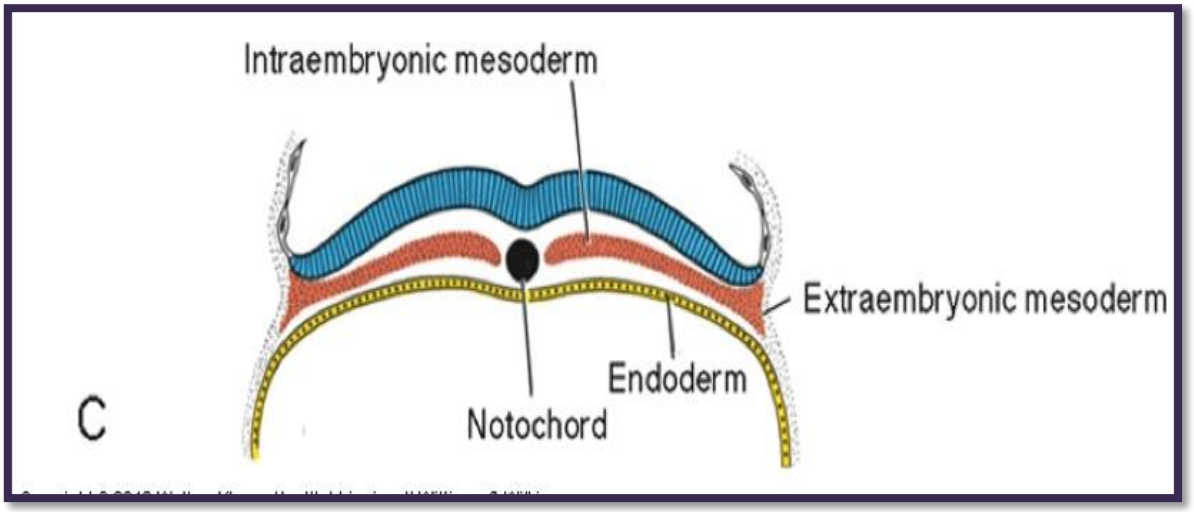


Sagittal section of the embryonic disc

- The floor of the notochordal canal, then, degenerates together with the underlying endoderm (which is the roof of the yolk) and as a result a communication called **neurenteric canal** is established between the yolk sac and the amniotic cavity.

- This communication is very transient as it rapidly closes by regeneration of the surrounding endoderm.





3. Formation of the definitive notochord:

- The roof of the notochordal canal is folded around its long axis to form a solid cord which lies exactly in the axis of the disc; this is the definitive notochord.

- The vertebral column is formed around the notochord which passes longitudinally through the centers of the vertebral bodies and inters-vertebral discs.

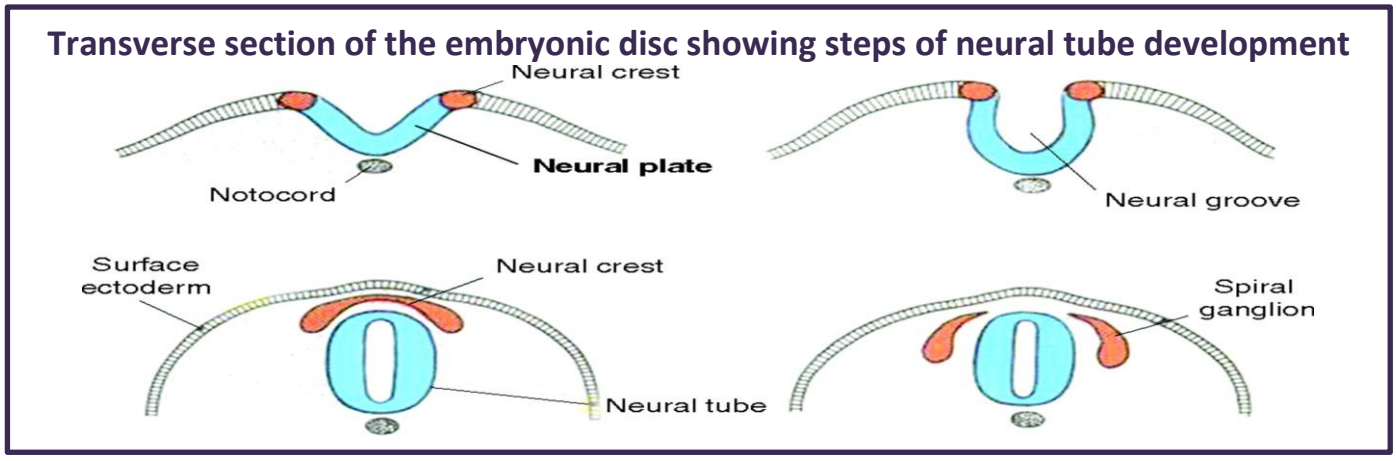
Neural tube

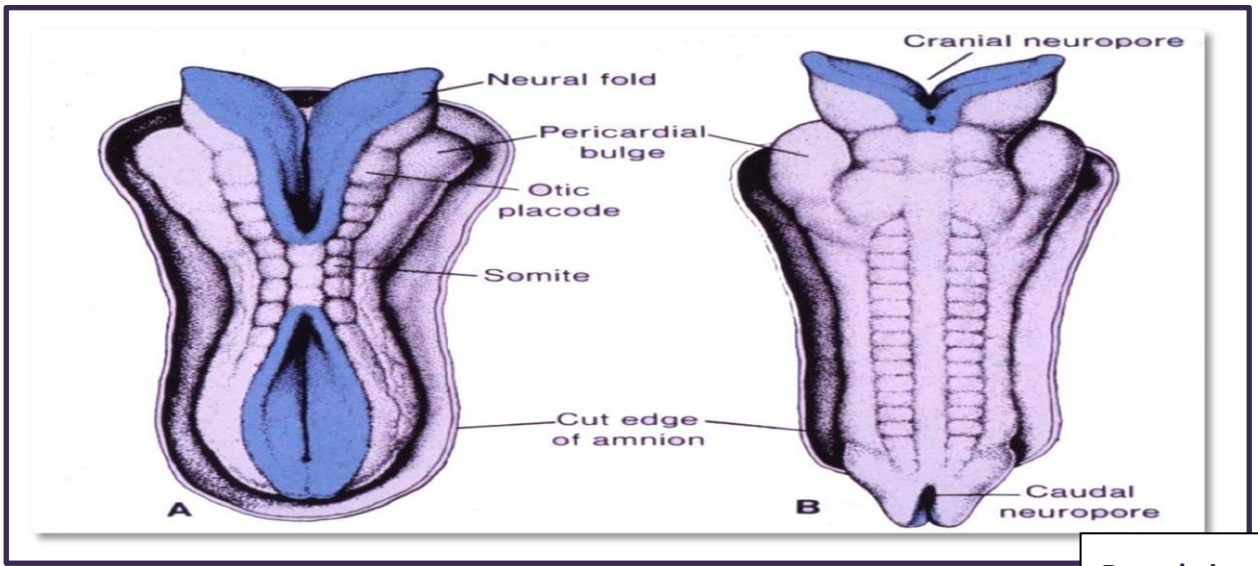
Definition: It is a dorsal tubular structure in the embryo that differentiates into the brain and spinal cord.

The neural tube develops from the neural plate, in the following steps:

1) Neural plate

- consists of thickening of the ectoderm in the midline of the disc between the primitive node and the buccopharyngeal membrane.
- Along the lateral margin of the neural plate (on each side), a longitudinal strip of ectodermal cells called **neural crest** is differentiated.

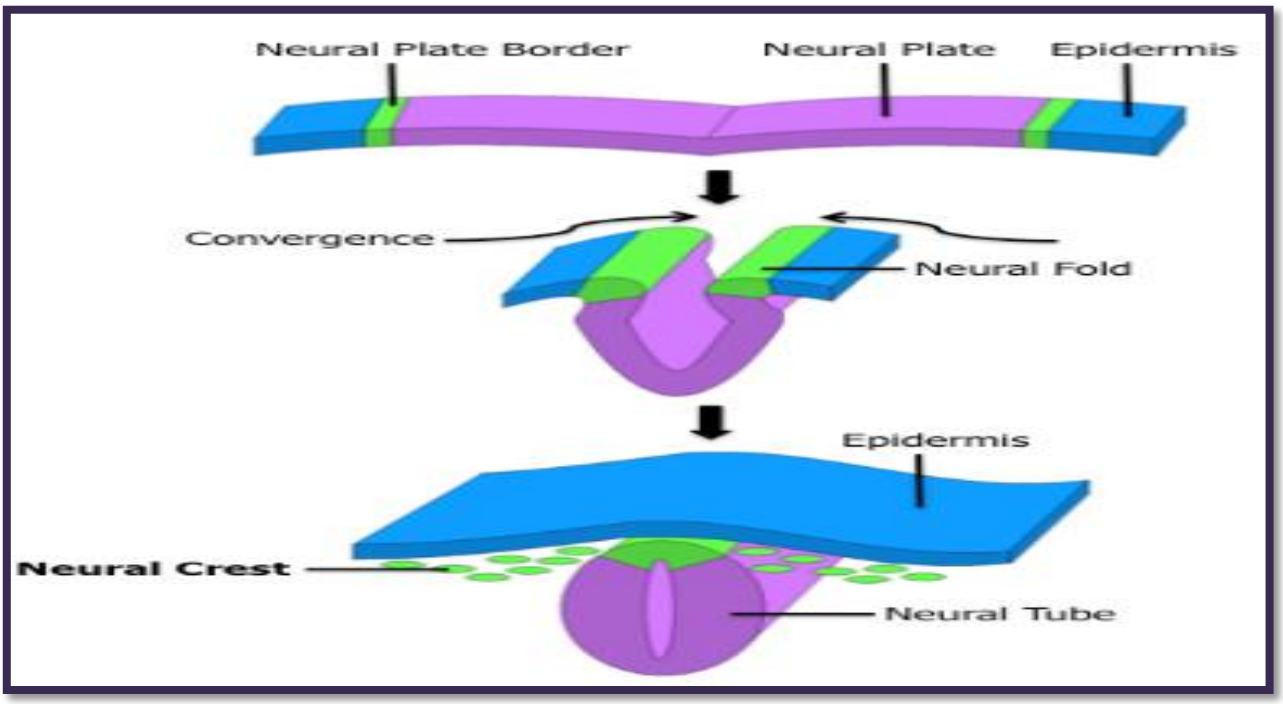




Dorsal view of neural tube

2) Neural groove

- In the midline of the neural plate, appears an antero-posterior groove.
- As the neural groove deepens, its edges are elevated to form the neural folds. These folds are approximated until they finally fuse together to form the neural tube.



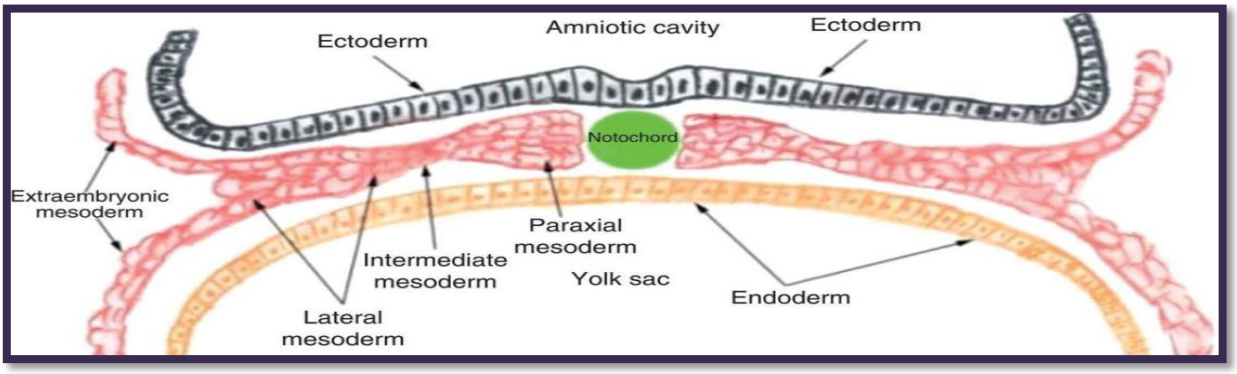
3) Neural tube

- The fully formed neural tube lies underneath the ectoderm, exactly in the midline and dorsal to the notochord.
- The anterior and posterior ends of the neural tube are at first open and are called **anterior and posterior neuropores**; they rapidly close.
- **The neural tube forms the brain from its cranial broad part and forms the spinal cord from its caudal narrower part.**

The Intra-embryonic mesoderm differentiation

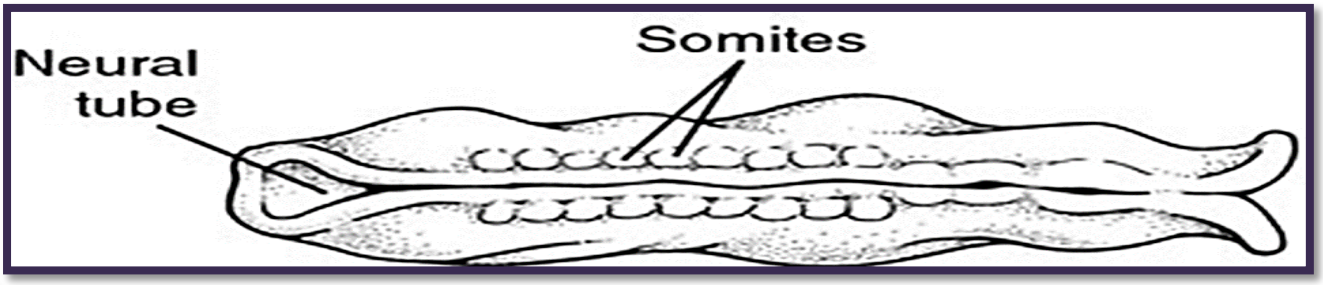
- The intra-embryonic mesoderm is formed inside the embryonic disc. It is also called secondary mesoderm because it appears later than the extraembryonic (primary) mesoderm.
- The intra-embryonic mesoderm is differentiated on each side of the notochord into 3 regions (from medial to lateral) as follows:

Paraxial mesoderm | **Intermediate mesoderm** | **Lateral plate mesoderm**



1) Paraxial mesoderm

- The most medial column which lies on each side of the notochord and parallel to it.
- It is segmented into 43-44: segments called somites.

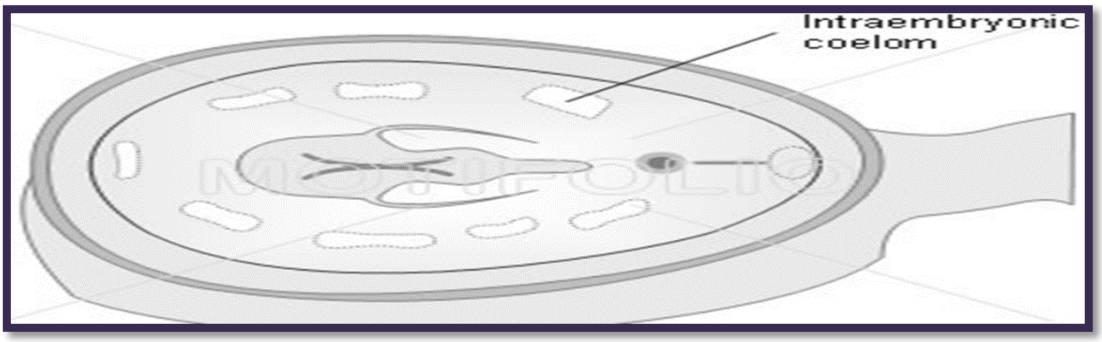


2) Intermediate mesoderm

- It is rapidly transformed into a cord-like mass called nephrogenic cord. It will give rise to the urogenital system.

3) Lateral plate mesoderm

- It is a flat plate which occupies the lateral part of the disc and extends as far laterally as the margin of the disc.
- Small cavities appear inside this lateral plate mesoderm and rapidly coalesce together to form one continuous cavity called intra-embryonic coelom.



- The coelom splits the lateral plate mesoderm into 2 layers: dorsal and ventral

1) Dorsal layer: lies dorsally in contact with the ectoderm of the disc, and forms together with the ectoderm the somatopleure (i.e. mesoderm + ectoderm).

2) Ventral layer: lies ventrally in contact with the endoderm of the disc, and forms together with the endoderm the splanchnopleure (i.e. mesoderm + endoderm).



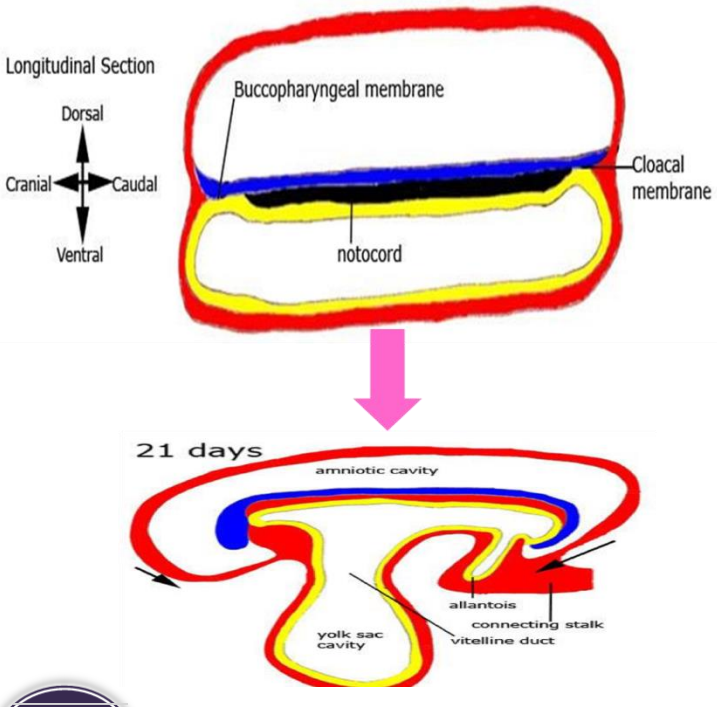
4 Fourth Week of Development

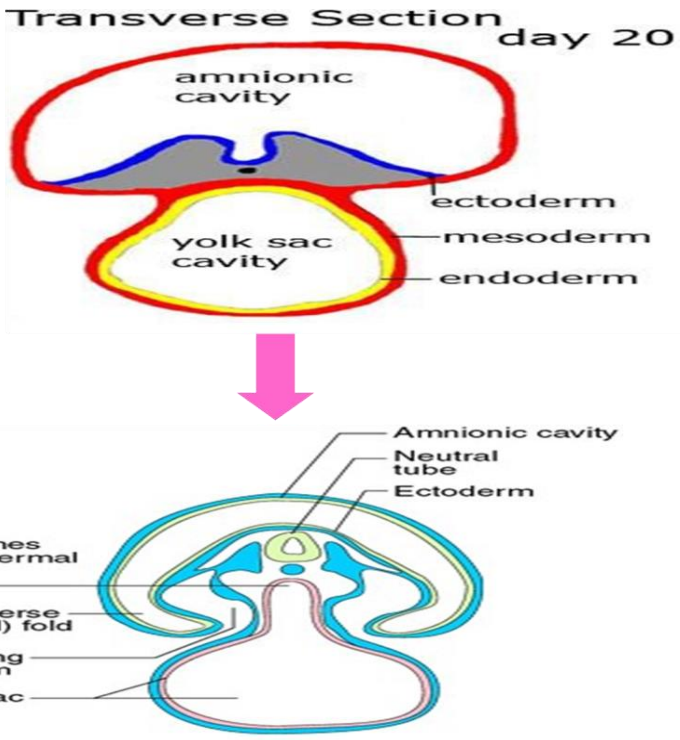
Folding of the disk

The flat trilaminar disc undergoes folding in both the antero-posterior axis and the transverse axis:

1-Folding in the antero-posterior(Craniocaudal) axis: leads to the formation of the head fold at the cranial end, and the tail fold at the caudal end.

2-Folding in the transverse axis: leads to the formation of 2 lateral , one on each side.





NOTE

Folding of the disc is due to the following mechanisms

*Progressive expansion of the amniotic cavity (cavity dorsal to the ectoderm).

*Rapid growth of the axial area (neural tube and the vertebral column) of the disc more than its periphery.

The notochord and the primitive streak are relatively firm structures, and accordingly they determine the limits of the head and tail folds as follows:

1. The head fold: lies just anterior to the notochord.

2. The tail fold: lies just behind the primitive streak.

Results of the folding

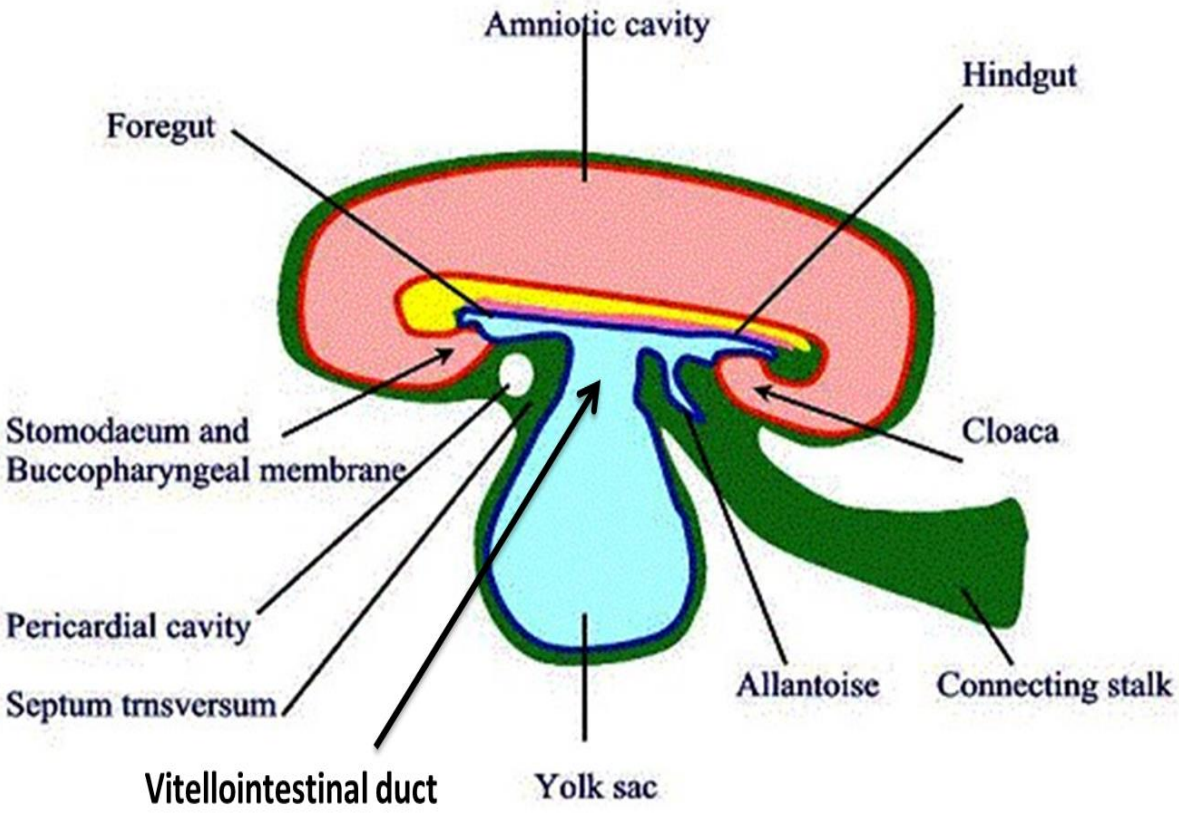
A)Formation of the gut:

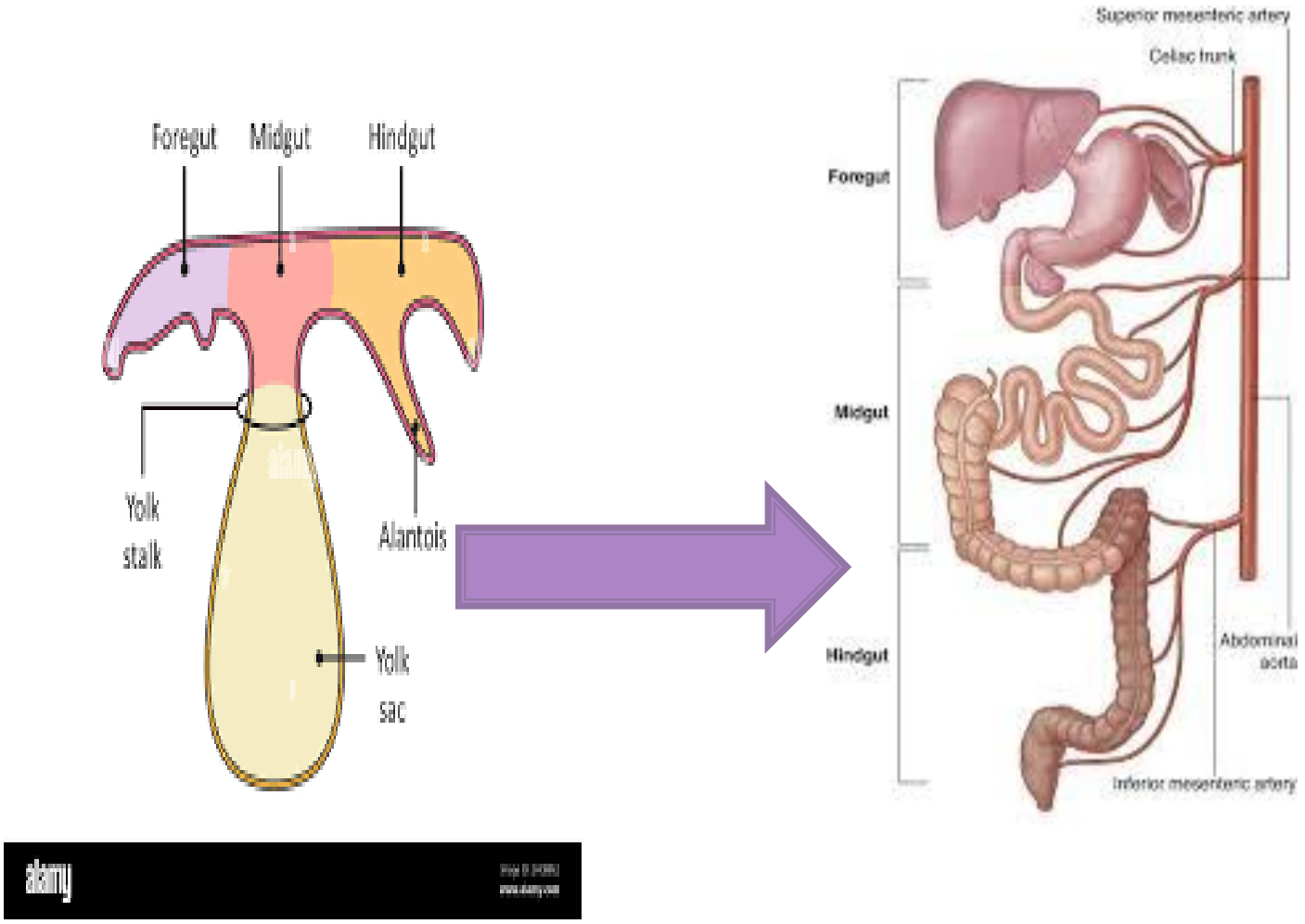
Folding of the disc is directed ventrally, and as a result a large part of the yolk sac (cavity ventral to endoderm) is enclosed within the folds to form the primitive gut.

Still, the rest of the yolk sac communicates with the gut through the vitello-intestinal duct.

The primitive gut is divided into 3 parts as follows:

1. **Foregut**: is the part within the head fold.
2. **Hindgut**: is the part within the tail fold.
3. **Midgut**: is the middle part, and communicates with the definitive yolk sac through the vitello-intestinal duct.





B)Reversal of positions:

As a result of folding the areas of the disc at the head and tail folds reverse their positions, i.e. what was cranial becomes caudal and what was caudal becomes cranial, as follows:

1. **Baccopharyngeal membrane:** comes to lie most cranial (cranial to the pericardial cavity, instead of being caudal to it before folding).

2. **Cloacal membrane:** comes to occupy the most caudal end of the disc(caudal to the connecting stalk, instead of being cranial to it before folding).

3. **Connecting stalk**: comes to lie ventral to the disc, instead of being the most caudal before folding.

C) Formation of the umbilical orifice:

As the head, tail and lateral folds approach each other ventrally, they surround the umbilical orifice. This orifice (the future umbilicus) is at 1st wide and contains the vitello-intestinal duct in addition to the allantois and connecting stalk.

Foetal Membranes

Definition

the structures which develop away from the embryonic disc, i.e. away from the tissues of the embryo proper.

At birth, the foetal membranes are expelled as one mass after the birth of the fetus.

The foetal membranes consist of the following components:

1) Chorion

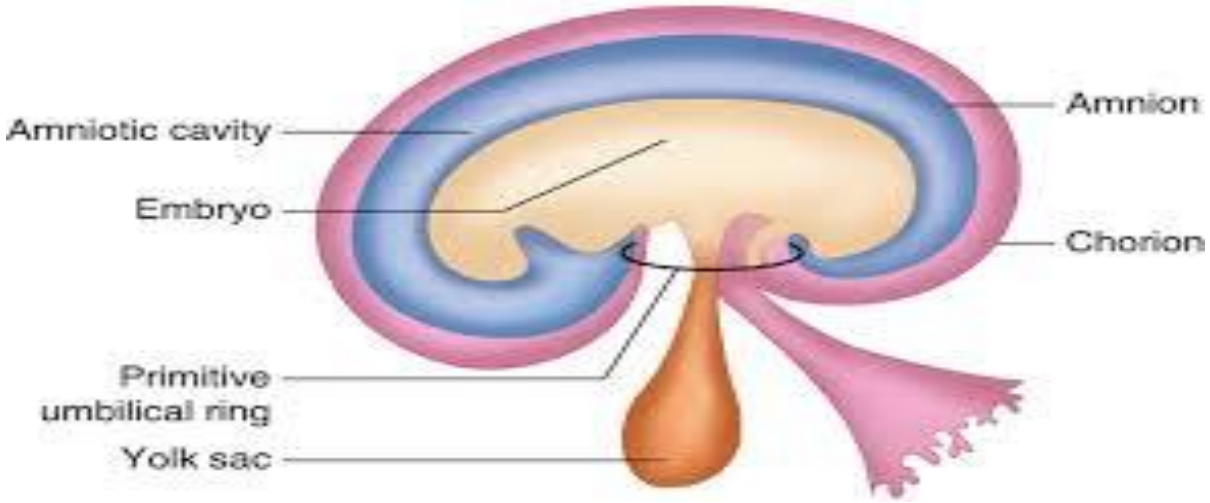
2) Amnion

3) Yolk sac

4) Allantois

5) Connecting stalk

6) Placenta



Decidua

Definition

The decidua is the endometrium after fertilization.

-The endometrium shows a decidual reaction which is an **exaggerated secretory change under the effect of the progesterone hormone**.

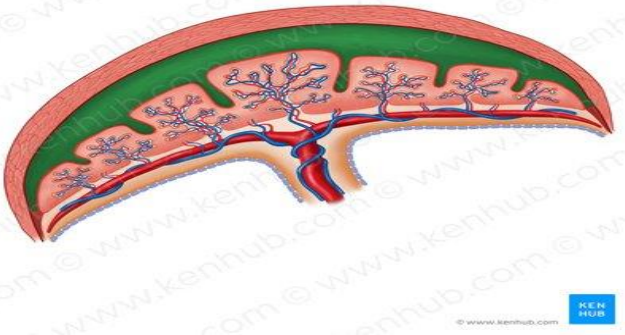
- This decidual change affects the main constituents of the endometrium as follows:

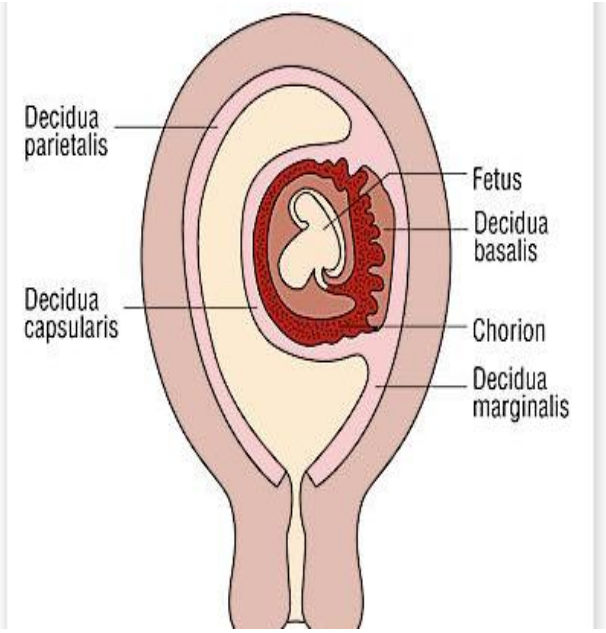
1. **Uterine glands**: become greatly distended with secretions rich in glycogen and mucin. The endometrium, thus, becomes very thick and oedematous

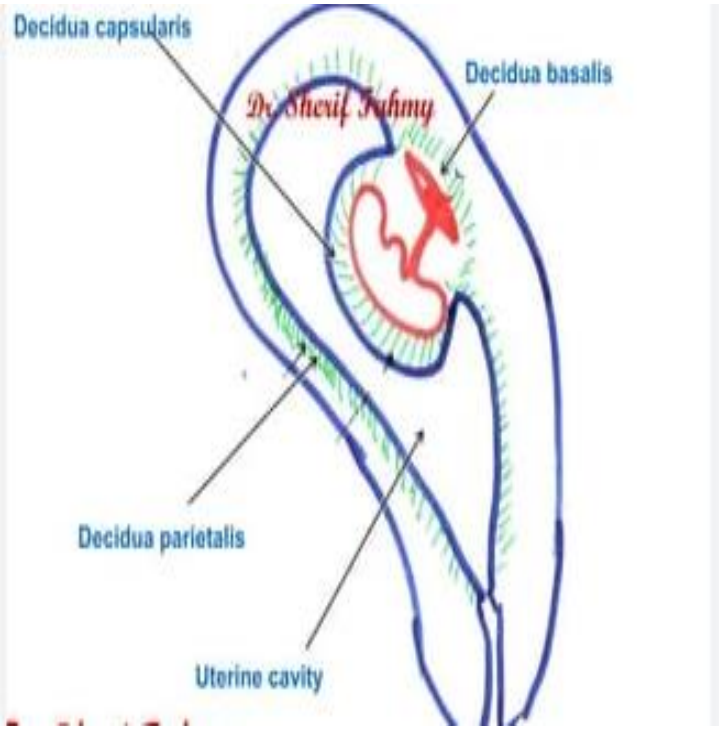
2.Blood vessels: become **much more spiral and show many arterio-venous shunts.**

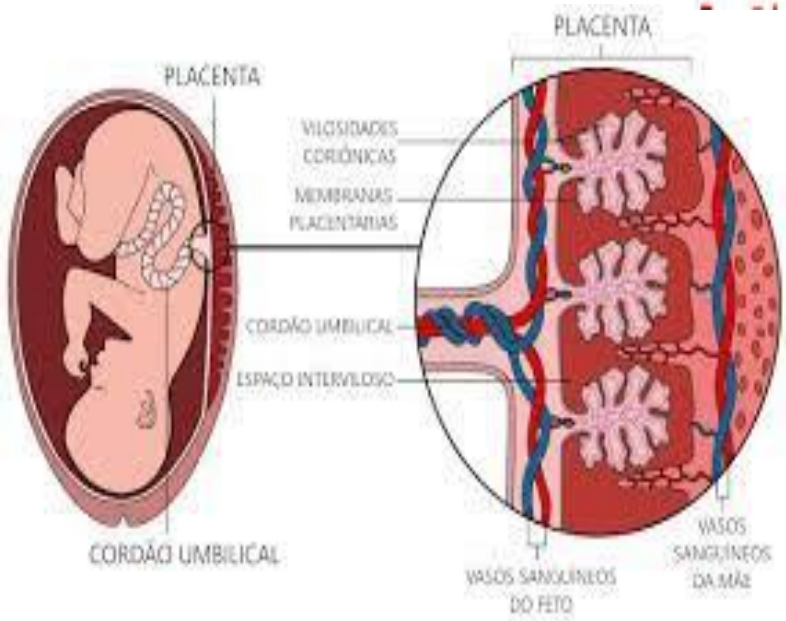
The endometrium, thus, becomes ready for implantation and embedding of the blastocyst.

3. Stroma cells: become much enlarged and show many vacuoles filled with glycogen and lipids; they are now called decidua cells.









Parts of decidua

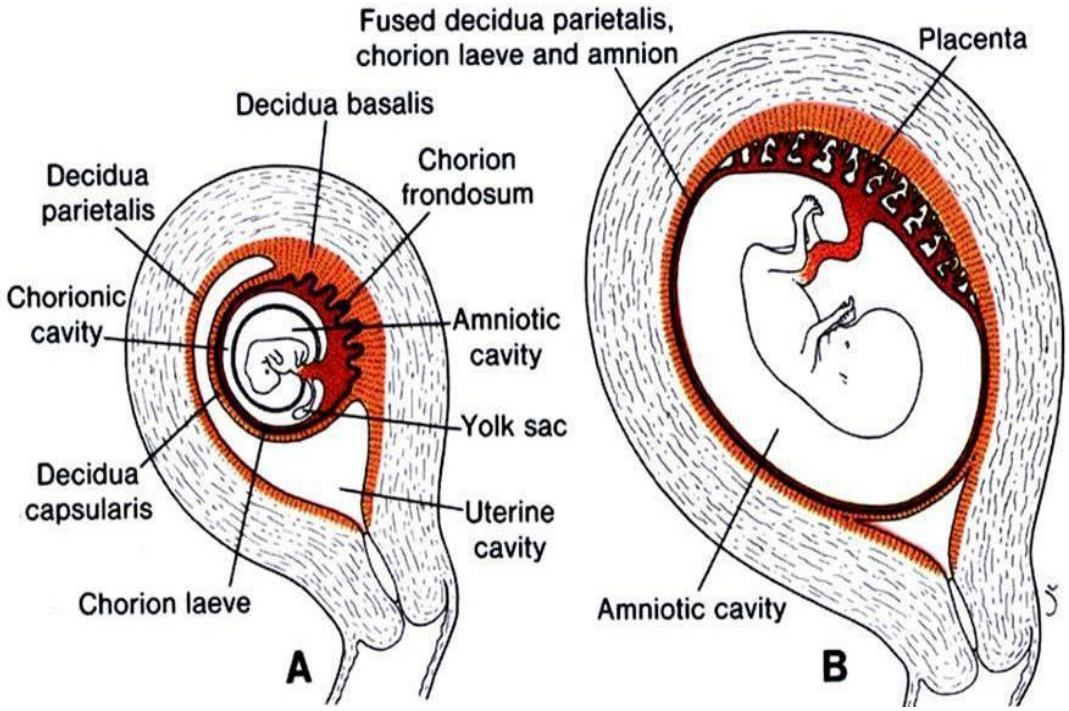
The decidua is divided in relation to the embedded blastocyst into parts:

1. **Decidua capsularis**: covers the superficial aspect of the blastocyst (like a capsule) separating the blastocyst from the uterine cavity.
2. **Decidua basalis**: lies deep to the blastocyst (at its base). This is the part which will share in formation of the placenta.
3. **Decidua parietalis**: is the part which lines the walls of the uterus away from the site of the blastocyst.

Growth of the decidua:

- At 1st, the decidua capsularis is separated from the decidua parietalis by the uterine cavity.

- With growth of the blastocyst the decidua capsularis bulges gradually into the uterine cavity until comes in direct contact with the decidua parietalis, thus obliterateing uterine cavity.
- At the same time, the decidua basalis thickens and shares in the formation of the placenta.



The splitting of the mesoderm by the extra-embryonic coelom is not complete throughout the whole circumference of the vesicle, but a mass of mesoderm

persist unsplitted at the caudal end of the bilaminar disc called the **connecting stalk**.

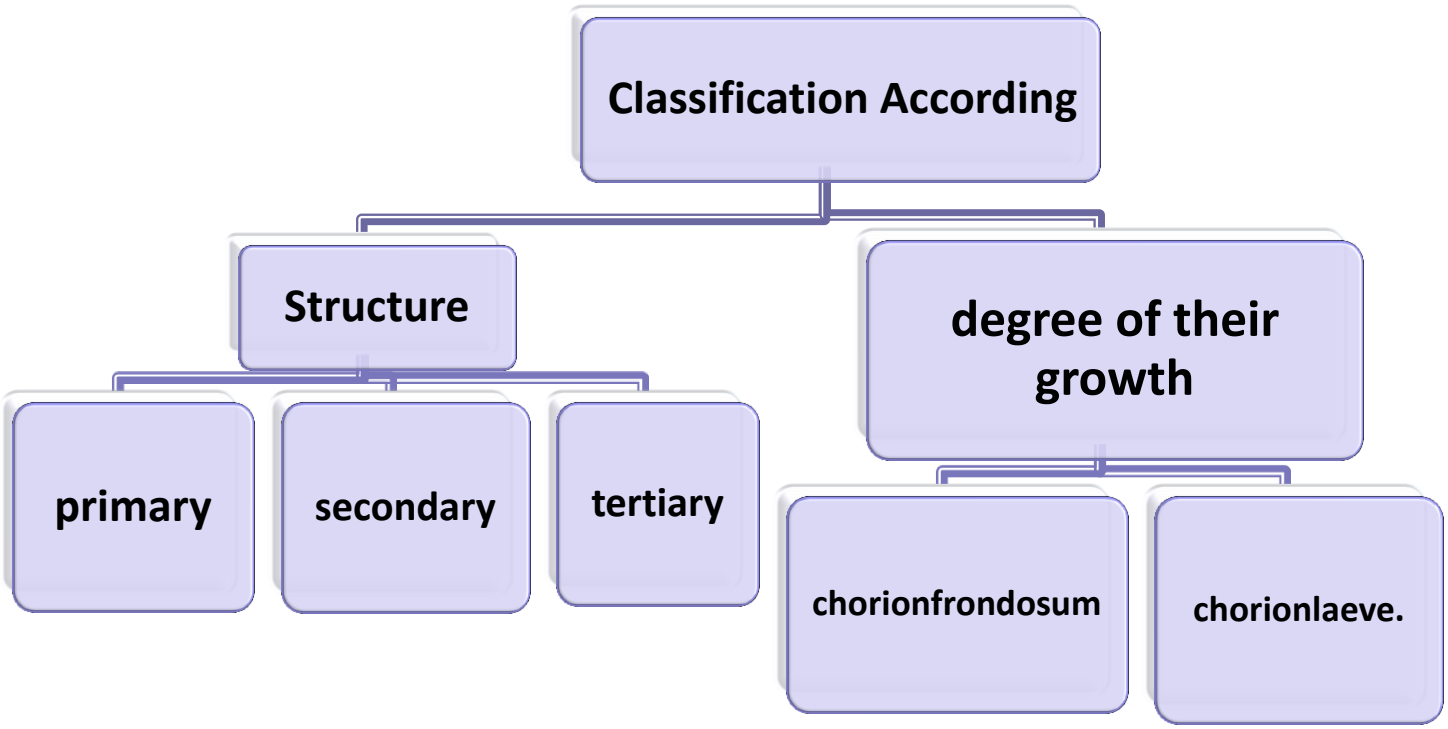
This stalk connects the embryonic disc to the wall of the chorionic vesicle and forms the basis of the future umbilical cord.

The trophoblast forming the wall of the blastocyst becomes transformed into the chorion by the appearance of the extra-embryonicmesoderm on the inner aspect of the trophoblast. Consequently, the blastocyst is transformed into a chorionic vesicle.

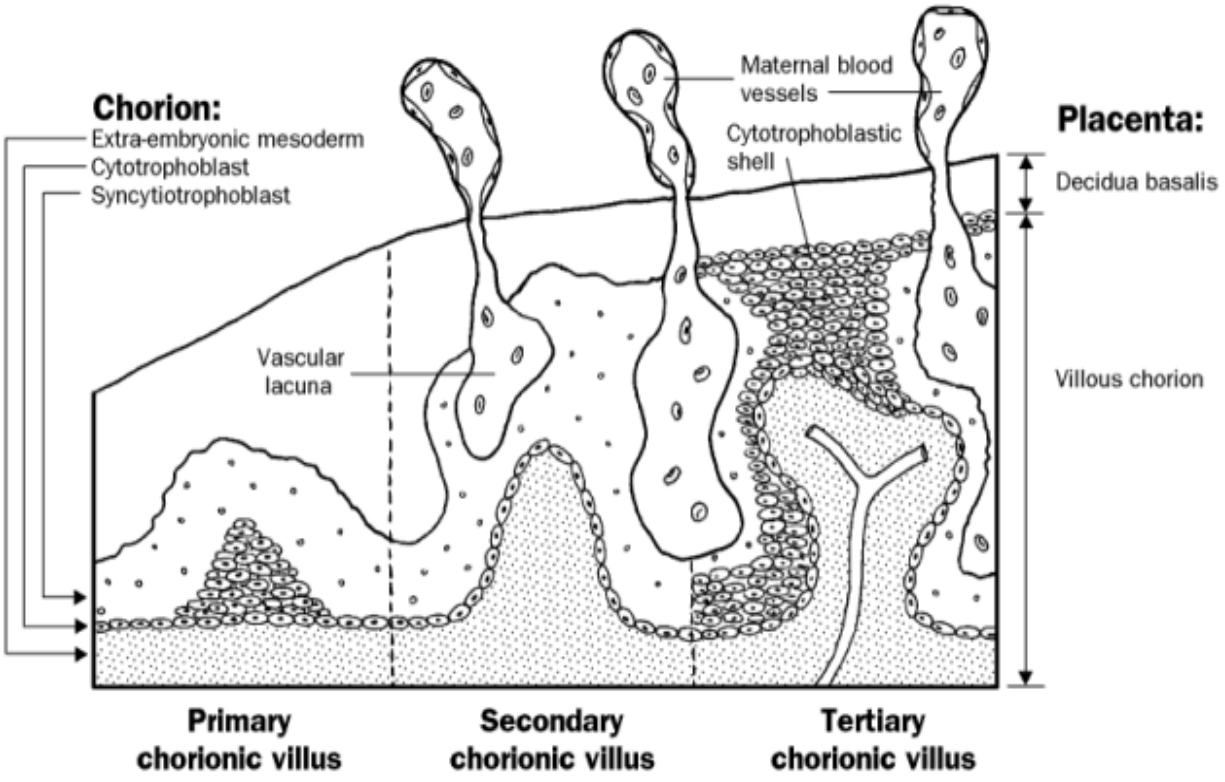
2 Chorionic villi

Definition

The chorionic villi are processes which project from the outer surface of the chorion, and are classified as follows;



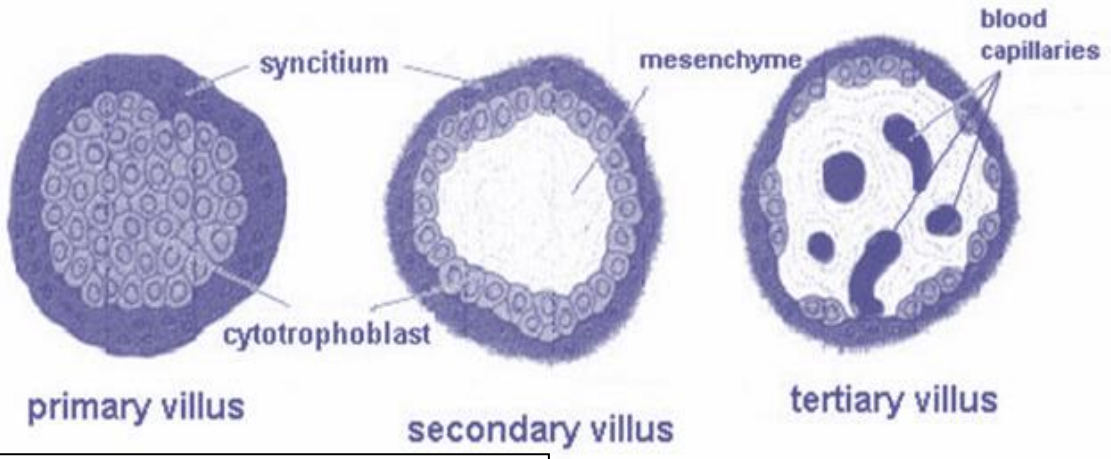
Types of villi (According to their structure:



1) Primary villi: These villi are developed from finger-like processes of sycytiotrophoblast called trabeculae, the cores of which are invaded by cytotrophoblast.

2) Secondary villi:where the cores of the primary villi are invaded by extra-embryonic mesoderm.

3) Tertiary villi: These are the mature and fully functioning villi. They characterized by the presence of blood vessels in the mesoderm filling the cores of the secondary villi.



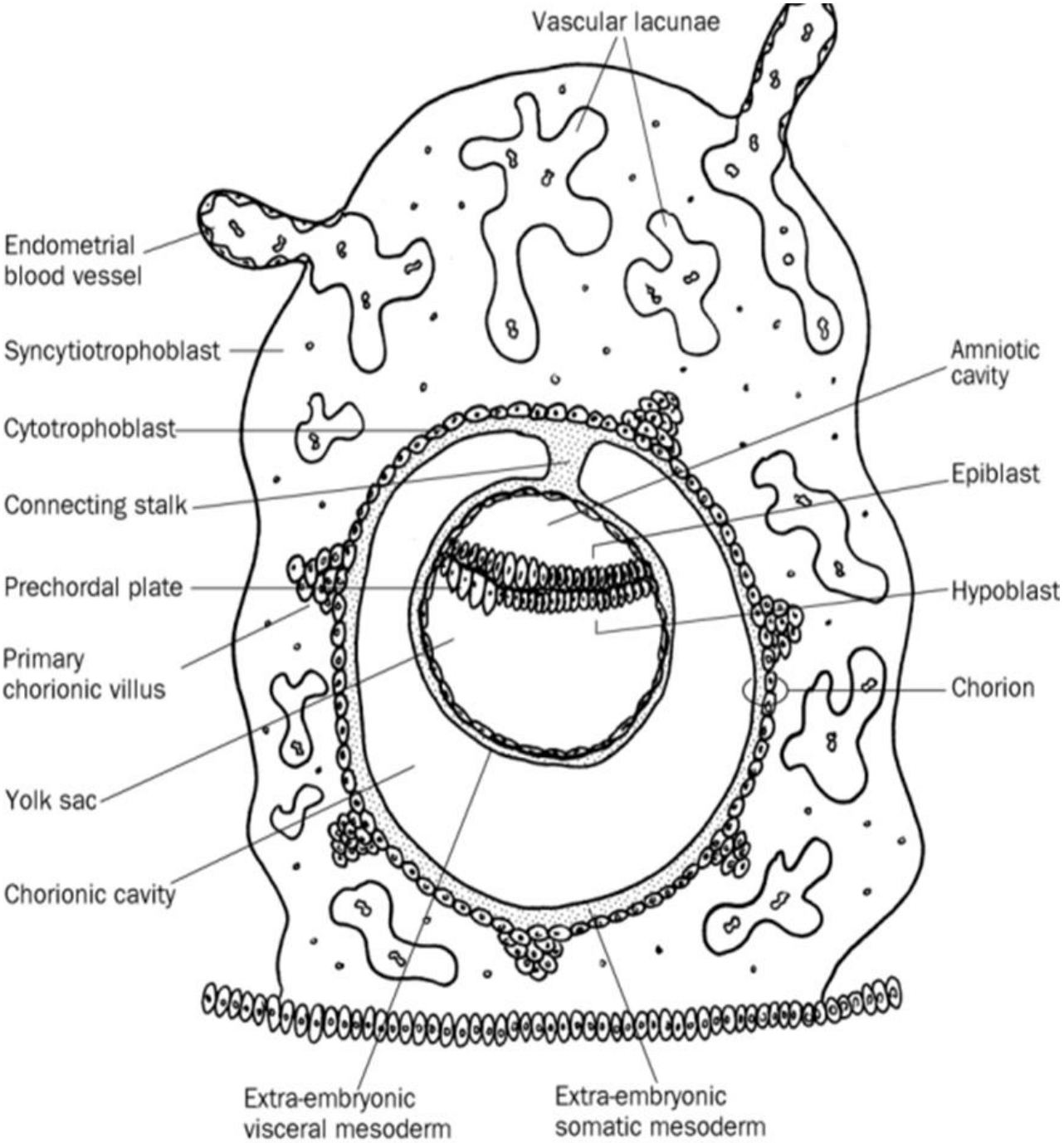
Transverse section showing structures of each type of villi

It should be noted that it is these tertiary villi which will enter in formation of the placenta. The presence of the foetalvessels inside the villi is a necessity for the exchange of aerials between the maternal blood and foetal blood.

Types of villi(According to the degree of their growth):

1. The villi at the site of the placenta: proliferate greatly and acquire numerous branches, and are called **chorion frondsum**(fronds = branches).

2. The villi away from the site of the placenta: shrink and degenerate forming the **chorion laeve** (laeve = with no, villi).


Vascular lacunae
Endometrial blood vessel
Syncytiotrophoblast
Cytotrophoblast
Connecting stalk
Prechordal plate
Primary chorionic villus
Yolk sac
Chorionic cavity
Amniotic cavity
Epiblast
Hypoblast
Chorion
Extra-embryonic visceral mesoderm
Extra-embryonic somatic mesoderm

3 Placenta

Definition

A temporary organ joining the mother and fetus, expelled during the birth process with the fetal membranes; together, these structures form the afterbirth.

It is formed of 2 parts (foetal and maternal):

1) Foetalpart: is the chorionfrondosum.

2) Maternal part: is the decidua basalis.

Gross morphology of the placenta

The full-term placenta has the 'shape of a circular disc with the following dimensions: (placenta = a flat cake).

1. Diameter: 15-25 cm.
2. Thickness: 3 cm.
3. Weight:300-600 gm.

The placenta has a rough irregular maternal surface, and ashiny fetal surface, as follows:

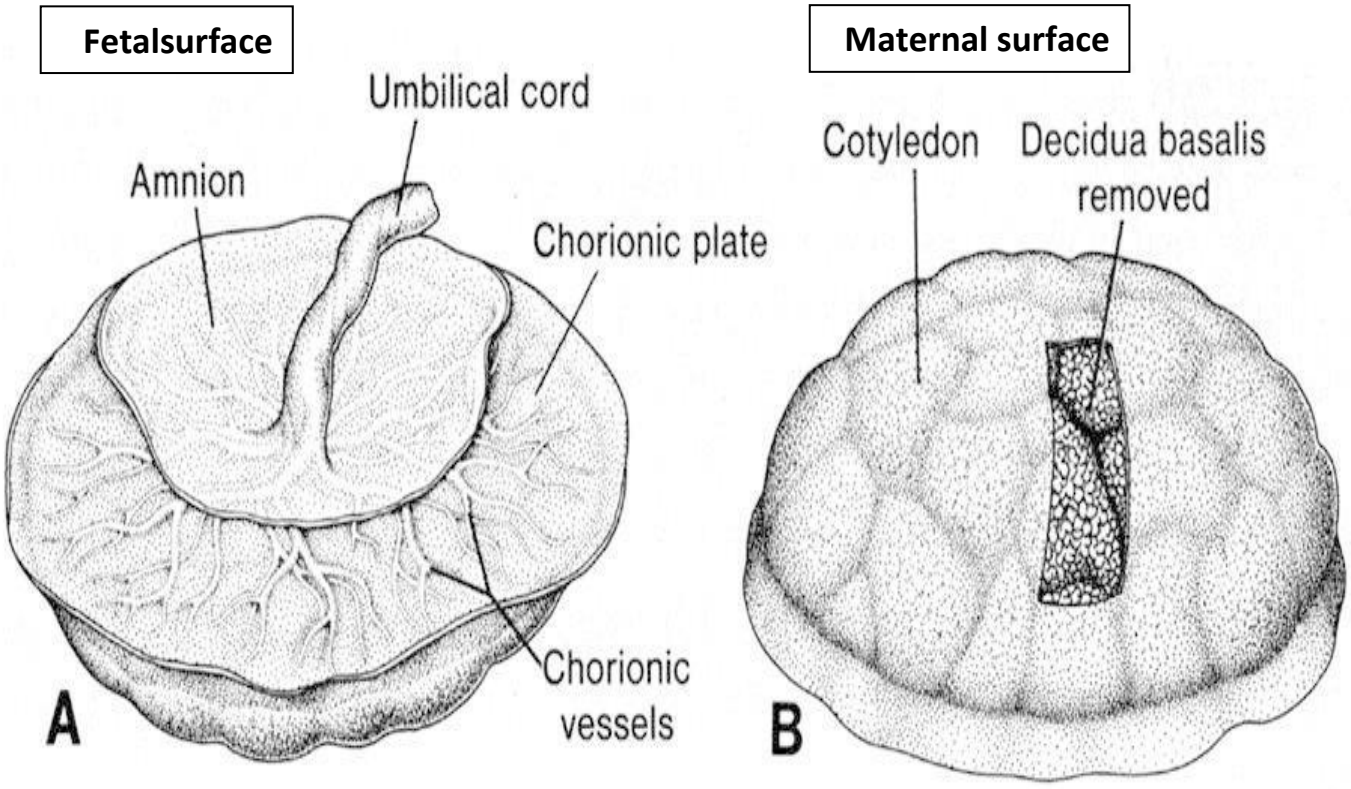
:

1. **Maternal surface**

is directed outwards towards the myometrium, and is formed by the decidua basalis. It show elevations called cotyledons and thus appears irregular, cotyledons are separated from each other by grooves which at the bases of the placental septa.

2. **Fetal surface**:

is directed inwards towards the fetus,formed by the chorionic plate. It is covered by the amnion thus appears shiny and smooth. Near its centre, it gives attachment to the umbilical cord.



Internal (microscopic) structure

The internal structure of the placenta consists of the following components

1) Chorionic villi.

2) Intervillous space.

3) Cytotrophoblastic shell.

4) Placental septa and cotyledons.

5) Placental barrier.

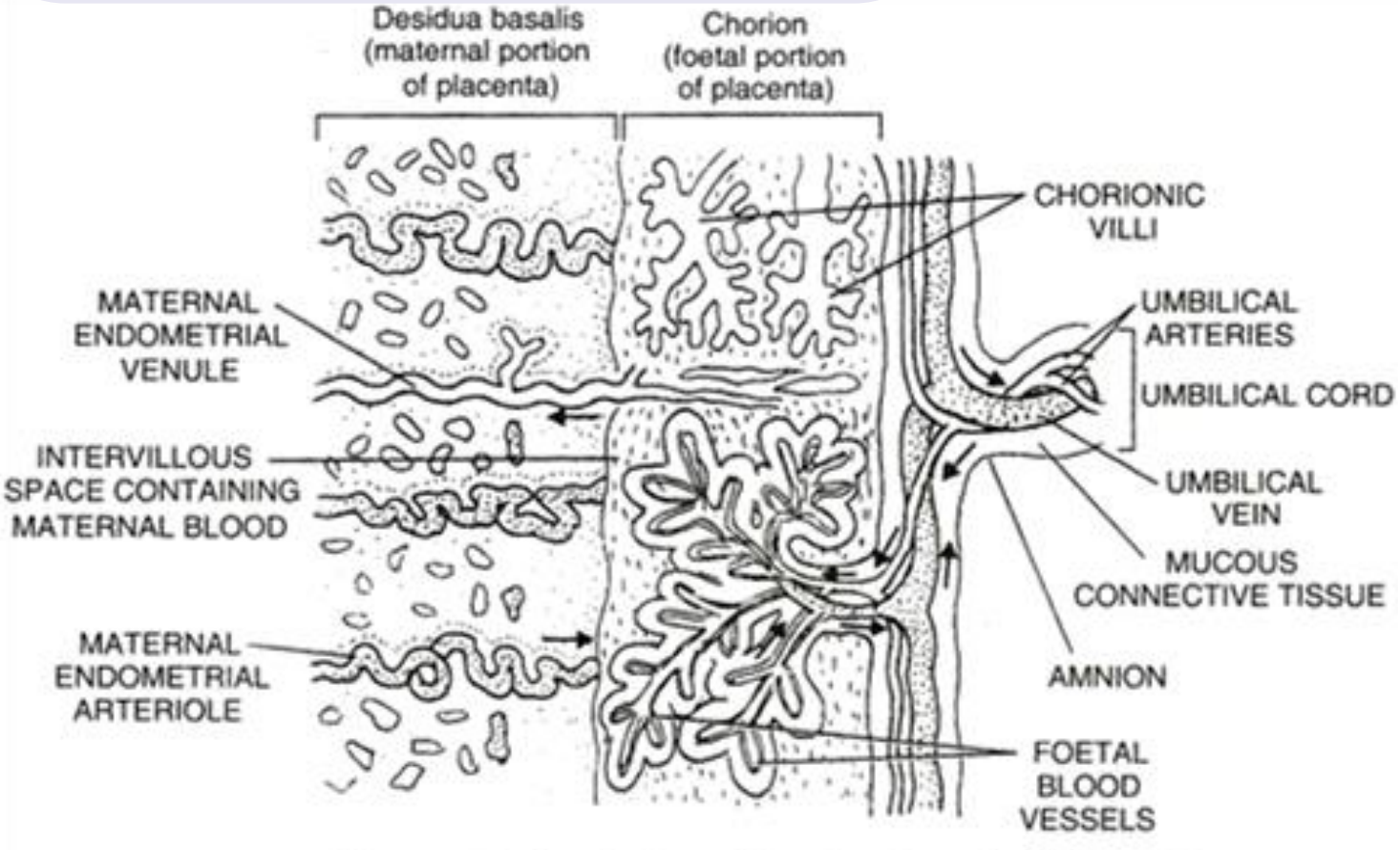
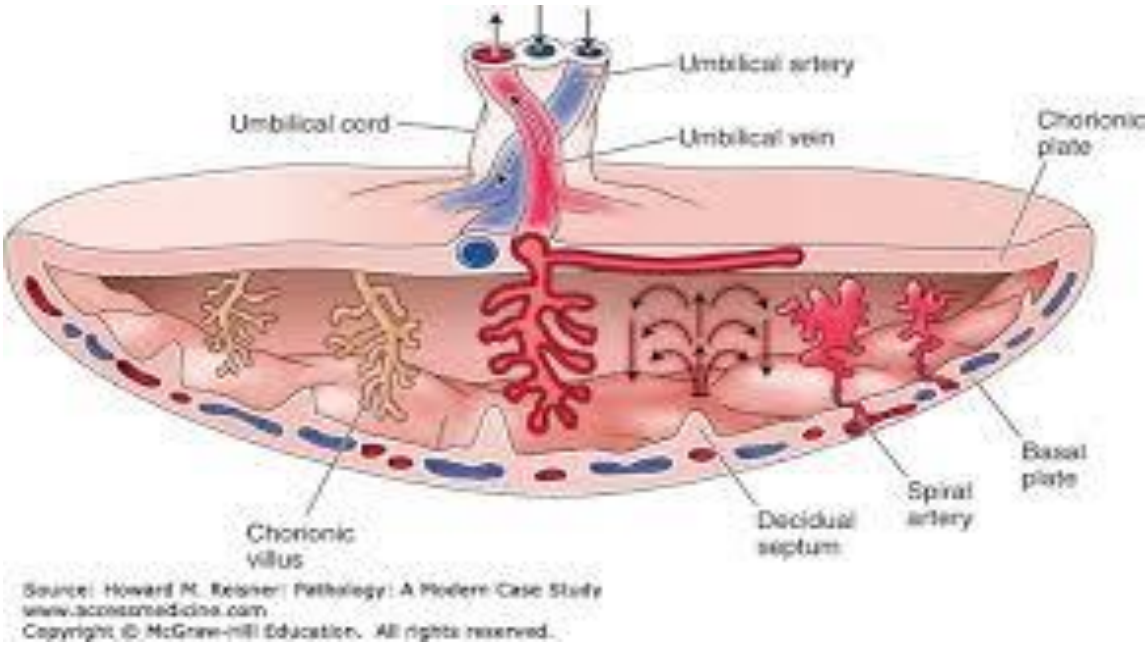


Diagram showing structure of the placenta and umbilical cord.



1 Chorionic villi:

- The villi sharing in the placenta belong to the **chorionfrondosum**, and are of the **tertiary type** which means that they possesscores of extra-embryonic mesoderm as well as foetal blood vessels.
- The area of the chorion to which these villi are attached is called **chorionic plate**.
- The villi spread from the Chorionic plate indirection to get anchored to the decidua basalis; these are **anchoring villi**.
- They are separated from each other by anintervillous space, which is filled with maternal blood.

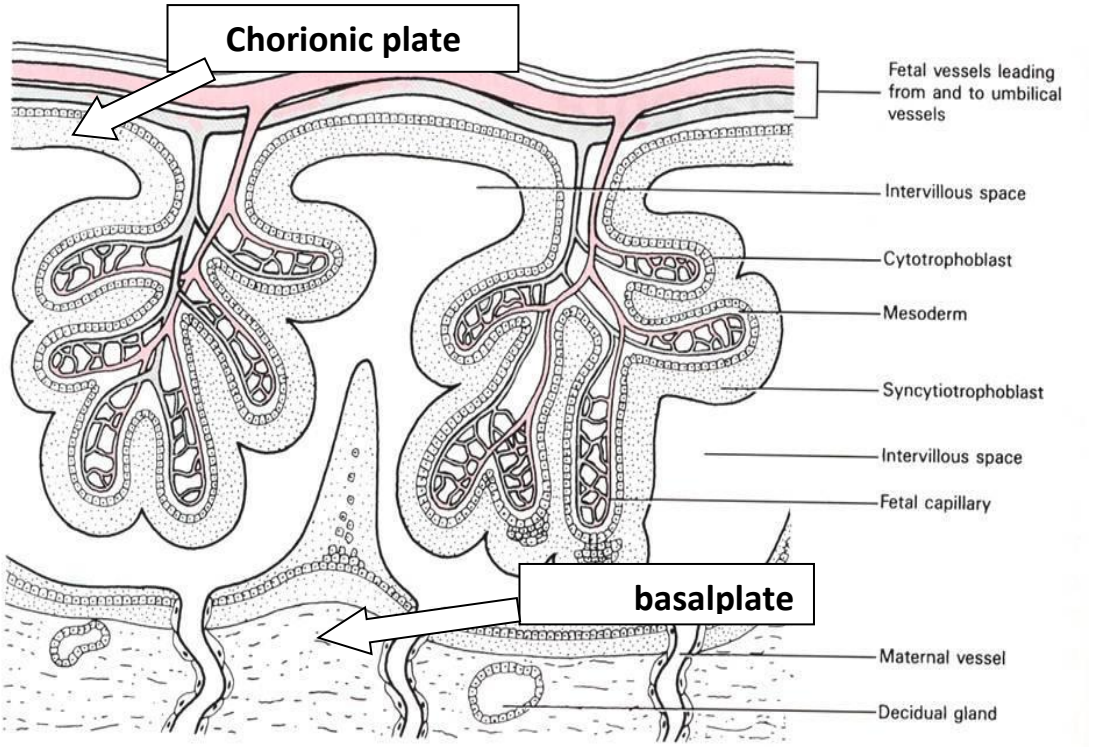
- The stem villi give off a number of **free villi** which float inside the intervillous space; they are called floating or absorbing villi.
- Through these absorbing villi exchange of materials takes place between the maternal blood and foetal blood.

2 Intervillous space:

It is the space between the villi, and is bounded towards the decidua

basalis by the **basal plate** (forming its roof), and bounded towards the chorion by the **chorionic plate** (forming its floor).

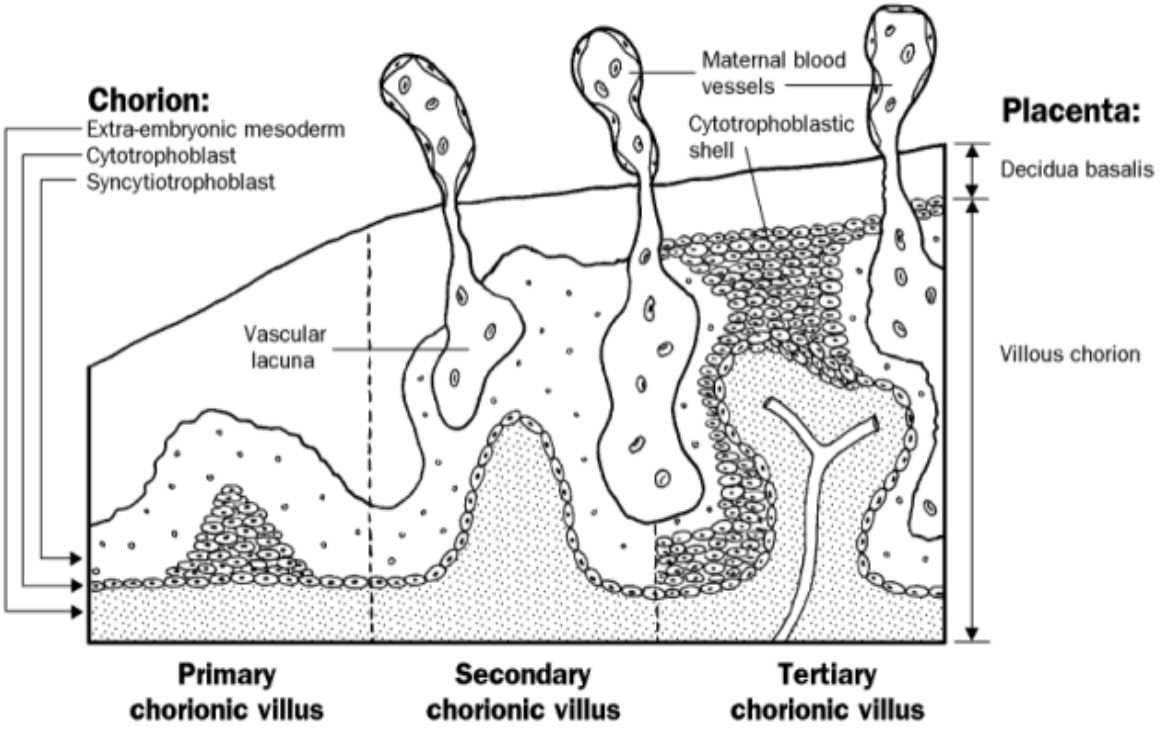
It is filled with maternal blood coming from the maternal arteries which perforate the basal plate. Its full capacity (at full-term) is about 160 cc, and its blood is in continuous renewal, circulating from the maternal arteries to the maternal veins.



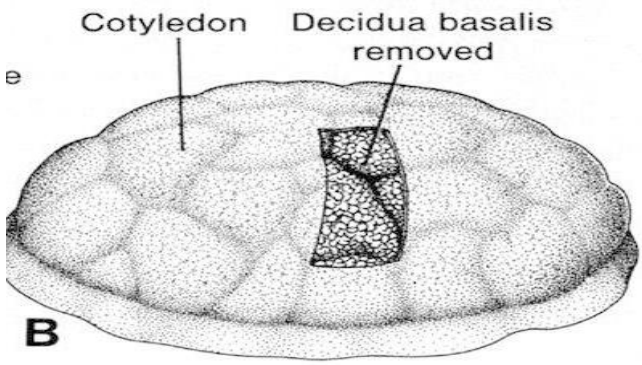
3) Cytotrophoblastic shell:

At the basal plate, the cytotrophoblast of the stem villi spreads side-wards to fuse with cytotrophoblast from neighbouring stem villi to form one continuous sheet called cytotrophoblastic shell.

It lies in the roof of the intervillous space but to the outside of the syncytiotrophoblast.



4 Placental septa and cotyledons:



✓ The placental septa are finger-like processes which project down-wards from the roof of the intervillous space, but their free ends do not reach the chorionic plate. each septum consists of syncytiotrophoblast, cytoptrophoblast and a core of decidua basalis.

✓ The base of each placental septum is marked by a fissure or groove in the decidua basalis. These fissures bound elevated areas on the maternal surface of the placenta called cotyledons.

5 Placental barrier(membrane):

The placental barrier is the part of the wall of the floating villus intervening between the maternal blood filling the intervillous space and the fetal blood in the vessels inside the floating villus.

This barrier controls the exchange of materials and gases betweenthe maternal blood and fetal blood.

It is thick in mid-pregnancy but it becomes thinner in late pregnancy. It should be noted throughout pregnancy the maternal blood never mixes with blood.

The placental barrier consists of the following layers:

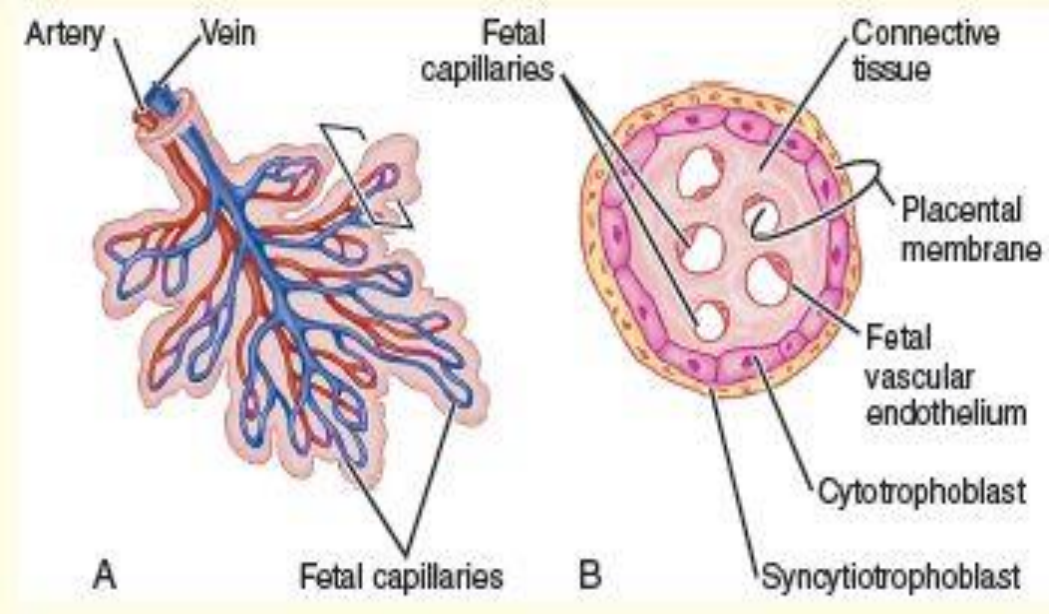
a) In mid-pregnancy:

1 -Syncytiotrophoblast covering the villus.

2 -Cytotrophoblast deep to the syncytiotrophoblast.

3 - Extra-embryonic mesoderm in the core of the villus

4 - Endothelium lining the foetal vessels.



b) In late pregnancy:

1 - The syncytiotrophoblast becomes very thin.

2 - The cytotrophoblast disappears completely.

3 - The mesodermal core becomes very thin.

4 - The endothelium of the foetal vessels persists.

Growth of the placenta

The placenta completes its formation by the 3rd month of pregnancy, and then grows in thickness as well as in surface area up to full-term, as follows:

1. It increases in thickness through elongation of the stem villi and deepening of the intervillous space.
2. It increases in surface area secondary to growth of the uterine wall.

Towards the end of pregnancy, the placenta undergoes aging changes result in thinning of the placenta as follows:

1. The villi show fibrosis resulting in reduced placental function.
2. The placenta starts to separate through a "fine of cleavage" which passes into the decidua basalis and extends into the decidua parietalis.

Functions of the placenta

1-Exchange of Gases: such as oxygen, carbon dioxide, and carbon monoxide is accomplished by simple diffusion.

***2-Exchange of Nutrients and Electrolytes*:** Exchange of nutrients and electrolytes, such as amino acids, free fatty acids, carbohydrates, and vitamins, is rapid and increases as pregnancy advances.

3-Transmission of Maternal Antibodies:

Immunoglobulins consist almost entirely of maternal immunoglobulin G (IgG), which begins to be transported from mother to fetus at approximately 14 weeks. In this manner, the fetus gains passive immunity against various infectious diseases.

4-Hormone Production:

During the first two months of pregnancy, the syncytiotrophoblast also produces **human chorionic gonadotropin (HCG)**, which maintains the corpus luteum. This hormone is excreted by the mother in the urine its presence is used as an indicator of pregnancy.

By the end of the fourth month, the placenta produces **progesterone** in sufficient amounts to maintain pregnancy if the corpus luteum is removed or fails to function properly.

In addition to progesterone, the placenta produces increasing amounts of **estrogenic** hormones. These high levels of estrogens stimulate uterine growth and development of the mammary glands.

Another hormone produced by the placenta is **somatomammotropin**. It is a growth-hormone–like substance that gives the fetus priority on maternal blood glucose .It also promotes breast development for milk production.

Congenital anomalies of the placenta

The abnormalities of the placenta are classified according to the following:

A) Anomalies due to abnormal implantation:

This is called placenta praevia. It is the attachment of the placenta to the lower part of the body of the uterus near the cervix.

B) Anomalies due to abnormal chorionic villi:



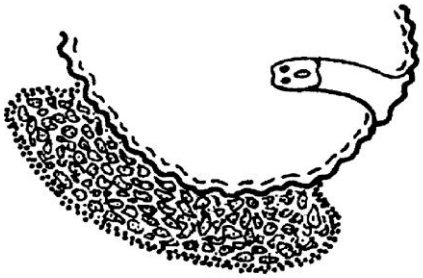
Membranaceous Placenta



Accessory Lobe



Normal Placenta



Velamentous Insertion



Bilobed Placenta

1. **Diffuse placenta**(membranous placenta): the placenta is more extensive than normal. This is due to the spread of the chorion frondosum over a wide area of the chorion.

2. **Accessory placenta** (accessory lobe): this is a separate small placenta, in addition to the main placenta

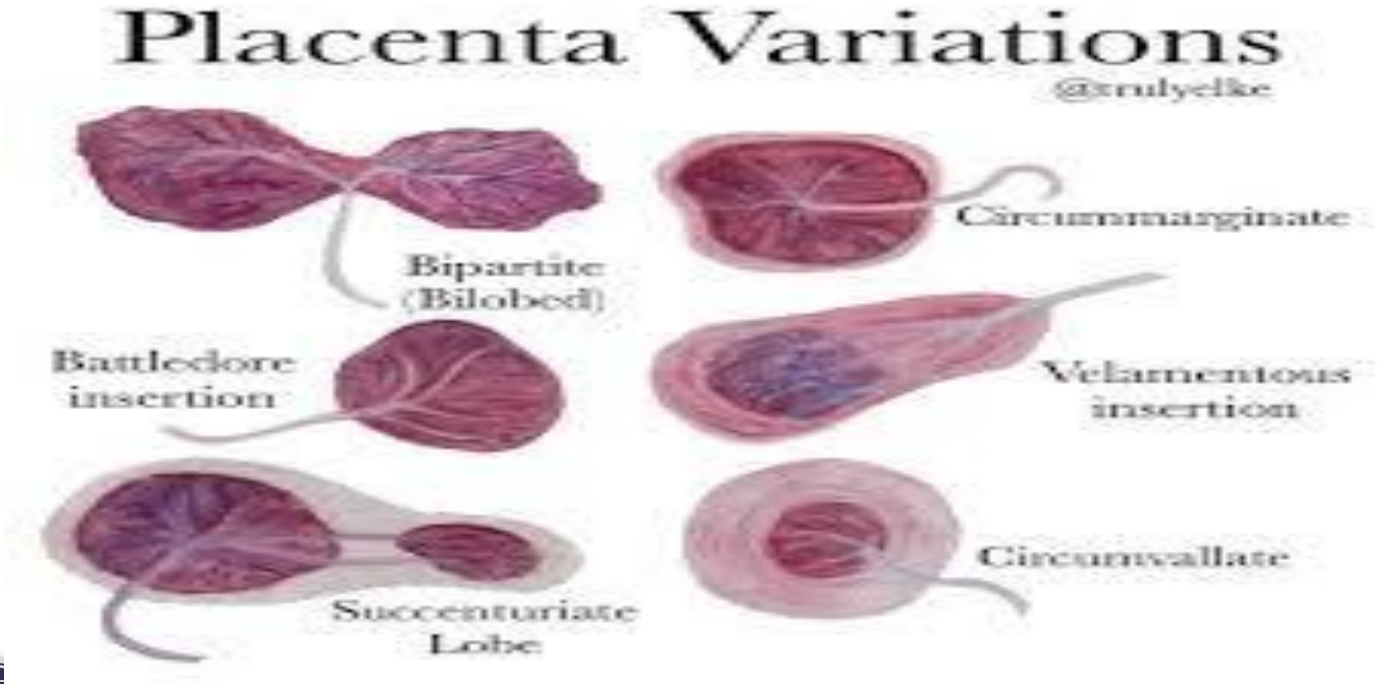
3. **Lobed placenta**(bilobed placenta): this is a placenta which is divided into 2disc-like equal parts, each of which receives a branch from the umbilical artery

4. Placenta accreta: this is a placenta which is abnormally fixed to the wall of the uterus. It is due to extensive anchorage of the stem villi to the myometrium, rather than to the decidua basalis

C) Anomalies/due to abnormal attachment of the umbilical cord:

1.**Battledore placenta**: in this type the umbilical cord is attached to the margin of the placenta, instead of the normal site near the centre of the placenta.

2. **Velamentous attachment of the cord**: the umbilical cord is attached to the fetal membranes some distance away from the margin of the placenta



4 Umbilical cord

The umbilical cord is formed as a result of the following factors:

1 Folding of the disc brings the following changes:

a - The connecting stalk together with the allantois come to lie ventrally in direct contact with the vitello-intestinal duct.

b - The umbilical' orifice is formed ventrally and is bounded by the 4 folds.

2 Expansion of the amniotic cavity results in the following changes:

a - The vitello-intestinal duct and connecting stalk elongate.

b - The extra-embryonic coelom is completely obliterated except for a few clefts left at the umbilical orifice.

c - The wall of the amniotic cavity is reflected into the outer aspect of the umbilical cord, to form a sheath around the cord.

Structure of the cord

Structure before full-term:

the cord consists of:

1. **Wharton's jel**

is loose mesenchymal tissue derived from the mesoderm of the connecting stalk as well as from the mesoderm around the vitello-intestinal duct. This tissue is rich in proteoglycans and functions as a protective layer for blood vessels.

This Wharton's jelly forms the main bulk of the cord and is surrounded from outside by a sheath derived from the amnion.

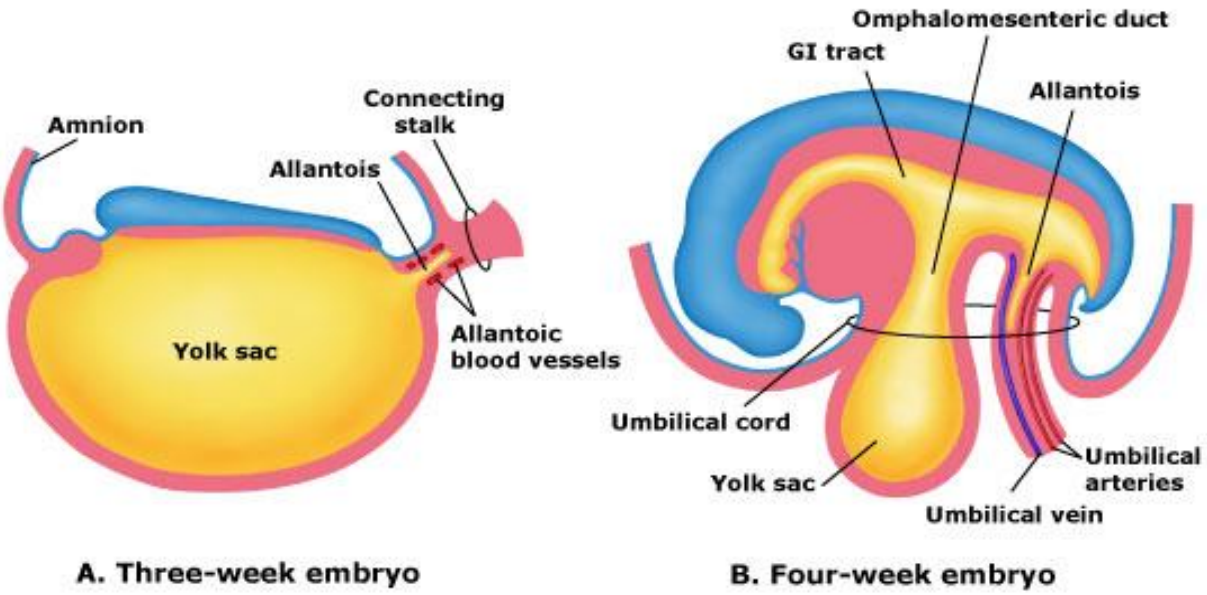
2. **vitello-intestinal duct**: connects the yolk sac with the midgut.

3. **Allantois: a duct which opens into the hindgut.**

4. **Umbilical vessels**: these are 2 umbilical arteries and 2 umbilicalveins which run in the connecting stalk to reach the placenta. Theright umbilical vein rapidly degenerates completely.

5. Vitelline vessels: run in the mesoderm around the vitello-intestinalduct, and drain the wall of the yolk sac.

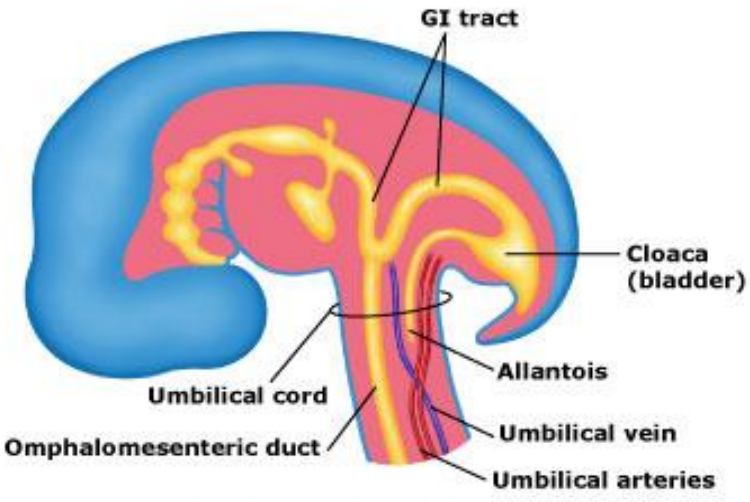
6. Clefts of extra-embryonic coelom: lie in the proximal part of the cord close to the umbilical orifice and receive the loop of midgut which herniates into these clefts.



A. Three-week embryo

B. Four-week embryo

Sagi



C. Five-week embryo

Structure at full-term

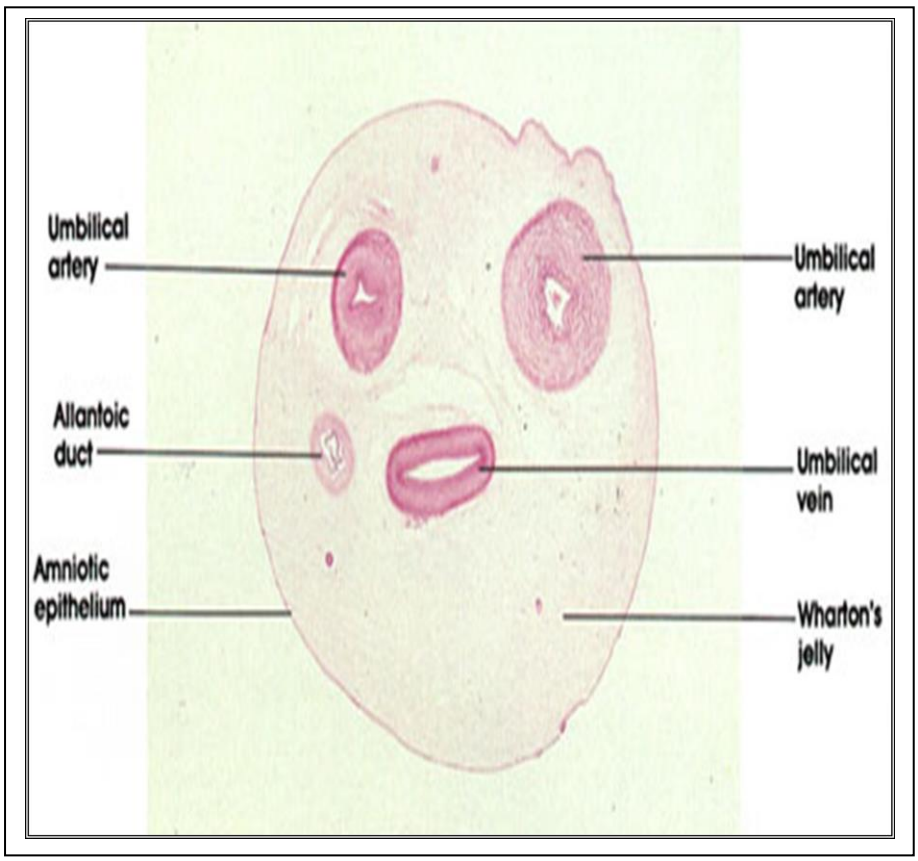
cord the has following feature:

1. It reaches the length of 50 cm.

2. It is normally attached to the fetal surface of the placenta, near its centre.

3. It shows disappearance of 2constituents: right umbilical vein and vitello-intestinal duct.

4. The cord always shows spiral torsion which may be due to unequal growth of the umbilical vessels.



Transverse section of the cord at full term

Changes after birth

At birth the cord is cut, and its constituents which lie in the body wall of the child become transformed into ligaments as follows:

1. **Left umbilical vein:** becomes the ligamentumteres of liver.
2. **Umbilical arteries:** become obliterated to form the medial umbilical

Ligaments (one on each side).

3. **Allantois:** forms the urachus which becomes fibrosed to form the median umbilical ligament.

Note

Importance of the umbilical cord:

1) It is the link between the foetus and the placenta, through which the umbilical vessels pass.

2) It gives the foetus full freedom to move within the amniotic cavity.

Congenital anomalies of the cord

A) Single umbilical artery (SUA), the most common anatomical abnormality of the umbilical cord, may association with stillbirths or fetal structural anomalies.

B) Anomalies of attachment:

1- Attachment to the margin of the placenta: this is called **battledore placenta.**

2- Attachment to the fetal membranes away from the placenta: this is called **velamentous attachment of the cord**.

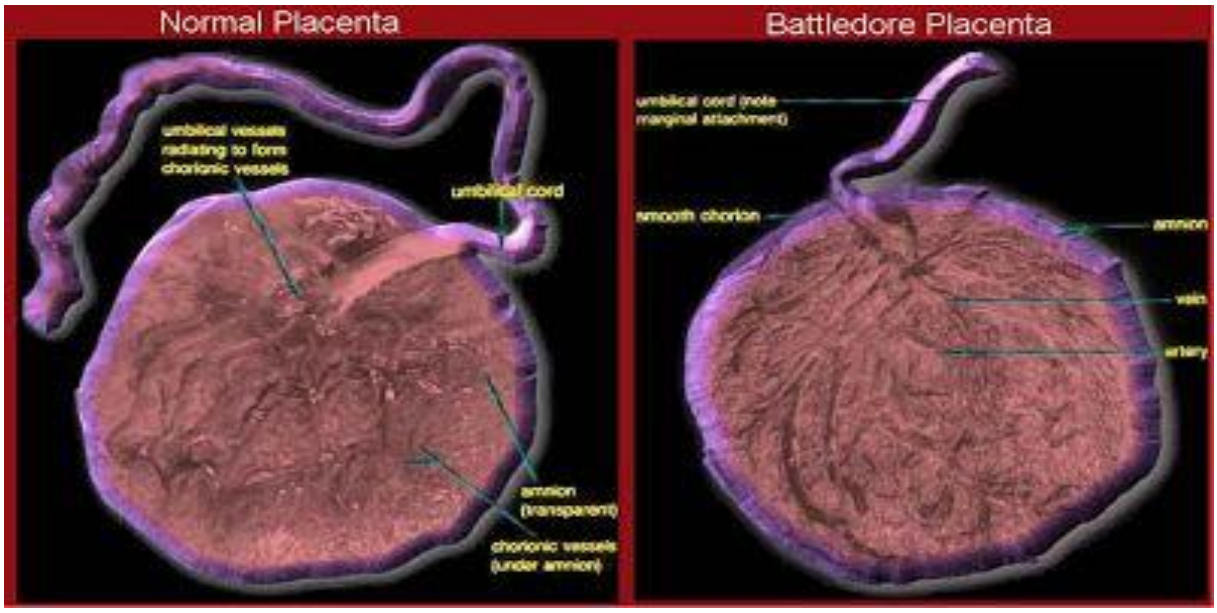
C) Anomalies of length:

1 - An **abnormally long cord** (more than 1 meter): carries the risk of forming a true knot which obstructs the blood flow, or may encircle the neck of the fetus leading to its asphyxia.

2 - An **abnormally short cord** (less than 40 cm): carries the risk of early detachment of the placenta during delivery.

D) Umbilical cord knots

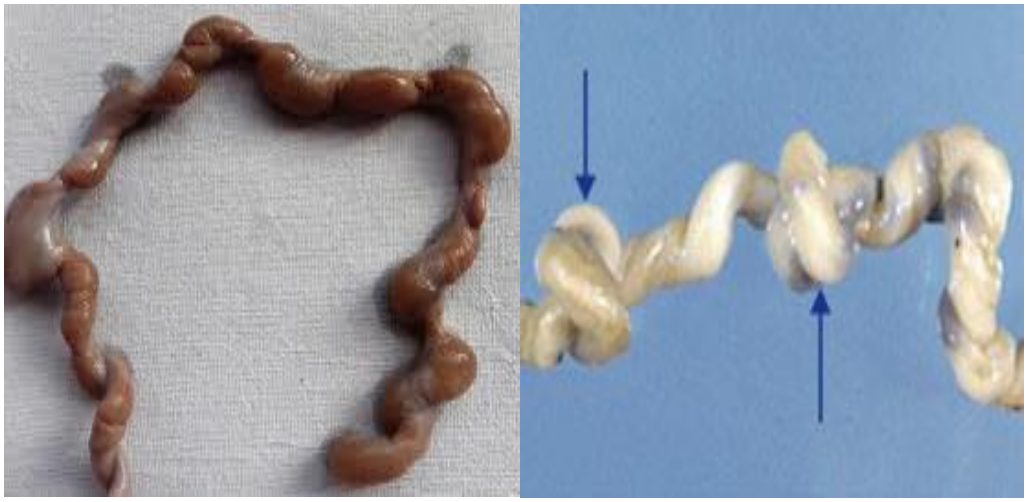
- **True knots** arise from fetal movements and are more likely to occur during early pregnancy
- **False knots** are due to umbilical artery loops within the cord.



Abnormalities of cord insertion

Velamentous insertion





False knots ***True knots***

5 Yolk sac

Definition

It is the cavity ventral to the embryonic disc.

It has 3 stage developments called

1- primitive (primary)

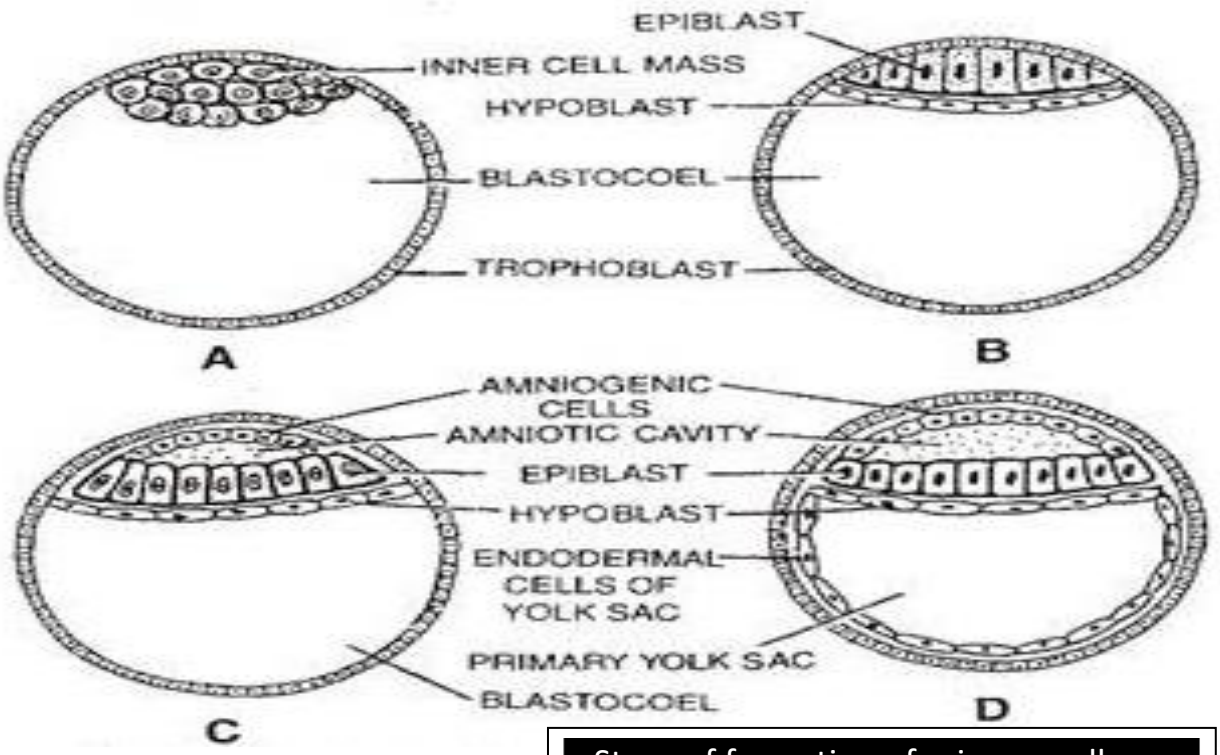
2- secondary

3- definitive

which appear one after the other.

1 *Primitive yolk sac:*

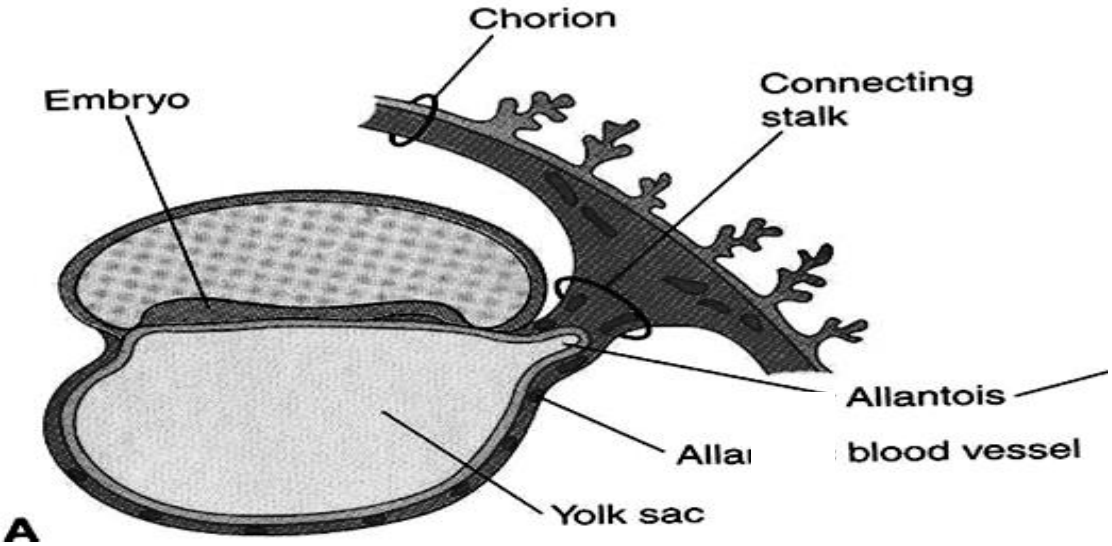
It is the earliest sac to appear. Its roof is formed by the endodermal layer of the embryonic disc, while the rest of its wall is lined with hypoblast cells in the form of a membrane called **Heuser'smembrane**.



Steps of formation of primary yolk sac

2 Secondary yolk sac:

It follows the primitive yolk sac, and is lined completely with endoderm migrating from its roof. The caudal part of the sac gives a diverticulum called **allantois** which is a diverticulum that protrudes into the mesoderm of the connecting stalk.



Secondary yolk

3 Definitive yolk sac:

It is the final stage which remains after folding of the embryonic disc. It is smaller in size and communicates with the gut through the vitellointestinal duct.

Functions of the yolk sac:

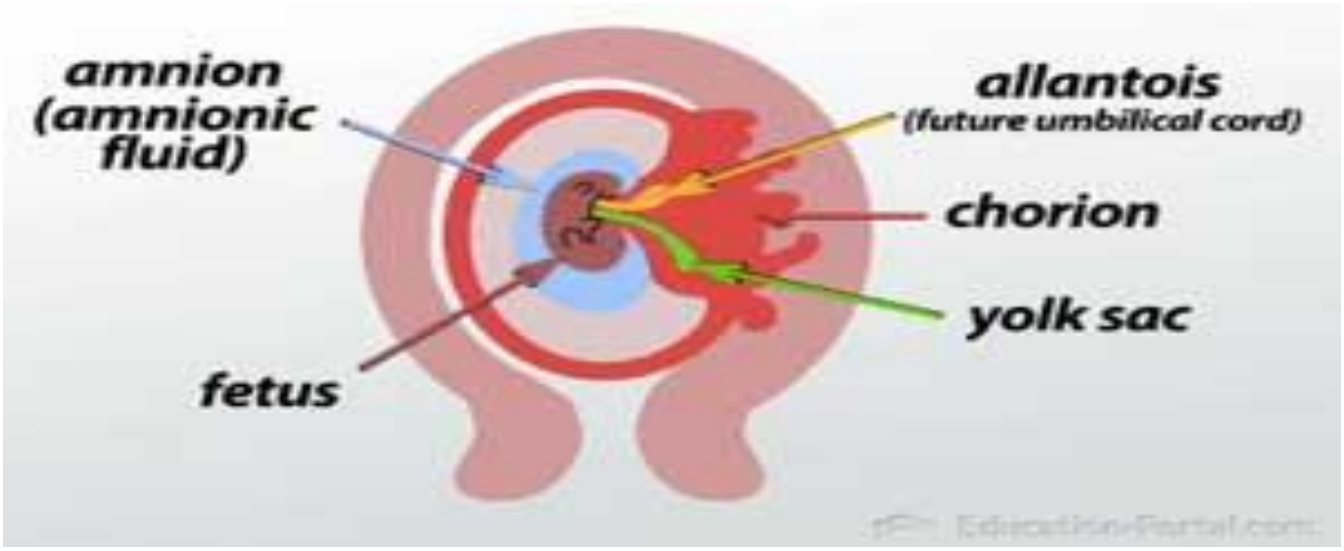
The yolk sac lining endoderm gives rise to the following derivatives:

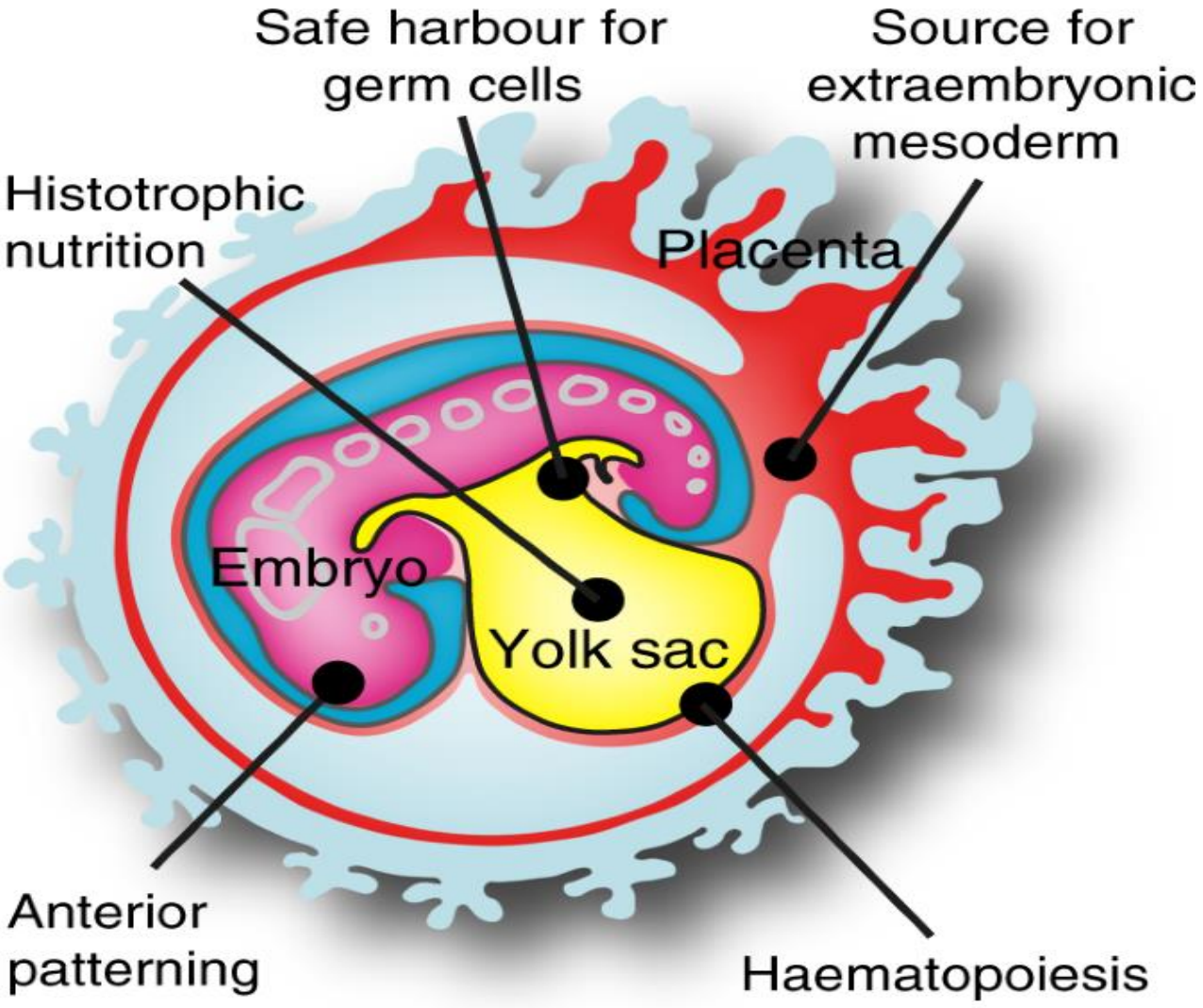
1) **The primitive gut**: derived from the roof of the yolk sac.

2) **Allantois**: derived from the caudal part of the roof of yolk sac and forms part of the urinary bladder.

3) **Primordial sex cells**: derived from caudal wall of the yolk sac and migrate to reach the gonads

4) **Blood**: develops in the mesoderm of the wall of the yolk sac.


amnion
(amnionic
fluid)
allantois
(future umbilical cord)
chorion
yolk sac
fetus


Safe harbour for
germ cells
Source for
extraembryonic
mesoderm
Histotrophic
nutrition
Placenta
Embryo
Yolk sac
Anterior
patterning
Haematopoiesis

6 Amnion

What is Amnion ?

The amnion is the "membrane" which is continuous with the ectoderm of the embryo and bounds the amniotic cavity.

- The amniotic cavity is filled with amniotic fluid which increases progressively with the growth of the fetus.

- As the amnion expands it encroaches on the cavity of the extraembryonic coelom which eventually becomes obliterated.
- The floor of the amniotic cavity is formed by columnar cells of the dorsal ectodermal layer of the embryonic disc, while the rest of its wall is formed by **flat amniogenic cells** derived from the Epiblastic cells.

characteristics of amniotic fluid

1) It is derived from 2 sources:

a -Secretions: by the amniogenic cells of the amnion

b -Excretions: by the kidneys of the fetus.

2) It is not a stagnant fluid, but is in a continuous state of renewal.

-It enters the gut of the fetus where it is absorbed into the fetal blood and from there to the maternal blood by passing through the placenta.

3) In full-term (age of 10 lunar months), the volume of the fluid normally measures 1000 cc(1 liter).

-However, **more than 2 liters** is abnormal and is called **polyhydramnios**,

- **less than 500 cc**. is also abnormal and is **called oligohydramnios**.

Functions of the amnion

1) It provides a fluid medium in which the fetus floats freely and grows evenly.

2)It prevents adherence of the fetus to the surrounding uterine walls.

3) It acts as a protective layer against the risk of injury to the fetus from external violence.

4) It maintains the body temperature of the fetus relatively constant.

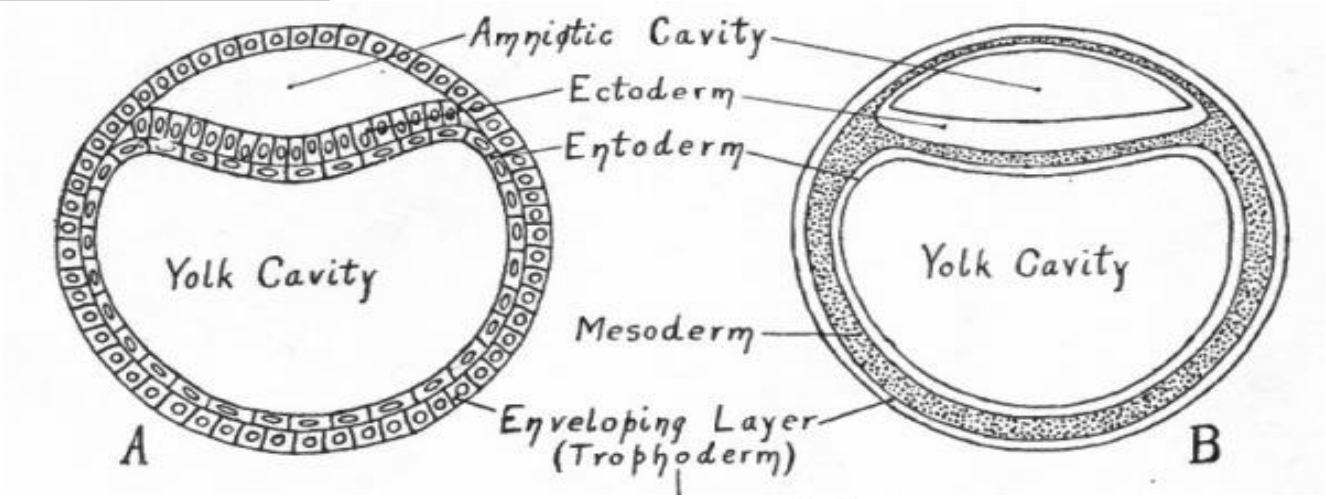
5) It helps in dilatation of the cervical at the start of delivery. This is achieved by pushing a pouch of amnion filled with fluid called “bag of waters” through the cervical canal.

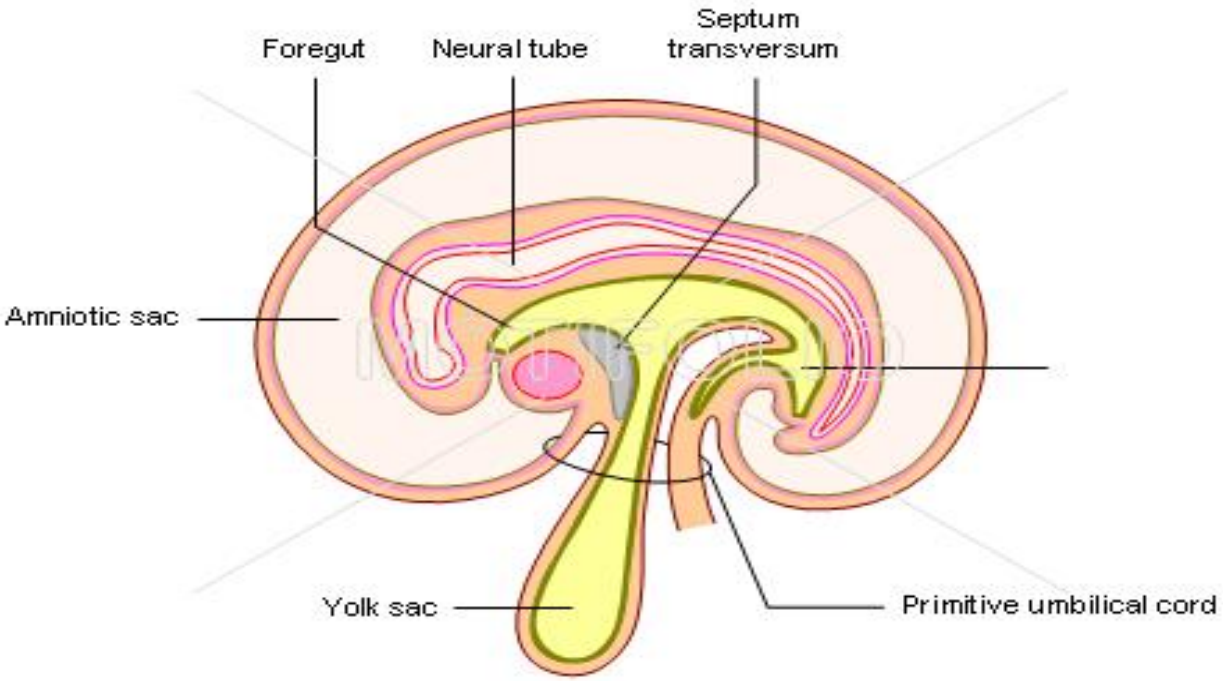
Rupture of this “bag of waters" leads to the escape of amniotic fluid; a condition called “premature rupture of membranes”.

NOTE

1- In a rare case an amniotic constriction bands are strands of the fluid-filled sac (amniotic sac) which surrounds a baby in the uterus. They may cause a congenital (present from birth) deformity of the face, arms, legs, fingers, or toes.

Before folding

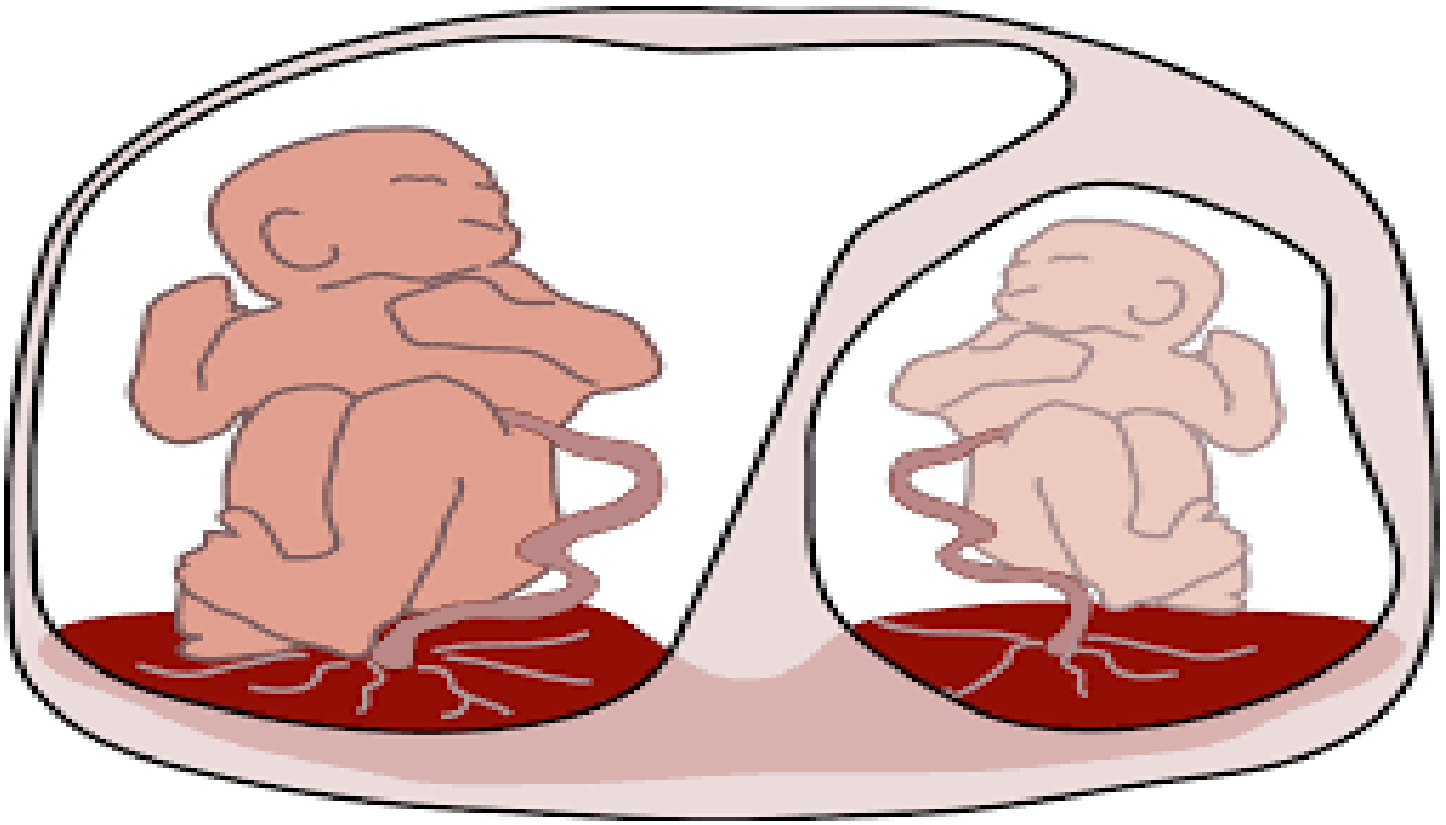




After folding

Chapter 4

Development of twins

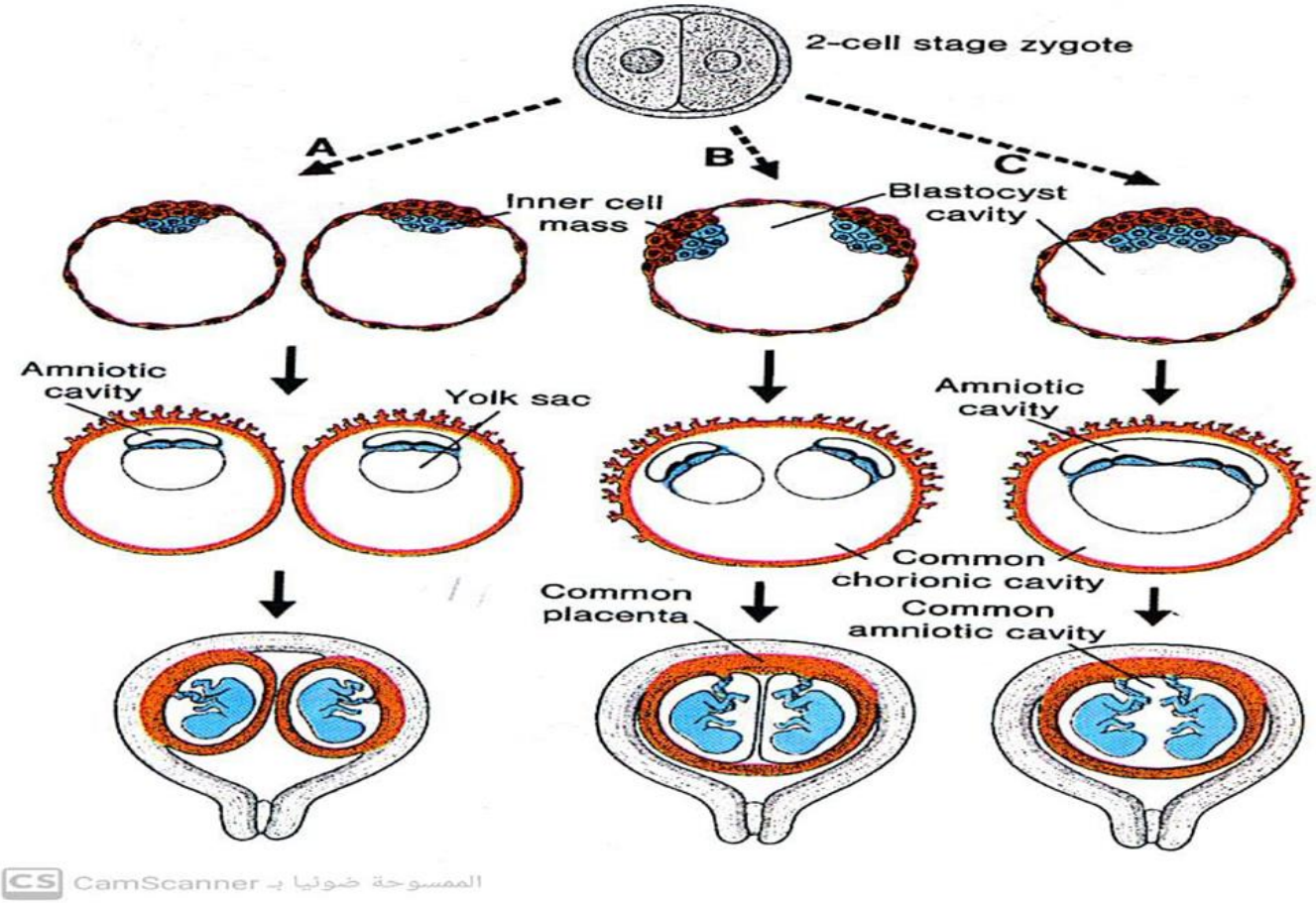


DEVELOPMENT OF TWINS

Types of twins

Dizygotic twins

Monozygotic twins



CamScanner الممسوحة ضوئيا بـ

1 Dizygotic twins:

-They result from simulataneous shedding of two oocytes and fertilization by two different spermatozoa.

-Each develops its own placenta, its own amnion and its own chorionic sac

2 Monozygotic twins:

They result from the division of a single zygote (also calledmon-ovular or identical twins). This type of twins may form at any of the following stages:

A- At the 2-cell stage of cleavage: each cell will develop into a separate embryo.

B-At the stage of blastocyst: the inner cell mass divides into 2 parts each of which forms an embryo.

C-At the stage of embryonic disc: the primitive streak splits longitudinally into 2 strands each of which induces the formation of an embryo.

Features of twins

A) Dizygotic twins

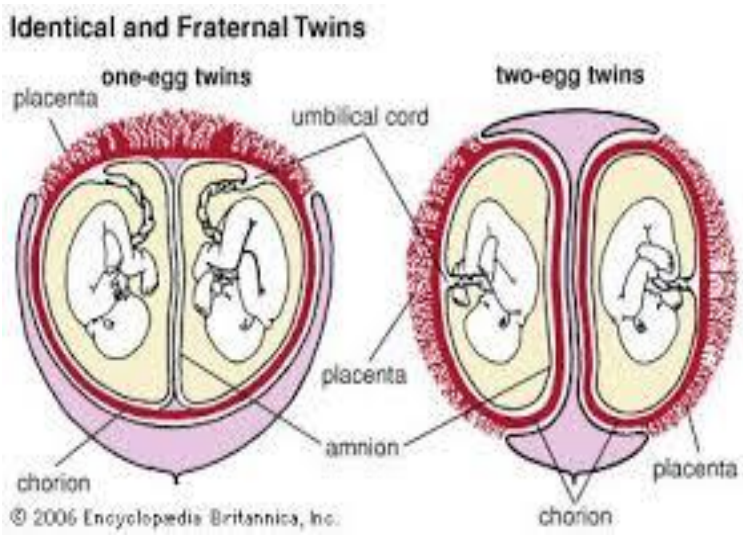
They are genetically non-identical and accordingly, the twins may be of the same or different sex.

The dizygotic twins implant as 2 separate blastocysts. Therefore, in dizygotic twins there are:

a - 2 separate placentae.

b - 2 separate amnions.

c - 2 separate chorions.



B) Monozygotic twins:

- They are always of the same sex.
- The number of placentae and foetal membranes differs accordingto the stage at which the twins are formed:

1) At the 2-cell stage: (common) each twin has its own placenta and membranes as in the case of dizygotic twins, as follows:

- 2 separate placentae.
- 2 separate amnions.
- 2 separate chorions.

2) At the stage of inner cell mass:

in this type there is oneblastocyst which is implanted (thus there is only one chorion),but the inner cell mass divides and each division have its amnionand its placenta. Accordingly, there are:

- 2 separate placentae (usually anastomose together).
- 2 separate amnions.
- One common chorion for the 2 twins.

3) At the stage of primitive streak: in this type there is only one blastocyst with one embryonic disc which is partially divided giving rise to conjoined twins. Accordingly, there are:

- One placenta for the 2 twins.
- One amnion for the 2 twins.
- One chorion for the 2 twins.

Conjoined Twins

* Conjoined twins:

These are fused monozygotic twins which occur as a result of partial splitting of the axial area of the embryonic disc.

*** Vanishing twin:** Refer to the death of one fetus (The fetal tissue is absorbed by the other twin- usually due to chromosomal abnormalities in the vanishing twin, while the surviving twin is usually healthy).

***Twin to twin transfusion syndrome (TTTS):** placenta vascular malarrangement are formed such that one twin receives most of blood flow while the other is compromised.

-Frequency of twins: The incidence of Dizygotic twins is 7-11/1000 births while the incidence of monozygotic twins is 3-4/1000

Dizygotic twins are more frequent (65%) than monozgotie twins (35%).



What Are Birth Defects?

While still in the womb, some babies have problems with how their organs and body parts form, how they work, or how their bodies turn food into energy. These health problems are called **birth defects**.

What Causes Birth Defects?

Environmental

Genetic

- **Environmental Causes**

If a mother has certain infections (such as **toxoplasmosis**) during pregnancy, her baby can have a birth defect. Other conditions that cause defects include rubella. Fortunately, many people get vaccinated against these diseases so these infections are rare.

Also, alcohol abuse by the mother may cause **fetal alcohol syndrome**, and certain medicines taken by the mother can cause birth defects.

- **Genetic Causes**

Every cell in the body has chromosomes containing genes that determine a person's unique characteristics. During conception, a child inherits one of each pair of chromosomes (and one of each pair of the genes they contain) from each parent.

An error during this process can cause a baby to be born with too few or too many chromosomes, or with a damaged chromosome.

One well-known birth defect caused by a chromosome problem is **Down syndrome**.

A baby develops Down syndrome after getting one extra chromosome. Other genetic defects happen when both parents pass along a faulty gene for the same disease.

How Are Birth Defects Diagnosed?

Many birth defects are diagnosed even before a baby is born through **prenatal tests**.



Moore KL ,Dalley AF and Agur AMR (2018): clinically oriented anatomy ,south Asian edition

Standering S, Ellis H, Healy JC ,Johanson and Williams A(2015):Gray'S Anatomy ,41th Ed. Edinburgh, London, Oxford ,Toronto. Elsevier

Mescher AL(2010): Junquira's Basic Histology ,12th Edition.New York Chicago ,London ,New Delhi ,Tornoto,Mcg Raw Hill Medical

Sadler, T. W.(2018): Langman's Medical Embryology ,14th Edition Wolters Kluwer Health